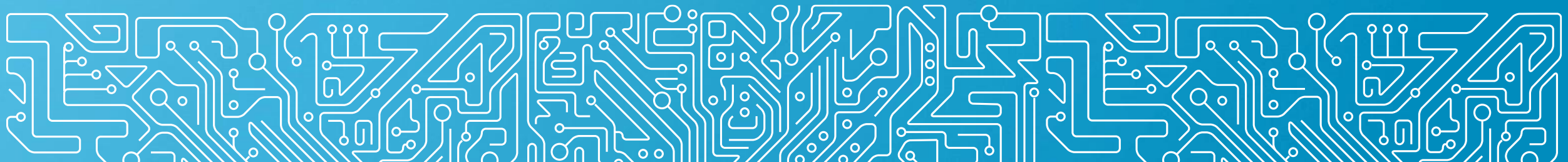
RicoSLAM: Localization and Mapping in Dynamic Environments



PhD Thesis | Ricardo Barbosa Sousa (https://sousarbarb.github.io/)
DEEC, I025, September 18, 2026

Doctoral Program in Electrical and Computer Engineering (PDEEC)
Faculty of Engineering, University of Porto (FEUP)
INESC TEC – Institute for Systems and Computer Engineering, Technology and Science

Supervisors: Prof. Dr. António Paulo Moreira (FEUP), Héber Miguel Sobreira, PhD (INESC TEC)



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PhD Thesis Jury Committee:
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(Referee) Prof. Dr. Robert Henry Deaves (University of Birmingham, UK)
(Referee) Prof. Dr. Carlos Baptista Cardeira (IST, Portugal)
(Referee) Prof. Dr. Paulo José Costa (FEUP, Portugal)
(Advisor) Prof. Dr. António Paulo Moreira (FEUP, Portugal)

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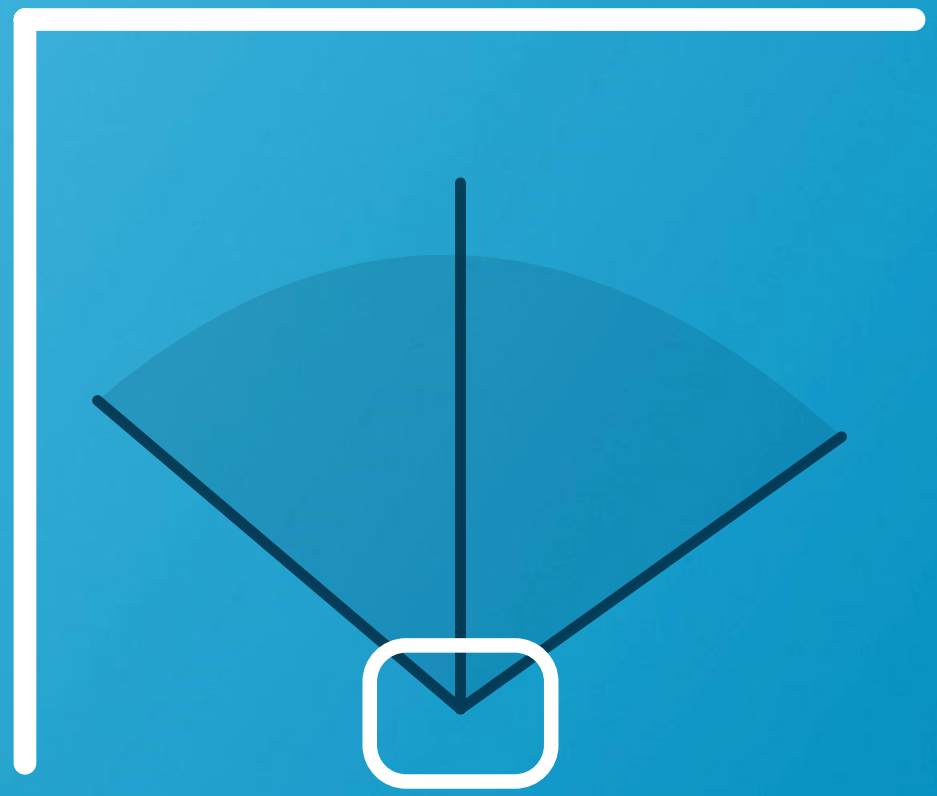

RICOSLAM FRAMEWORK


CONCLUSIONS & FUTURE WORK

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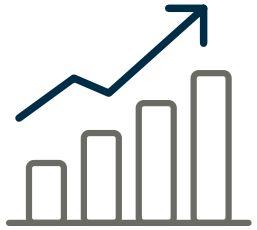
Introduction



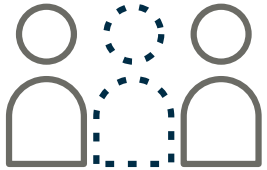
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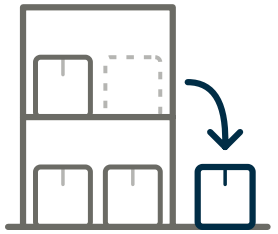
Context & Motivation



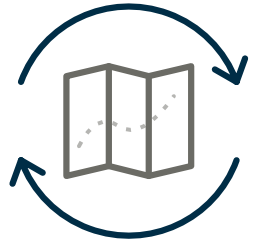
Robotics market growth



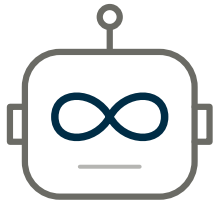
Labor shortage



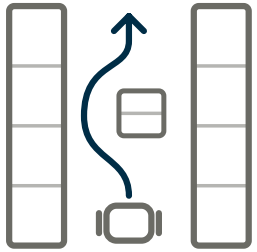
Industrial floors never stay still



Static maps become outdated



Autonomy still short-term



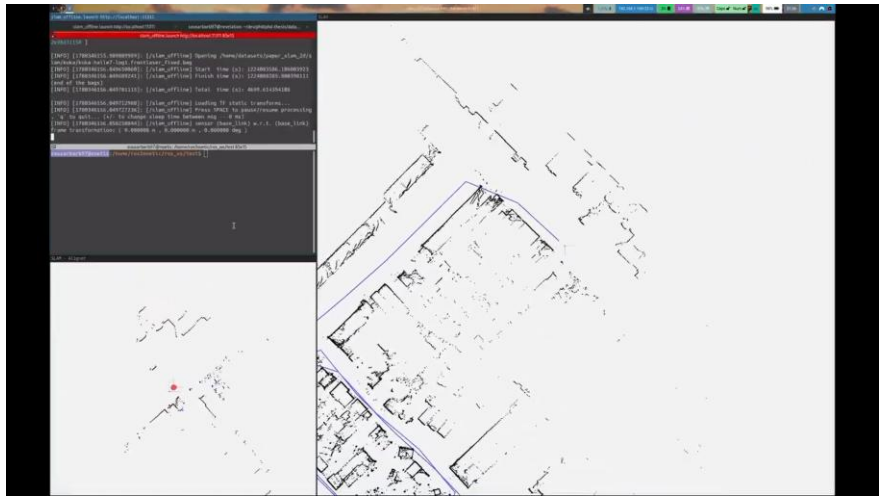
Industrial floors need robust navigation



2D on-board lasers required for safety

Problem Statement & Scope

- **Problem Formulation**
  - Localization and mapping in dynamic indoor environments with ground mobile robots
  - Not restricted to one class of dynamics:
    - Highly dynamic – people, moving objects and platforms
    - Semi-static – change outside sensor Field-Of-View (FOV), appears static when re-observed but the element moved



RicoSLAM in local map mode on kuka dataset (https://youtu.be/8hL7B-FI4G8)

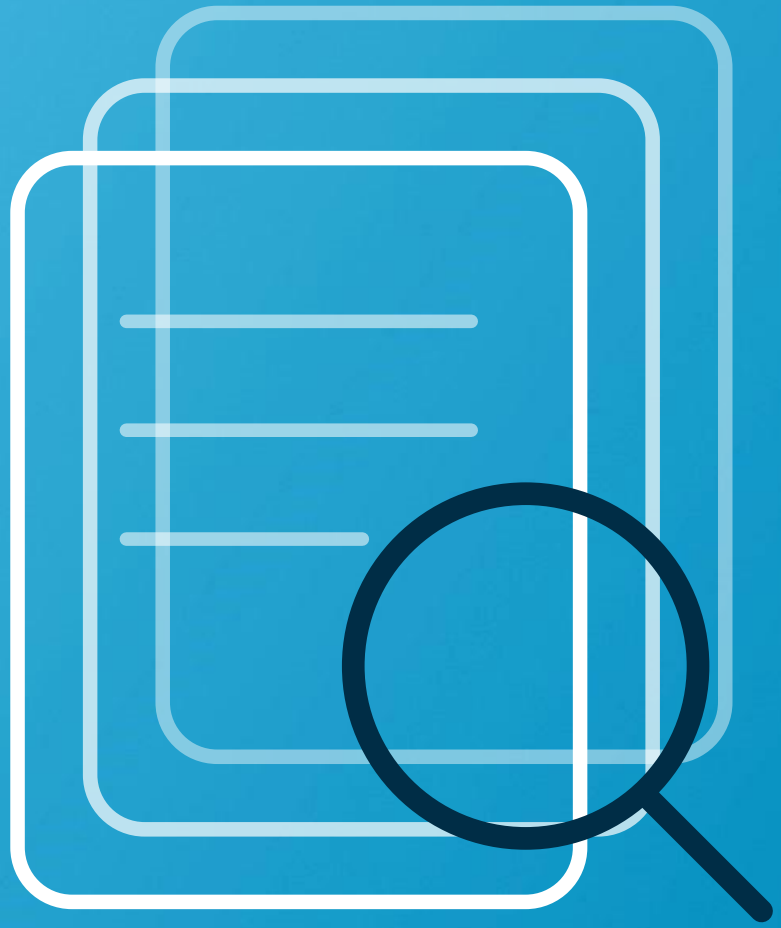
- **Scope**
  - 2D laser scanners + (optional) wheel odometry
  - 2D lasers already mandated as safety devices (ISO 10218-1:2025, ANSI/A3 R15.06-2025, ISO 3691-4:2023)
  - Planar plant floor → $SE(2)$ pose estimation
- **Context**
  - CRIIS – Centre for Robotics in Industry and Intelligent Systems, iilab – industry and innovation lab @ INESC TEC
  - GreenAuto / WP10 / PPS18 – 3D navigation system for mobile robotic equipment (PPS Leader: Flowbotic)

Research Questions (RQ) & Hypotheses (H)

- **RQ1 – How to improve mapping results in dynamic indoor environments?**
  - **H1.1:** ray tracing to update local maps online with latest observations
  - **H1.2:** relocalization into existing graph to avoid redundant information
  - **H1.3:** offline removal of dynamic points during map generation
  - **H1.4:** offline Global Bundle Adjustment (GBA) / global refinement to remove map distortions

- **RQ2 – How to integrate environment dynamics into the localization process?**
  - **H2.1:** geometric feature extraction (corners) instead of raw range data
  - **H2.2:** distance maps for correspondence-free registration + robust kernels
  - **H2.3:** continuously updated local maps for stability and improve loop closure with scan-to-map alignment

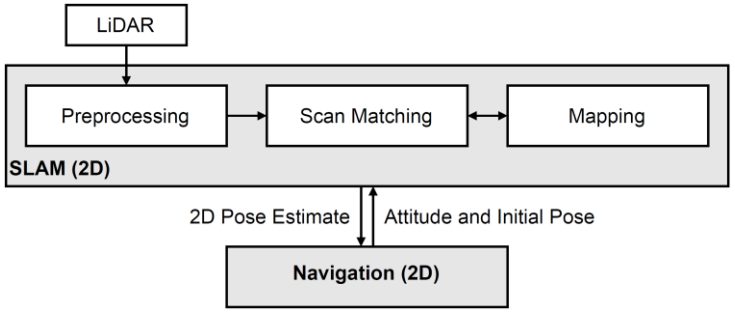
Literature Review



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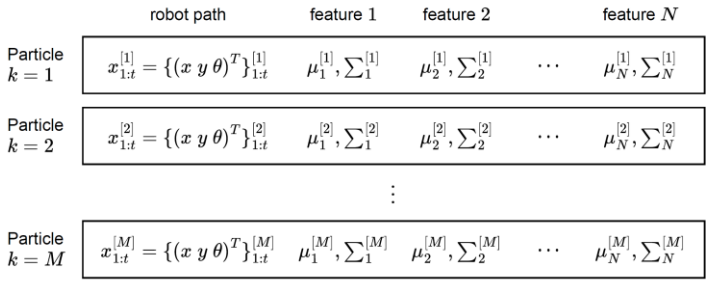
Simultaneous Localization & Mapping (SLAM)

Scan Matching



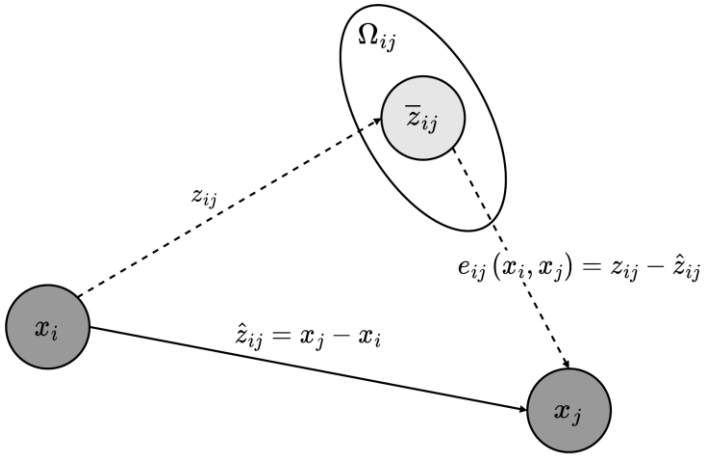
- **HectorSLAM**
- Low computational cost
- No loop closure

Particle Filters (RBPF)



- **FastSLAM, GMapping**
- Per-particle data association (multi-hypothesis)
- Each particle holds its own map
- Loop closure relies on particle diversity, sensitivity to the motion model, particle depletion

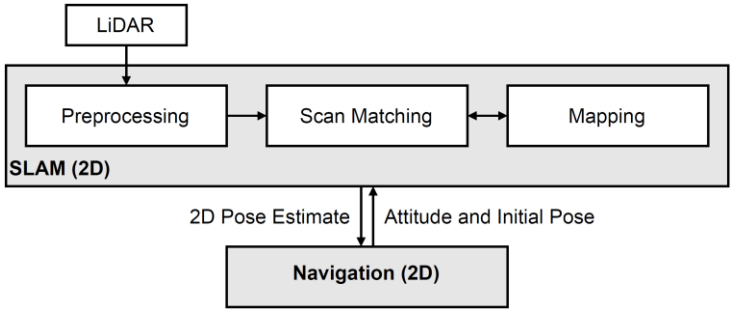
Pose Graph



- **Cartographer, SLAM Toolbox**
- Full trajectory kept (relinearize, revise associations, correct past poses on loop closure)
- Sparse Cholesky solvers (Ceres, g2o, GTSAM, srrg2_solver)
- Graph grows linearly over path length

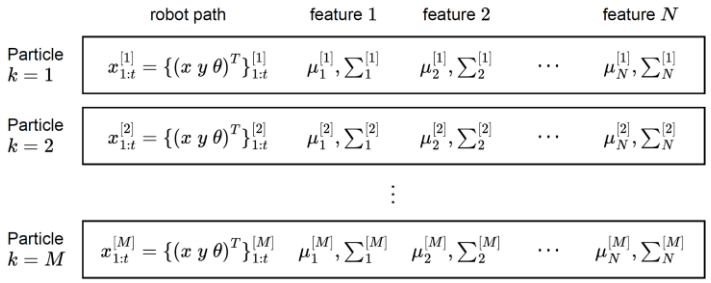
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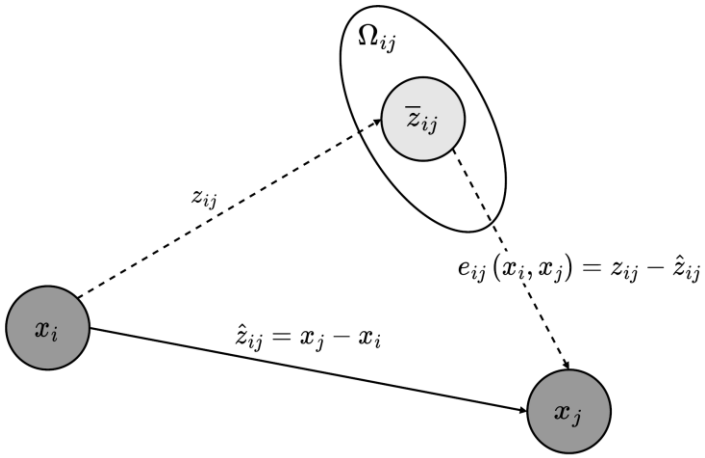
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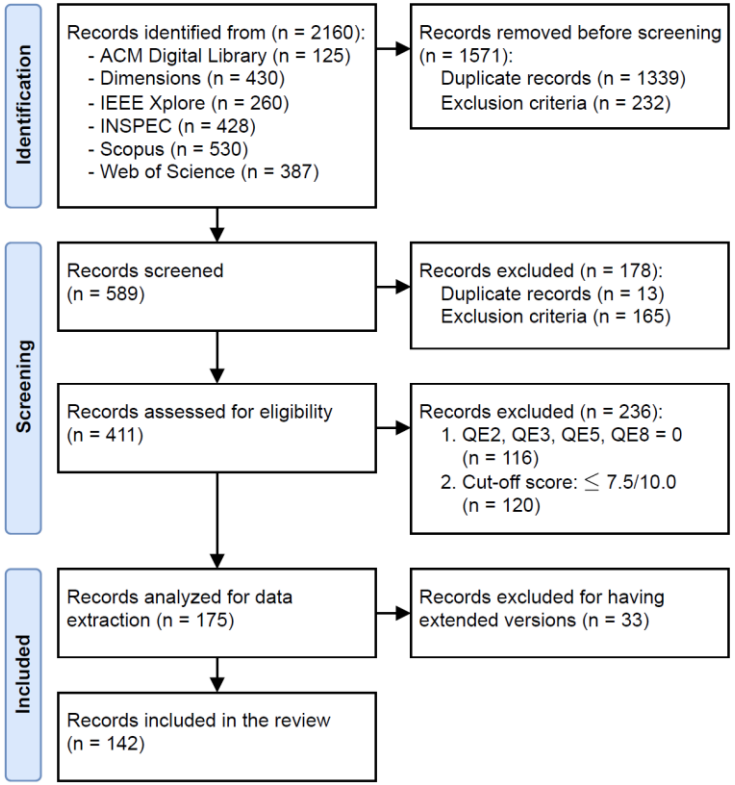
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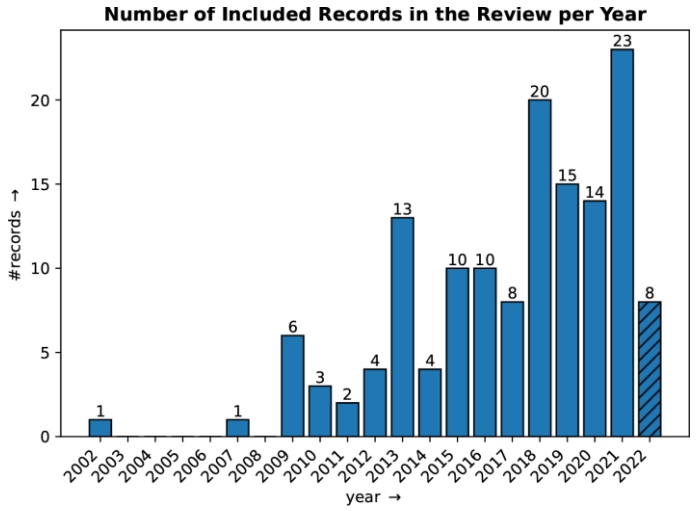
SYSTEMATIC LITERATURE REVIEW (SLR) (JFR 2023)

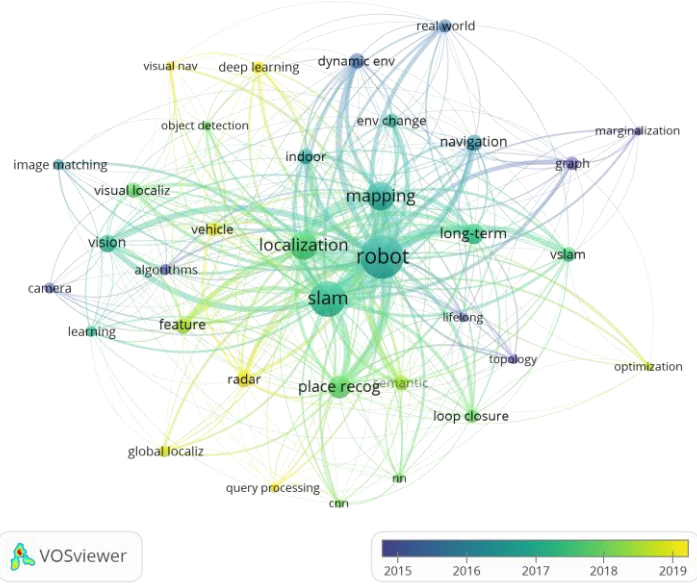



```
(robot* OR vehicle*) AND ((locali*
AND map*) OR "slam") AND ("long
term" OR "life long" OR lifelong)
```

Title + Abstract + Keywords

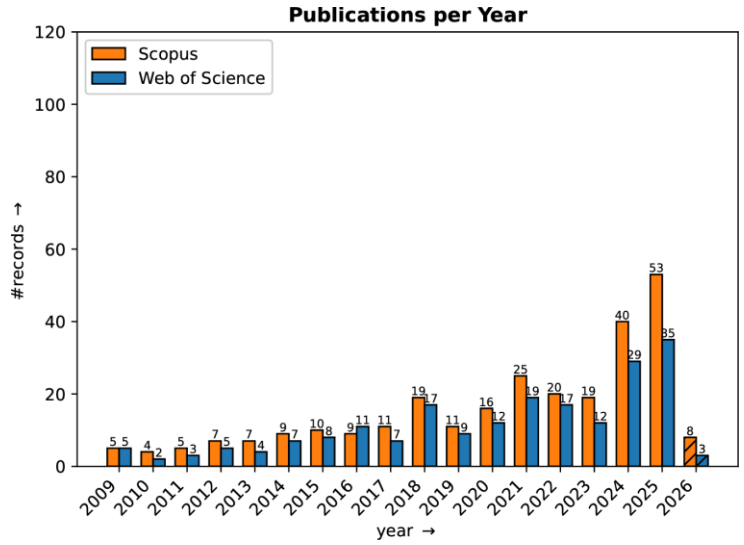
Last search: May 17, 2022





- **Why systematic?**
  None of prior surveys reported how the works were selected
- **PRISMA Statement**
- **Goals:** long-term challenges, strategies, datasets, metrics
- **142 included works**

Long-term SLAM in Dynamic Environments



(robot* OR vehicle*) AND ((locali* AND map*) OR "slam") AND ("long term" OR "life long" OR lifelong) AND (dynamic* OR mov*)

Only Scopus & Web of Science

Last search: February 13, 2026

53 included works in PhD Thesis
(32 from the SLR 2023 review + 21 new works)

Sensors camera (32), 3D LiDAR (16), 2D laser (16), radar (1)

Map Representation

- Layers (static / semi-static / dyn., permanent, movable, dynamic)
- Memory-inspired (sensory → short-term → long-term)

Map Matching

- Scan-to-map / scan-to-scan, reprojection, feature motion
- Ray tracing (free-space, multi-session differences)

Prediction Modeling

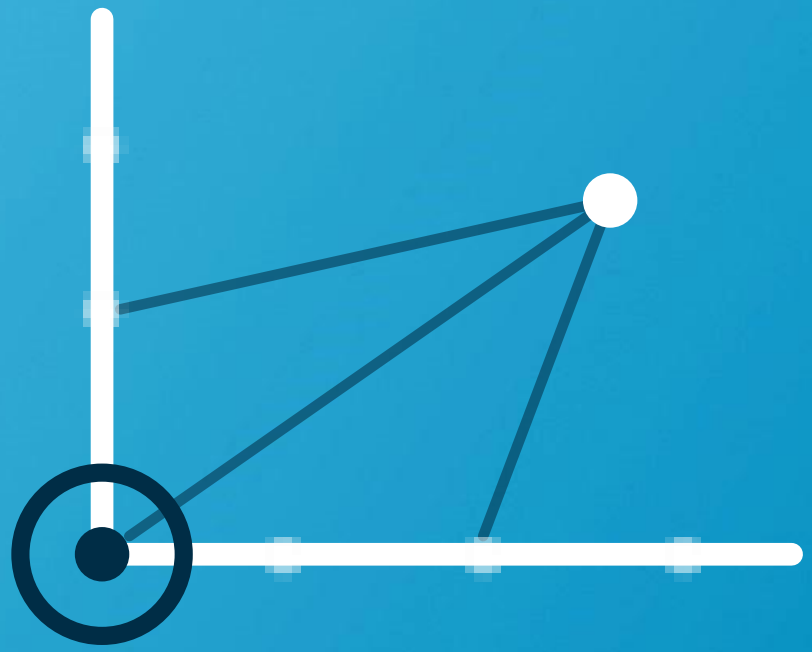
- Observation counts (hit/miss, change scores, stability histograms)
- Temporal models (HMM, ARMA, FreMen, ...)
- Output: persistence score!

Dynamic Object Detection

- Object detection, pixel-wise segmentation (camera / 3D LiDAR)
- Clustering (outliers → dynamic)

Literature is static map-oriented!
Dynamics are removed before map insertion, down-weighted, or masked...

Corner-Like Detector for 2D Lasers



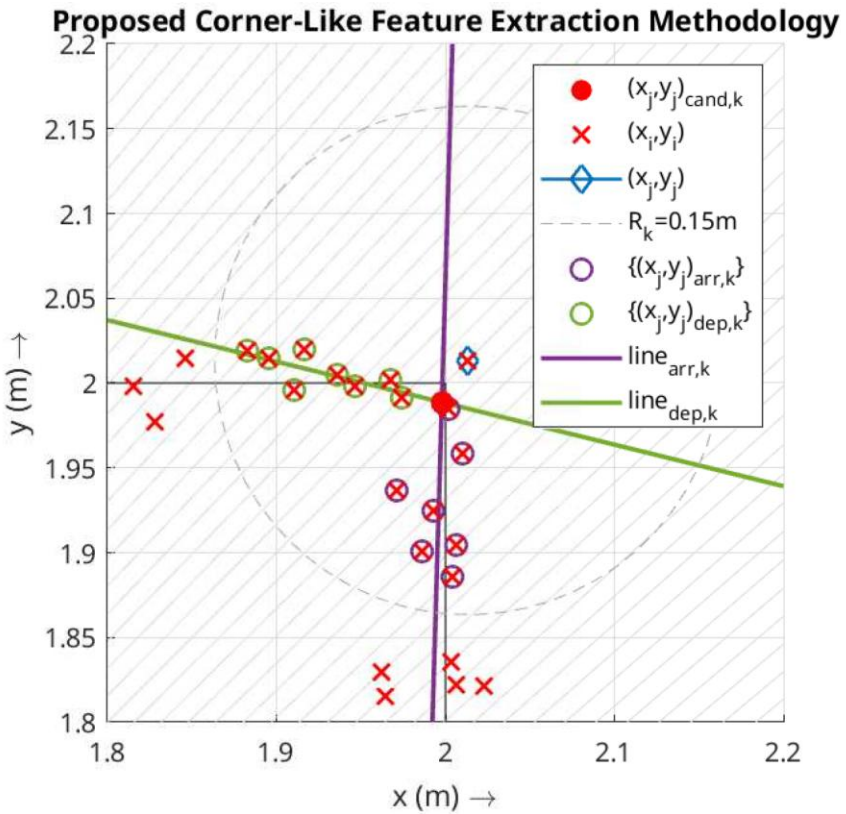
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Motivation & Research Gap

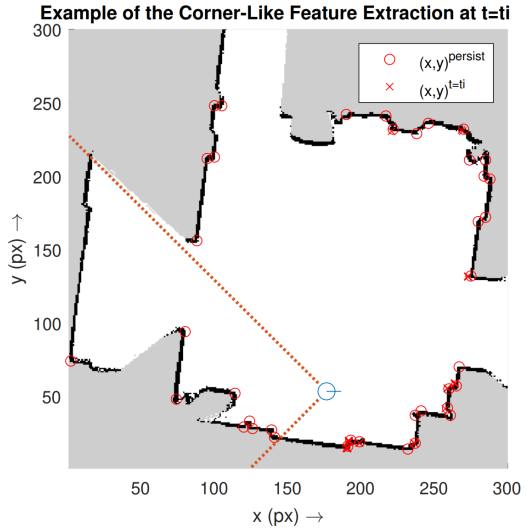
- **Why features? (H2.1)**
  - Grid maps: arbitrary shapes, but memory-heavy and semantically opaque
  - Features: compact, physical correspondence, interpretable to the human operator
  - A landmark can carry persistence → per-landmark reasoning about change
- **Literature on Corner Features Extraction**
  - Line-segment intersection → inherits the failure modes of the global line extractor; limited to well-structured scenes
  - Curvature estimation → noise-sensitive; feature keypoint anchored to a laser return (position varies with range)
- **Proposal: Line Fitting-Based Corner-Like Detector for 2D Laser Scanners Data**
  - Closed-form line parameters and fitting error (no RANSAC, no iterative optimization)
  - Local fitting over multiple distance scales R_k
  - Keypoint = intersection centroid across valid scales → may not coincide with any laser return

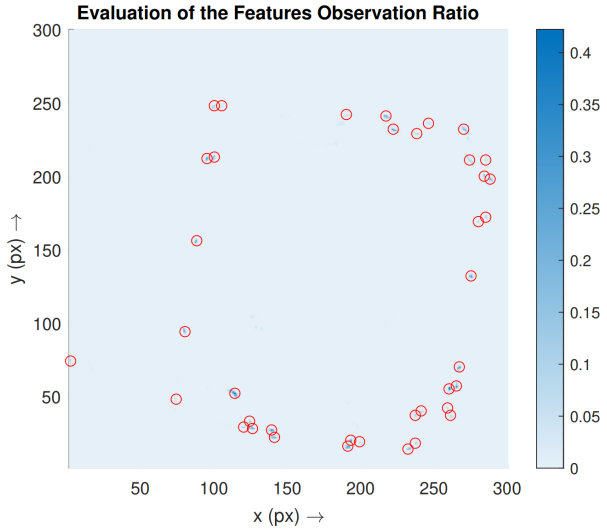
FEATURE EXTRACTION ALGORITHM (ICARA 2024)

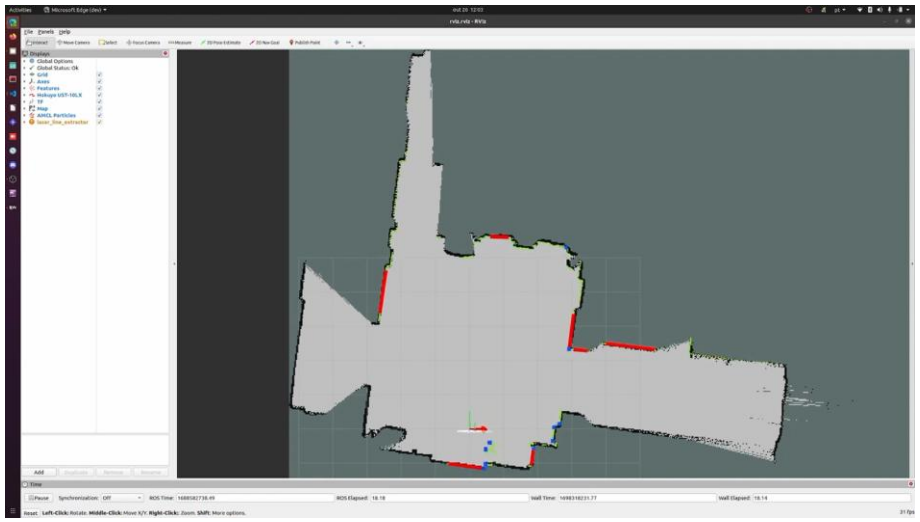
1. **Evaluate each of the laser points (x_i, y_i) as a candidate keypoint to a corner feature:**
   - Retrieve the arrival $\{(x_j, y_j)\}_{arr,k}$ and departure $\{(x_j, y_j)\}_{dep,k}$ neighborhoods considering a search distance radius R_k
   - Compute the fitted lines to each neighborhood using the best 2D line model according to the ε_{RMSE} ($y = \beta_0 + \beta_1 x$ versus $x = \alpha_0 + \alpha_1 y$)
   - Estimate the intersection point between the two lines
   - Repeat the previous steps for different consecutive search radius R_k
2. **Determine if the candidate point is a corner feature by evaluating the following indicators:**
   - Minimum number of points in each neighborhood and scale R_k
   - Absolute angle between arrival and departure fitted lines
   - Maximum ε_{RMSE} in line fitting
   - Maximum distance of the intersection point on multiple scales R_k to its centroid
   - Maximum distance of intersection point to current laser point (x_i, y_i)
   - Validation of the previous indicators in a minimum number of consecutive scales R_k
3. **If the candidate point is a corner feature, define its keypoint as the centroid of the intersection points in the valid scales**



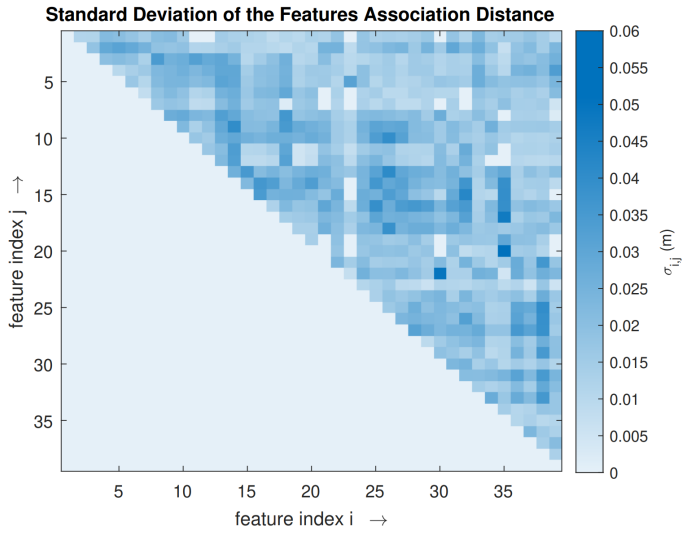
EXPERIMENTAL RESULTS

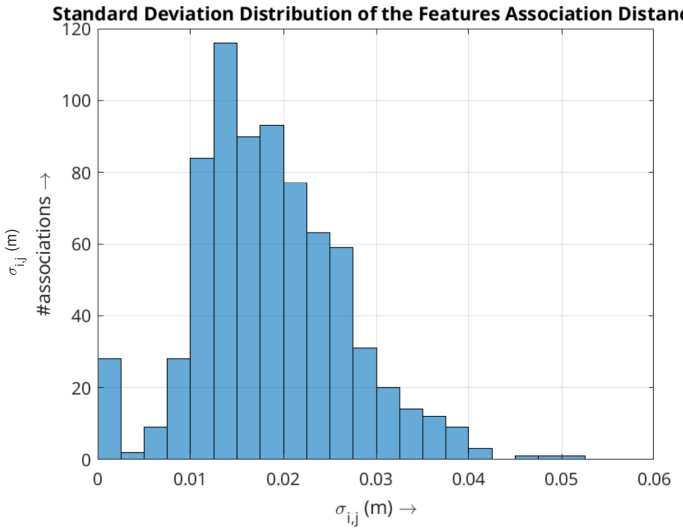






Integration of the proposed corner-like detector with https://github.com/kam3k/laser_line_extraction for 2D laser scanners (https://youtu.be/7xMKtlvd0wU)





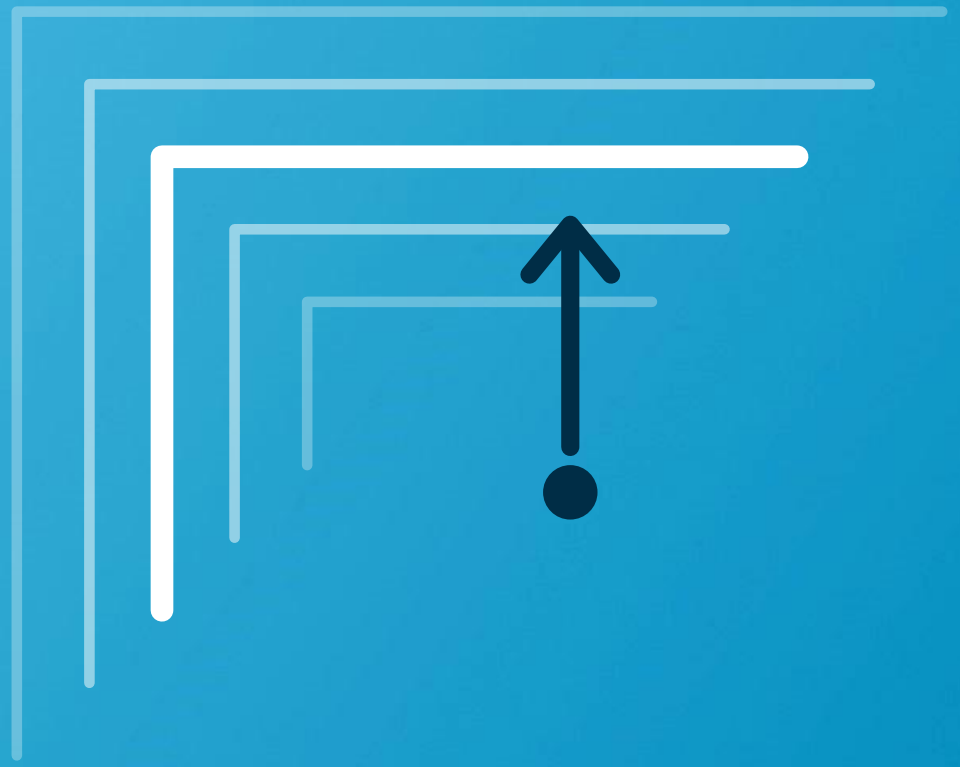
- **Observability ratio (viewpoint invariance)**
  39 persistent keypoints manually selected;
  ≥ 50% (20) → ≥ 75% (16) → 100% (6) observability ratio considering $\mathcal{N}_8$ neighborhood;
  elsewhere stays below 1% → few false positives!
- **Association distance (geometric consistency)**
  91.5% of the physically possible associations have σ ≤ 0.03 m, i.e., Hokuyo UST-10LX noise

TAKEAWAYS & DESIGN DECISION

- Viewpoint invariance
- Geometric invariance below the sensor noise floor
- Low false positives
- Independent of a global line extractor

- **Why not using the corner detector in the RicoSLAM framework?**
  - Features do not remove the dynamics problem
  - A detector is not yet an estimator (covariance, association, descriptors)
  - Distance maps deliver data association + cost at $O(1)$ lookups
  - Start with a SLAM framework focused on a single data type → evolve to hybrid raw-scan / feature / beacon SLAM

UDF-Based Front-End



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Motivation

- Classical **Iterative Closest Point (ICP)** algorithms have two steps on every iteration of the optimization:
  - Data association: Nearest-Neighbor (NN) search (kd-tree, $O(\log n)$ per point)
  - Error minimization

- A **distance map** $\mathcal{D}(\mathbf{y}) = \{\mathbb{R}^2 \rightarrow \mathbb{R}_0^+ : \|\mathbf{y} - \mathbf{p}^*(\mathbf{y})\|_2\}$ is built once over the reference cloud $\mathcal{P}$
- **Data association solved inside the error evaluation with an $O(1)$ grid lookup per point, per iteration**
- **Unsigned Distance Field (UDF)** is continuous everywhere, differentiable almost everywhere, and satisfies the Eikonal property $\|\nabla\mathcal{D}\|_2 = 1$ of Euclidean distance fields

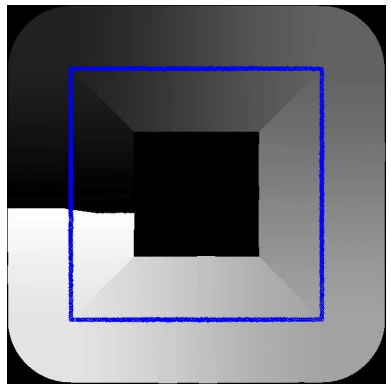
- **How to handle outliers on UDFs?**
  - M-estimator re-weighting (Cauchy) inside Gauss-Newton
  - Truncation at $d_{\max}$ (constant $\mathcal{D}$, zero gradient, like a truncated L^1 / L^2 robust kernel)

Unsigned Distance Field (UDF) Data Structure

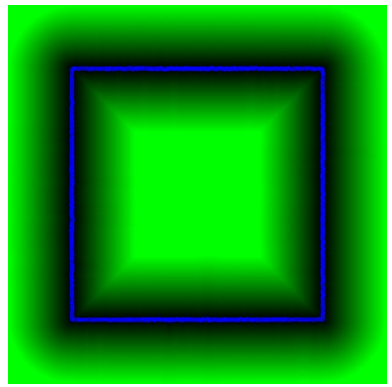
- $\mathcal{M}_0(\mathbf{y}) = k^*(\mathbf{y})$
  - Only the index of the NN point in $\mathcal{P}$
- $\mathcal{M}_1(\mathbf{y}) = \left(\mathcal{D}(\mathbf{y}), \frac{\partial \mathcal{D}}{\partial y_1}(\mathbf{y}), \frac{\partial \mathcal{D}}{\partial y_2}(\mathbf{y})\right)$
- $\mathcal{M}_2(\mathbf{y})$ is $\mathcal{M}_1$ plus all 4 Hessian entries

- **UDF Computation**
  - Best-first wavefront (Dijkstra-like) over $\mathcal{N}_8$
  - Seeds are the occupied cells with $\mathcal{D} = 0$
  - Propagation stops at $d_{\max}$
  - Finite-difference post-processing steps compute the 1st / 2nd-order derivatives

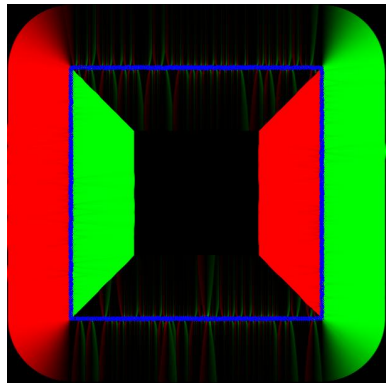
Variant	Storage	#val./cell	Query	Truncation
Nano $\mathcal{M}_0$	Dense array	1 (int)	$O(1)$	Optional
Standard $\mathcal{M}_{1\|2}$	Dense array	3 – 8	$O(1)$	Optional
Sparse $\mathcal{M}_{1\|2}$	`std::unordered_map`	3 – 8	$O(1)$ avg.	Advisable



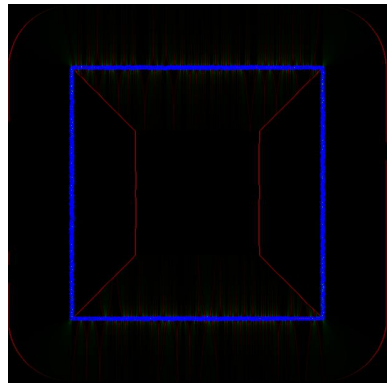
$k^*(\mathbf{y})$



$\mathcal{D}(\mathbf{y})$



$\frac{\partial \mathcal{D}}{\partial y_1}(\mathbf{y})$



$\frac{\partial^2 \mathcal{D}}{\partial y_1^2}(\mathbf{y})$

Optimization & Error Formulations (ROBOT 2026)

Algorithm .1: Gauss-Newton minimization algorithm with robust estimators for non-Euclidean measurement and state spaces (a single iteration)

Input: Current state estimation $\mathbf{x}_k \in \mathcal{M}$, measurements $\mathcal{C} = \{\langle \mathbf{z}^{[i]}, \mathbf{\Omega}^{[i]} \rangle\}$, robust loss function ρ

Output: State estimation for the next iteration step $\mathbf{x}_{k+1} \in \mathcal{M}$

$\tilde{\mathbf{b}} \leftarrow \mathbf{0}_{n \times 1}$

$\tilde{\mathbf{H}} \leftarrow \mathbf{0}_{n \times n}$

forall $i = 1, \ldots, M$ **do**

  $\mathbf{e}^{[i]} \leftarrow \mathbf{h}^{[i]}(\mathbf{x}_k) \ominus \mathbf{z}^{[i]}$ // $\in \mathcal{T}_{\mathbf{z}^{[i]}}(\mathcal{M})$

  $\mathbf{J}^{[i]} \leftarrow \left. \frac{\partial \mathbf{e}^{[i]}(\mathbf{x}_k \oplus \mathbf{\Delta x})}{\partial \mathbf{\Delta x}} \right|_{\mathbf{\Delta x}=\mathbf{0}}$ // $\in R^{m_i \times n}$

  $s^{[i]} \leftarrow \sqrt{\mathbf{e}^{[i]^T} \mathbf{\Omega}^{[i]} \mathbf{e}^{[i]}}$

  $w^{[i]} \leftarrow \frac{\rho'(s^{[i]})}{2 \cdot s^{[i]}}$

  $\tilde{\mathbf{\Omega}}^{[i]} \leftarrow w^{[i]} \mathbf{\Omega}^{[i]}$

  $\tilde{\mathbf{H}} \leftarrow \tilde{\mathbf{H}} + \mathbf{J}^{[i]^T} \tilde{\mathbf{\Omega}}^{[i]} \mathbf{J}^{[i]}$

  $\tilde{\mathbf{b}} \leftarrow \tilde{\mathbf{b}} + \mathbf{J}^{[i]^T} \tilde{\mathbf{\Omega}}^{[i]} \mathbf{e}^{[i]}$

$\mathbf{\Delta x}_k \leftarrow \text{solve}\left(\tilde{\mathbf{H}} \mathbf{\Delta x}_k = -\tilde{\mathbf{b}}\right)$ // $\in \mathcal{T}_{\mathbf{x}_k}(\mathcal{M}) \cong R^n$

$\mathbf{x}_{k+1} \leftarrow \mathbf{x}_k \oplus \mathbf{\Delta x}_k = \exp_{\mathbf{x}_k}(\mathbf{\Delta x}_k)$

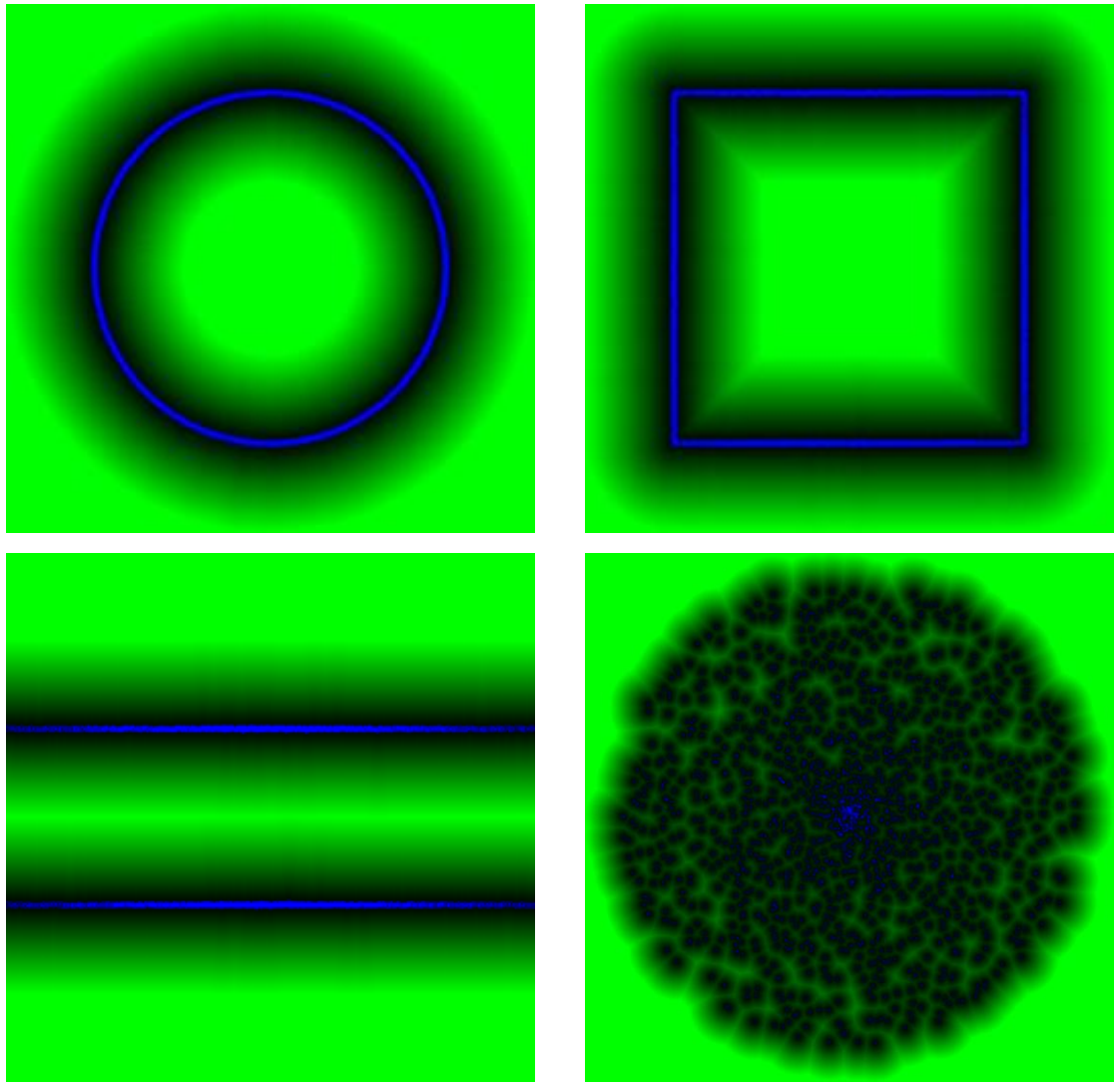
return $\mathbf{x}_{k+1}$

- **State**
  - $\mathbf{x} = [\mathbf{R}(\theta) \mid \mathbf{t}] \in SE(2)$
  - $\mathbf{x} \oplus \Delta\mathbf{x} = \mathbf{x} \cdot \exp(\Delta\mathbf{x})$
- Projected moving point: $\tilde{\mathbf{z}}_i = \mathbf{T}_R^S \, \mathbf{x} \, \mathbf{T}_S^R \, \mathbf{z}_i$

- **Error Metrics**
  - ICP p2p: $\tilde{\mathbf{z}}_i - \mathbf{p}^*$
  - ICP p2pl: $\mathbf{n}_{z_i}^T (\tilde{\mathbf{z}}_i - \mathbf{p}^*)$
  - $\mathcal{D}_{p2p}$: $e_i = \mathcal{D}(\tilde{\mathbf{z}}_i)$
  - $\mathcal{D}_{p2pl}$: $e_i = \left(\mathbf{R}(\theta)\mathbf{n}_{z_i}\right)^T \hat{\nabla}\mathcal{D}(\tilde{\mathbf{z}}_i) \, \mathcal{D}(\tilde{\mathbf{z}}_i)$

- With $\mathcal{M}_0$, $\hat{\nabla}\mathcal{D} \cdot \mathcal{D} = \tilde{\mathrm{z}}_i - \mathrm{p}^* \rightarrow \mathcal{D}_{\mathrm{p2pl}}(\mathcal{M}_0) \equiv$ ICP p2pl
- Normals by Principal Component Analysis (PCA)

Synthetic Benchmark: Setup



- Laser model based on the Hokuyo UST-10LX
- **Scenes**
  - **Circle** (r = 10 m): heading is unobservable
  - **Square** (20 m side): all 3 Degrees-of-Freedom (DoF) are constrained
  - **Corridor** (100 m walls, 10 m apart): x is unobservable within the 30 m sensor maximum range
  - **Random** (range from uniform distribution over laser rays)
- **Ablation study**
  - Sensor noise $\sigma \in \{0, 0.03\}$ m
  - Distance map resolution: $\delta \in \{0.01, 0.03, 0.05\}$ m
  - W/ or w/o voxelization & bilinear interpolation
- **Metrics:** ε_t, ε_θ as RMSE; $\chi^2_{\boldsymbol{\varepsilon}} = \boldsymbol{\varepsilon}^T \boldsymbol{\Omega} \boldsymbol{\varepsilon}$ as the mean, with $\boldsymbol{\Omega} = \mathbf{H}$

Synthetic Benchmark: Results

Method	*Circle* (ε_t / χ^2)	*Square* (ε_t / ε_θ / χ^2)	*Corridor* (ε_y / ε_θ / χ^2)	*Random* (ε_t / ε_θ / χ^2)
$\mathcal{D}_{\mathrm{p2p}}(\mathcal{M}_0)$	0.003 / 3100.86	0.003 / 0.08 / 0.19	0.003 / 0.12 / 453.71	0.005 / 0.02 / 0.03
$\mathcal{D}_{\mathrm{p2p}}(\mathcal{M}_1)$	0.003 / 934.92	0.003 / **0.01** / **0.00**	0.005 / 0.20 / 158.98	0.022 / 0.03 / 0.40
$\mathcal{D}_{\mathrm{p2pl}}(\mathcal{M}_0)$	0.003 / **189.38**	**0.002** / 0.02 / 0.01	**0.001** / 0.02 / **13.83**	–
$\mathcal{D}_{\mathrm{p2pl}}(\mathcal{M}_2)$	0.003 / 1716.85	0.003 / 0.02 / 0.01	**0.001** / **0.01** / 129.01	–
$\mathrm{ICP}_{\mathcal{D},\mathrm{p2p}}$	0.003 / 6479.07	0.003 / 0.08 / 0.39	**0.001** / 0.02 / 854.64	**0.001** / **0.00** / **0.00**
$\mathrm{ICP}_{\mathcal{D},\mathrm{p2pl}}$	0.003 / **189.38**	**0.002** / 0.02 / 0.01	**0.001** / 0.02 / **13.83**	–

Example: *$\delta = 0.03\ m$, $\sigma = 0.03\ m$ Gaussian noise, no voxelization, no bi-linear interpolation ($\varepsilon_{y\,/\,t}$ [m] / ε_θ [°] / χ^2 [–])*

- **Accuracy** Analytical point-to-plane at least as accurate as point-to-point variants
- **Equivalence** $\mathcal{D}_{\mathrm{p2pl}}(\mathcal{M}_0)$ similar to ICP p2pl
- **Best uncertainty** Analytical point-to-plane with $\mathcal{M}_0$ ($\mathcal{D}$, ICP) has the lowest χ^2 (structured scenes)
- **Overconfidence of analytical point-to-point formulations**
- $\mathcal{D}_{p2p}(\mathcal{M}_1)$ χ^2 Uncertainty more calibrated than its analytical counterparts
- $\mathcal{D}_{p2pl}(\mathcal{M}_2)$ **and noise** Two-step finite differences…

IILABS 3D DATASET

MODIFIED DISCOVERY Q2 PLATFORM (ICARSC 2025)

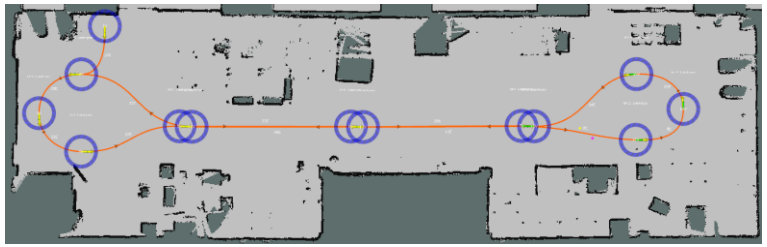


DATASET & BENCHMARK (IEEE ACCESS 2025)

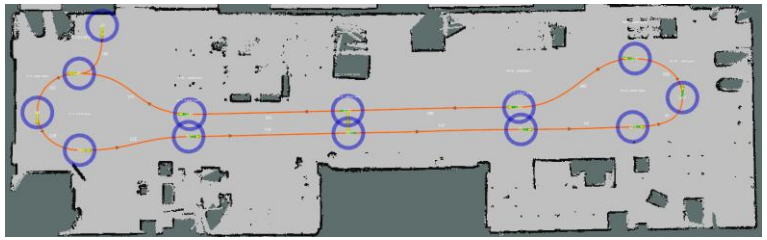
- **Sensors**
  - Wheel odometry (motor encoders)
  - IMU: Xsens MTi-630
  - 2D laser scanner: Hokuyo UST-10LX
  - 3D LiDARs: Livox Mid-360, Ouster OS1-64, RoboSense RS-Helios-5515, Velodyne VLP-16
- **Ground truth**
  - OptiTrack with 24 x PrimeX 22 cameras at 240 Hz
- **Data Sequences:** Nav A Diff, Nav A Omni, Loop, Slippage, Ramp and Elevator
- **Public dataset (10.25747/VHNJ-WM80) & iilabs3d-toolkit**

Real-World Laser Odometry on IILABS 3D

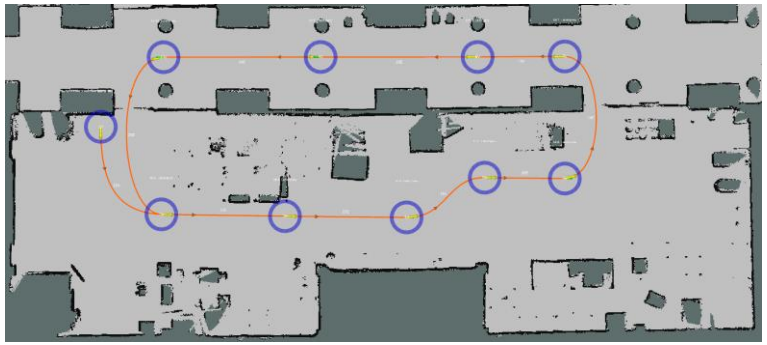
Nav A Diff



Nav A Omni



Loop



Slippage



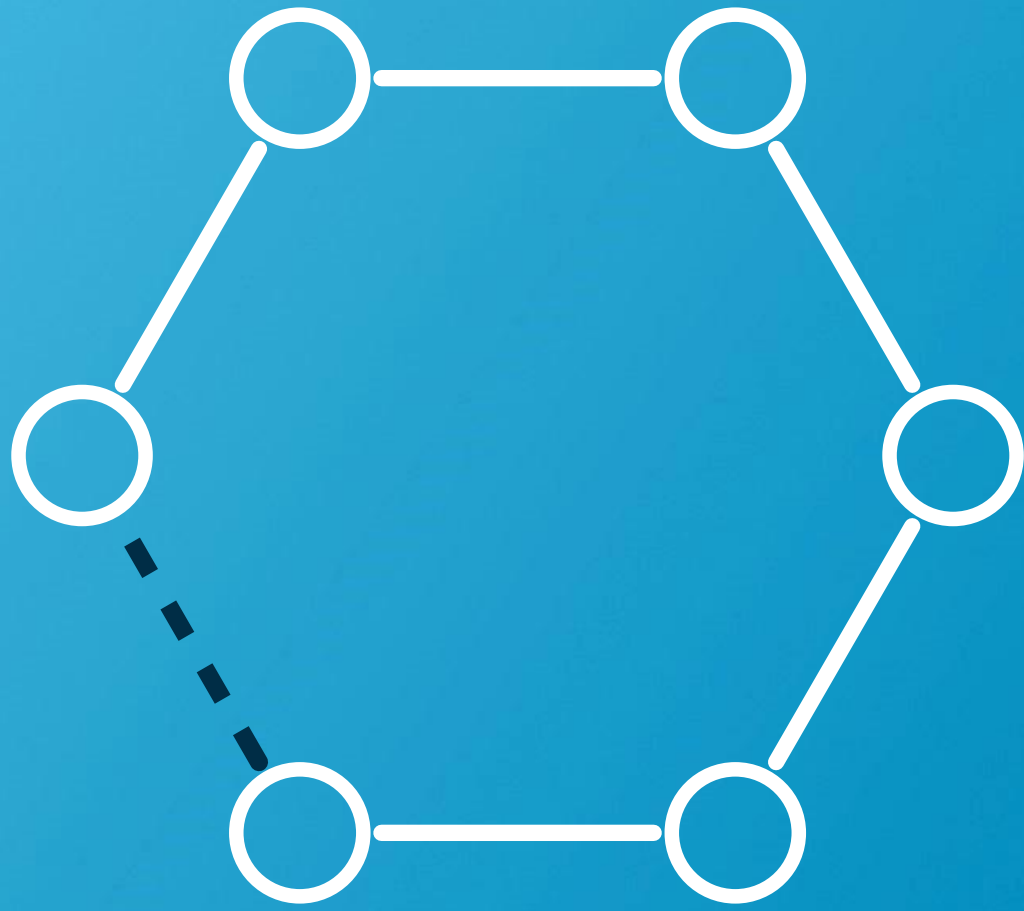
- **Equivalence holds in real-world** Over 144 cases, the mean absolute difference between $\mathcal{D}_{\mathrm{p2pl}}(\mathcal{M}_0)$ and ICP p2pl is 0.04% RTE and 0.003°/m RRE
- **Point-to-plane beats point-to-point** $\mathcal{D}_{\mathrm{p2pl}}(\mathcal{M}_0)$ lower RTE than $\mathcal{D}_{\mathrm{p2p}}(\mathcal{M}_0)$ in 37/30/34 of 48 cases
- **$\mathcal{D}_{p2p}(\mathcal{M}_1)$ is the best point-to-point variant**
  - Lower RTE than $\mathcal{D}_{\mathrm{p2p}}(\mathcal{M}_0)$ in 42/36/43 cases (median 22.2 / 28.1 / 28.6%)
  - Lower RTE than ICP p2p in 37/37/39 cases
- **$\mathcal{D}_{p2p}(\mathcal{M}_1)$ is competitive vs point-to-plane without normals**
  - Beats $\mathcal{D}_{\mathrm{p2pl}}(\mathcal{M}_0)$ on RTE in 26/29/30 cases and on RRE in 31/34/39
- **Keyframe splitting criteria** Inlier-ratio policy at 40–60% matches or beats the best translation threshold (which is 2.0–2.5 m)

TAKEAWAYS

- **A single framework covers point-to-point and point-to-plane, dense or sparse storage, and even support ICP baselines**
- **Precomputed $\mathcal{D}_{p2p}(\mathcal{M}_1)$:**
  - **Lowest drift among point-to-point methods**
  - **Competitive with point-to-plane without needing normals computation**
  - **Has lower χ^2 in degenerate scenes than its analytical point-to-point counterparts**
- **Keyframe splitting criteria driven by the inlier ratio beats fixed-distance splitting**

- **Next steps…**
  - Runtime against kd-tree ICP is still pending
  - Alternative Hessian computation / improve precomputed point-to-plane formulation
  - Explore using the uncertainty estimates in pose graphs
  - **UDF-based tracker (ROBOT 2026) becomes the RicoSLAM front-end**

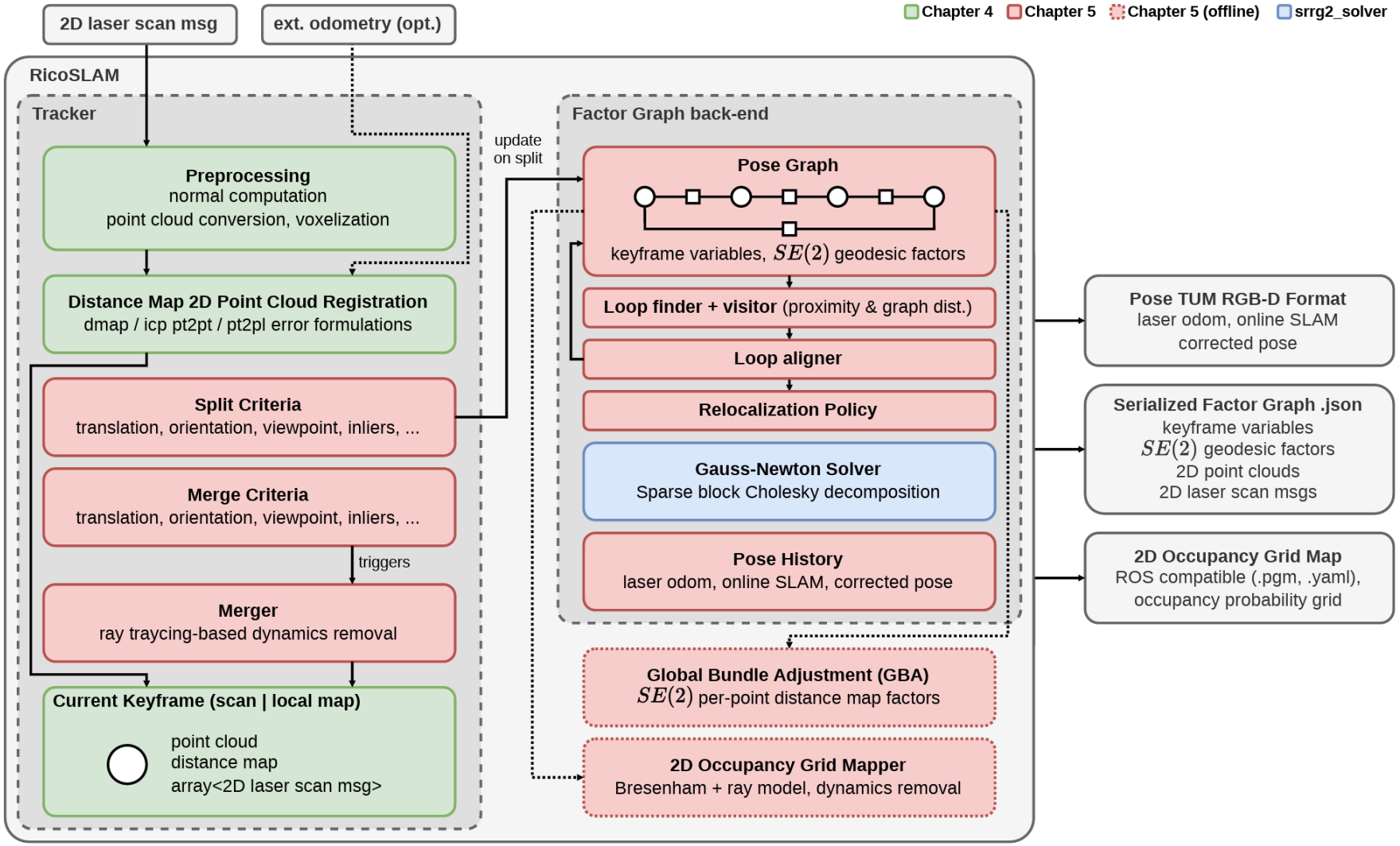
RicoSLAM Framework



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Framework Overview



Online SLAM: Keyframe Maintenance

- **Split:** commit the current keyframe to the graph and start a new one
- **Merge:** fuse the scan into the local map through the ray-tracing merger, then rebuild the distance map
- **Split & Merge criteria types**
  - translation, orientation, viewpoint, grid cell ($\boldsymbol{\Delta} = \log\left(\mathbf{x}_{R,\mathrm{ref}}^{-1}\mathbf{x}_{R,\mathrm{kf}}\right)$), time (latest scan message vs reference one)
  - **inlier ratio (UDF-based tracker)**
  - empty (never fires on its own)
- Every type can also be **force-triggered** through a ROS service! (ICARA 2026; IEEE T-ASE, under review)
- **Operating modes**
  - **scan:** merge = empty, so every committed scan is a keyframe
  - **local maps:** both criteria active, with merge thresholds tighter than split criteria

Online SLAM: Loop Closure & Relocalization

Loop Closure

- **Stage 1 (proximity)**
  - Nodes within $d_{\ell,\max}$ of $\mathbf{x}_{R,W}$ form the short-loop set $\mathcal{H}_s$
- **Stage 2 (topology)**
  - Best-first graph visitor keeps a candidate only if its graph distance is at least $L_{\min}$
  - Rejects trivial loops along the odometry chain
- **Validation –** the loop aligner may try two guesses:
  - Pose-graph guess $\hat{\mathbf{x}}_{R,c} = \mathbf{x}_c^{-1}\mathbf{x}_{\mathrm{kf}}\mathbf{x}_{R,\mathrm{kf}}$
  - Same-place guess $\exp\left((0,0,\theta)^T\right)$
- **Acceptance (example)**
  - $\eta_{\mathrm{inliers}} \geq 70\%$
  - $\chi^2_{\mathrm{inliers}} \leq 0.001$

Relocalization

- For the closest successfully aligned candidate $h_s^* \in \mathcal{H}_s$:
  - $d_{\mathrm{loop}} = \left\|\left(\mathbf{x}_{h_s^*}^{-1}\mathbf{x}_{R,W}\right).\mathbf{t}\right\|$
  - $d_{\mathrm{local}} = \left\|\mathbf{t}_{R,\mathrm{kf}}\right\|$
- If $d_{\mathrm{loop}} < d_{\mathrm{local}}$, the tracker switches its keyframe to h_s^* instead of creating a new node
- **Effect:** graph size bounded by explored area, not by path length

Ray Tracing Dynamics Removal

Online Merging

- Update local map representation
- Single-return ray tracing model
- **With $\Delta\rho_k = \rho_{S,k} - \rho_{\mathcal{M},k}$:**

Condition	Action
no map point on ray	add $\mathbf{z}_k$
$\Delta\rho_k < -\tau_{\text{new}}$ (in front)	add $\mathbf{z}_k$
$\Delta\rho_k > +\tau_{\text{through}}$ (see-through)	replace map point (dynamics removal)
otherwise	keep

- After any update, the distance map is rebuilt, so the next alignment sees the current map

Offline Removal

- The same merger runs over all retained scans, using the **optimized sensor poses**

$$\mathbf{x}_{S,W} = \mathbf{x}_{\text{kf}}\, \mathbf{x}_{R,\text{kf}}\, \mathbf{T}_S^R$$

- Outputs a cleaned global point cloud plus, for each scan, the list of endpoints that survived
- **Bresenham-like rasterization → 2D occupancy grid map representation of the environment**

Global Bundle Adjustment (GBA)

- **Why:** online factor graph compresses hundreds of point correspondences into one $SE(2)$ factor measurement with a fixed $\mathbf{\Omega}$ → inter-keyframe misalignment such as double or fat walls!
- **GBA** replaces those constraints with per-point residuals against another keyframe's distance map

- **Error Metrics**
  - $e_i^{\text{p2p}} = \mathcal{D}_{\text{to}}(\mathbf{T}_R^S\ \mathbf{x}_{\text{to}}^{-1}\mathbf{x}_{\text{from}}\mathbf{T}_S^R\ \mathbf{z}_i)$
  - $e_i^{\text{p2pl}} = \left(\mathbf{R}_{\text{to}}^T\mathbf{R}_{\text{from}}\mathbf{n}_{z_i}\right)^T \widehat{\nabla}\mathcal{D}_{\text{to}}\ \mathcal{D}_{\text{to}}$

RESULTS: IILABS 3D – ACCURACY

Pose streams:

Online: $\mathbf{x}_{\mathrm{kf}}(t)$ $\mathbf{x}_{R,\mathrm{kf}}(t)$

Offline corrected stream (†): $\mathbf{x}^{*}_{\mathrm{id}_{\mathrm{kf}}(t)}$ $\mathbf{x}_{R,\mathrm{kf}}(t)$,

so past poses inherit later loop closures!

System (ATE / RTE / RRE)	*Nav A Diff*	*Nav A Omni*	*Loop*	*Slippage*
GMapping	0.115 / 1.76 / 0.109	0.121 / 1.89 / 0.104	0.107 / 2.54 / 0.137	0.041 / 1.95 / 0.118
SLAM Toolbox	0.053 / 0.86 / 0.084	0.061 / 1.40 / 0.105	0.040 / 1.68 / **0.055**	0.042 / 5.13 / 0.326
RicoSLAM (online)	**0.047 / 0.81 / 0.072**	**0.030 / 0.71 / 0.057**	**0.033 / 0.94** / 0.068	**0.014 / 0.53 / 0.057**
RicoSLAM† (offline)	**0.022 / 0.74 / 0.059**	**0.023 / 0.71 / 0.063**	**0.023 / 0.99** / 0.058	**0.014 / 0.57 / 0.057**

Best of each system's variants (ATE [m] / RTE [%] / RRE [°/m])

- All variants of online RicoSLAM beat GMapping and SLAM Toolbox on every ATE and RTE.
  Only exception is Loop RRE, where SLAM Toolbox 0.5 m/rad gets 0.055 against 0.068 °/m
- **Offline stream**
  - ATE drops by 53% on Nav A Diff, 23% on Omni and 43% on Loop
  - Slippage is unchanged, since it is a straight out-and-back path
- **Scan versus local maps**
  - Best offline accuracy comes from **scan mode**, because merging adds residual errors that cannot be undone later
  - **Local map w/ Ω update mode** is the best online variant on Omni and Loop, and within 0.005 m of scan offline

Results: IILABS 3D – Scalability

Example: Local Map Mode w/ Relocalization

Nav A Diff

Nav A Omni

Loop

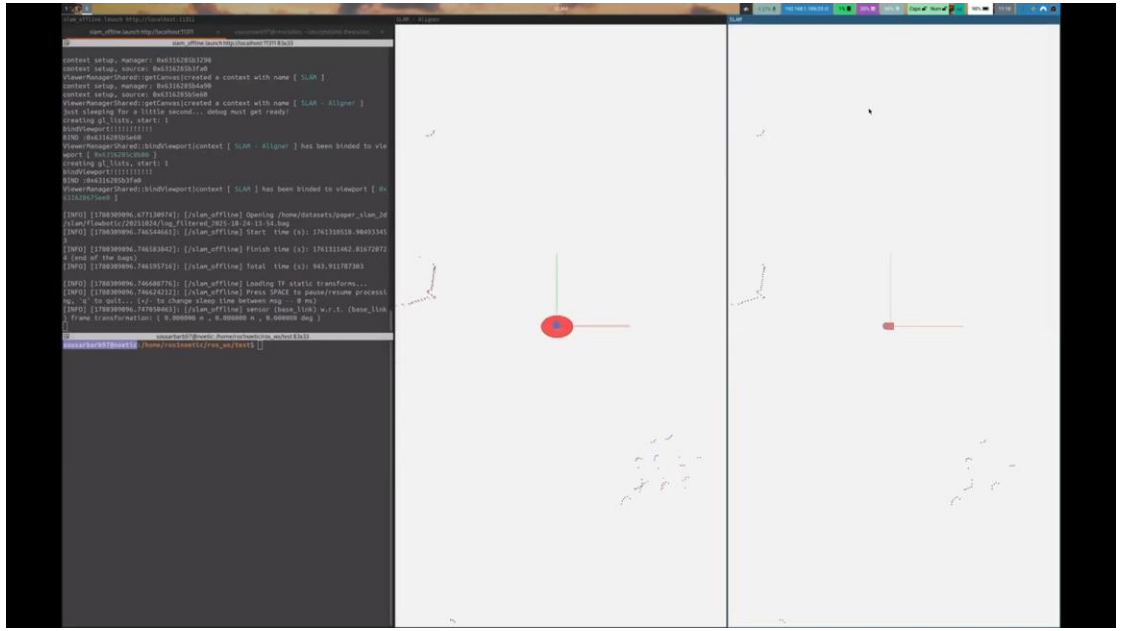
Slippage



- **Inlier-ratio split** Scan mode uses 3.1–4.6× fewer nodes than the most compact SLAM Toolbox configuration, at better accuracy
- **Relocalization** Scan w/ reloc. cuts nodes by 55% on Diff (75 → 34) and 77% on Loop (83 → 19)
- **Local Map Mode w/ Relocalization**
  - 17 / 17 / 18 / 8 nodes
  - Graph size follows the explored area, not the run duration nor path length
  - Its offline ATE is at or below the best SLAM Toolbox ATE on all four sequences, with 5.8–18× fewer nodes

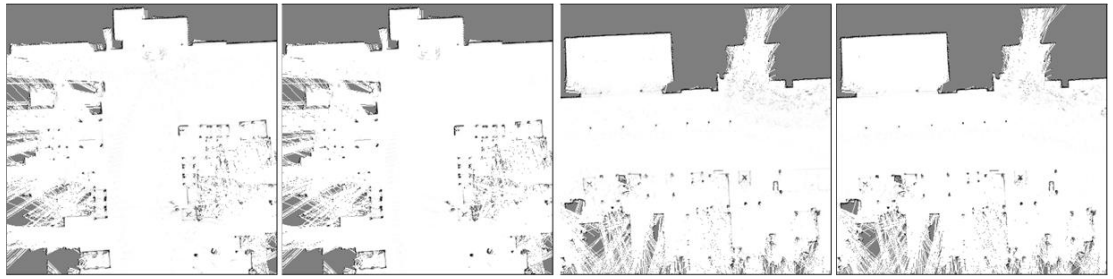
Results: Dynamics Removal

Online Local Map Update

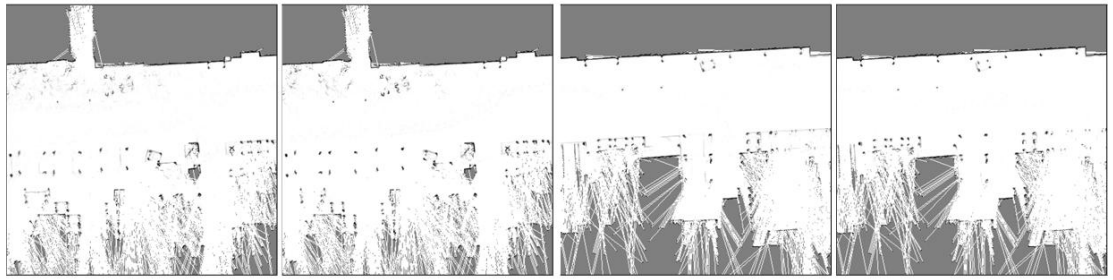


Flowbotic 20251024 sequence with RicoSLAM on local map mode (https://youtu.be/fdY6bE3Vyt8)

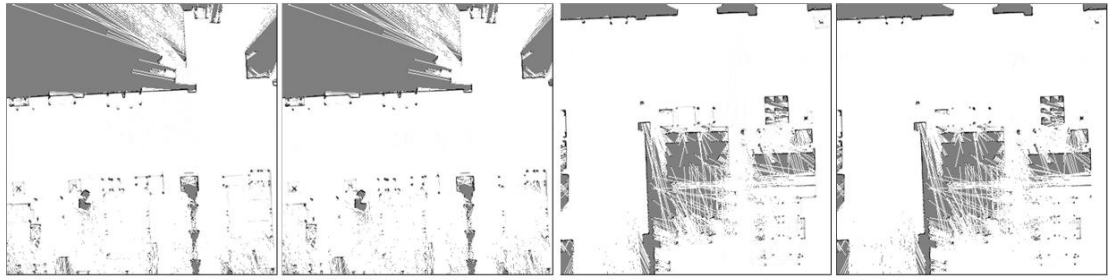
Offline Dynamics Removal



(a) 20251024, region 1 (b) 20251024, region 2

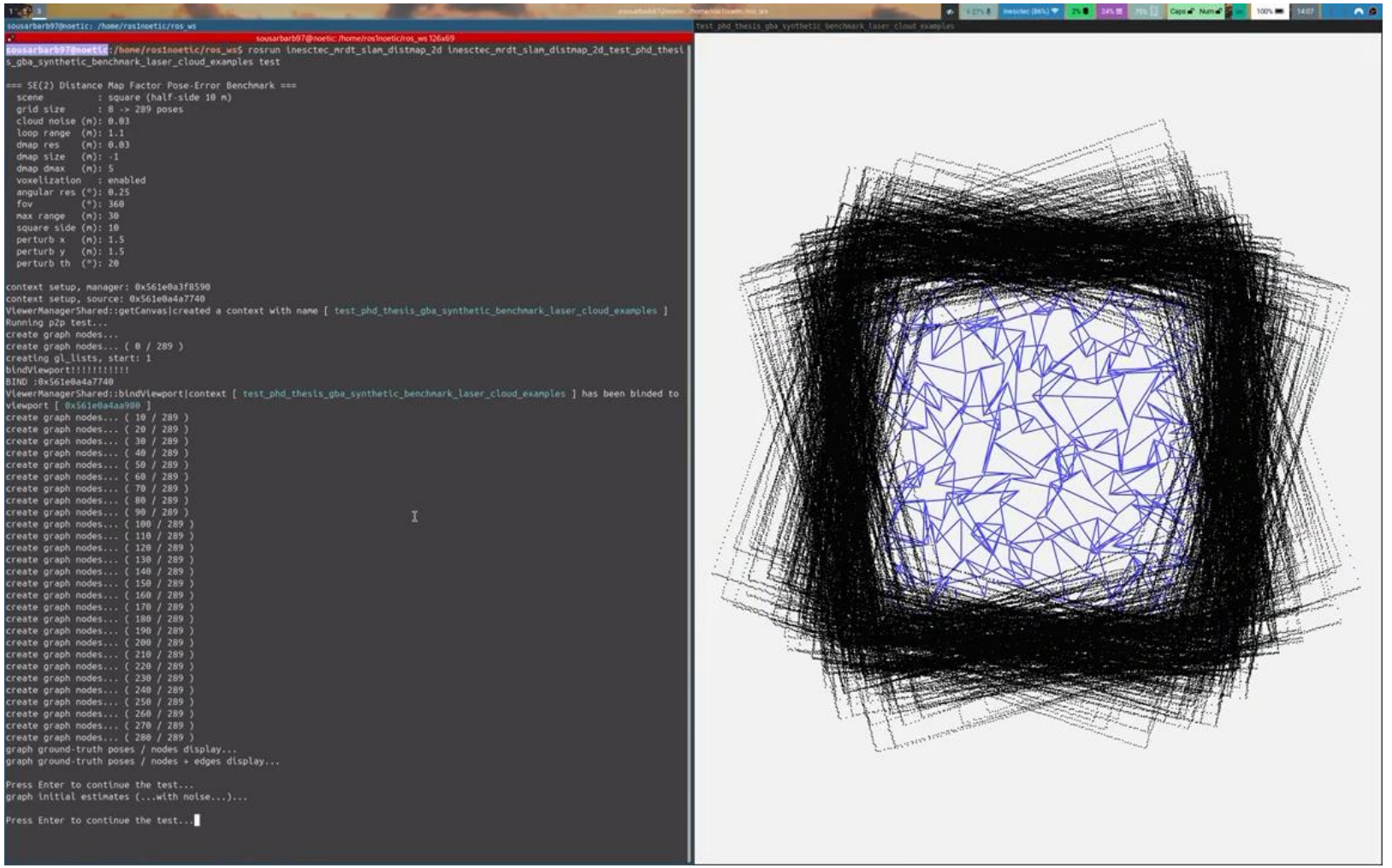


(c) 20251024, region 3 (d) 20251024, region 4



(e) 20251024, region 5 (f) 20251024, region 6

RESULTS: GBA ON SYNTHETIC DATA

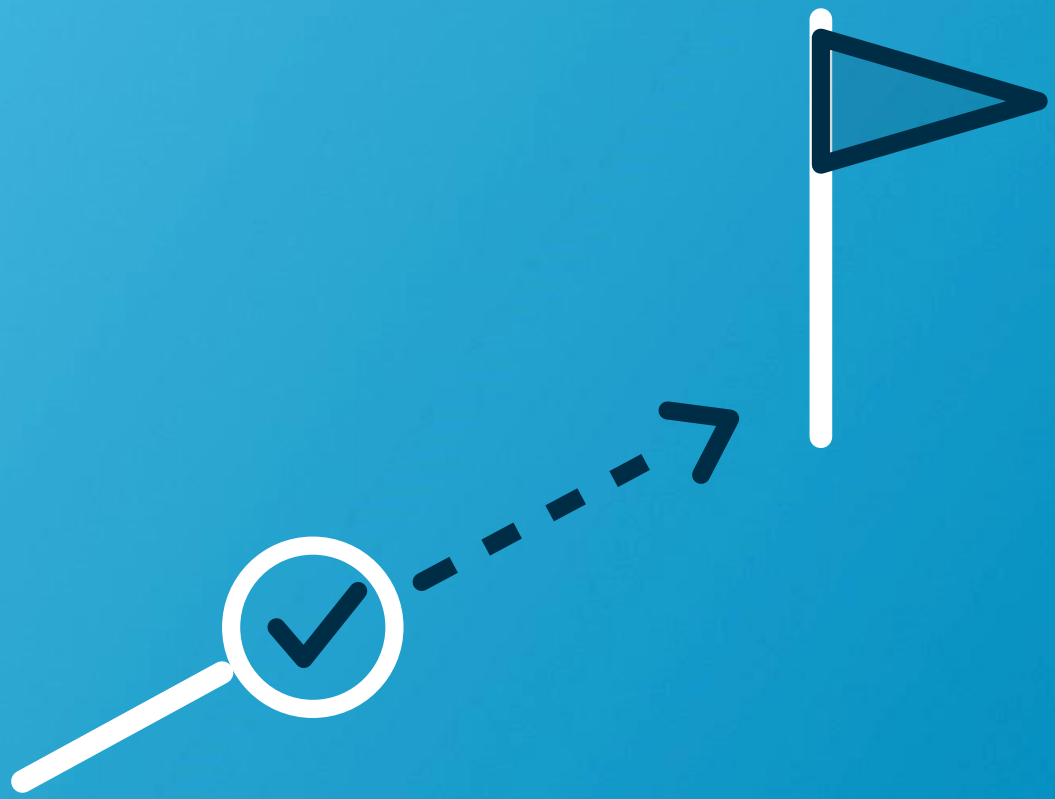


*GBA $\mathcal{D}_{p2p}(\mathcal{M}_1)$ on Square Scene w/ 0.03 m Gaussian noise added to range data **w/ voxelization** (https://youtu.be/AOuAD12Ga6g)*

TAKEAWAYS

- **One distance map per keyframe serves as the registration target, the ray-tracing reference and the relocalization target**
- **Accuracy** lowest online ATE on all four IILABS 3D sequences against SLAM Toolbox and GMapping; offline corrected pose improves the ATE results even further to around 2 cm
- **Scalability** relocalization bounds the graph by the explored area
- **Maps:** offline dynamics removal sharpens the occupancy distribution
- **Limitations**
  - Single-return ray model does not clean all semi-static objects
  - GBA not yet validated on real-world data
  - No timing results in the PhD Thesis
- **Going further:** *RicoSLAM: Framework for Localization and Mapping in Dynamic Environments*, IEEE RA-L [in preparation], **adds the Cartographer baseline and total processing time.**

Conclusions & Future Work



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Conclusions

RQ1 – Mapping in dynamic indoor environments

- **H1.1 ◐ Ray tracing local map update**
  Online merger runs, but no on/off ablation study
- **H1.2 ✓ Relocalization into the existing graph**
  Local w/ reloc. results in 5.8–18× fewer nodes than SLAM Toolbox at equal or better ATE; graph size follows the explored area, not run duration or path length
- **H1.3 ✓ Offline dynamics removal**
  Sharpens the occupancy distribution in every sequence, mode, and threshold across 5 datasets and demotes transient-only cells to *unknown*
- **H1.4 ◐ Distance map-based GBA**
  Factors recover a 289-pose Manhattan graph perturbed by ±1.5 m / ±20°; $\mathcal{D}_{p2p}(\mathcal{M}_1)$ with voxelization is best under 0.03 m noise; integration with the online pipeline pending

RQ2 – Dynamics in the localization process

- **H2.1 ✓ Corner-like features**
  Viewpoint invariance (6 features at 100% observability, 20 ≥ 50%, 16 ≥ 75%) and geometric invariance (91.5% of pairs at σ ≤ 0.03 m)
- **H2.2 ✓ Distance maps on 2D point cloud registration**
  One framework spans p2p / p2pl, dense / sparse, and the ICP baselines; NN search replaced by O(1) lookups; $\mathcal{D}_{p2p}(\mathcal{M}_1)$ competitive with point-to-plane without ever estimating normals
- **H2.3 ✓ Continuously updated local maps**
  Local map mode improves online accuracy on 2/4 sequences

Conclusions

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- **H2.3 ✓ Continuously updated local maps**
  Local map mode improves online accuracy on 2/4 sequences

RicoSLAM unifies distance map-based registration, dynamics-aware local mapping, and offline global refinement in a single SE(2) graph SLAM framework | ~2 cm ATE, graphs bounded by explored area, cleaner static maps

Contributions



8

Scientific Contributions

5 main + 3 complementary | 6 published + 1 review + 1 preparation | 4 journal + 4 conference



1

Public Dataset

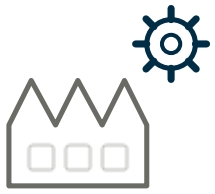
IILABS 3D



8

Open-Source Repositories

4 new + 4 extending existing projects



3

Products, Processes, and Services (PPS)

PPS18 (GreenAuto/WP10) | PPS16 + PPS29 (PRODUTECH R3/WP6 & WP9)

Contributions: Main & Complementary



- **Main contributions**
  - R.B. Sousa, H.M. Sobreira, and A.P. Moreira. “A systematic literature review on long-term localization and mapping for mobile robots”. In: **Journal of Field Robotics** 40.5 (2023), pp. 1245–1322. doi: 10.1002/rob.22170.
  - R.B. Sousa, H.M. Sobreira, M.F. Silva, and A.P. Moreira. “Line fitting-based corner-like detector for 2D laser scanners data”. In: **2024 10th International Conference on Automation, Robotics and Applications (ICARA)**. Athens, Greece, 2024, pp. 210–216. doi: 10.1109/ICARA60736.2024.10553006.
  - R.B. Sousa, H.M. Sobreira, J.G. Martins, P.G. Costa, M.F. Silva, and A.P. Moreira. “Integrating multimodal perception into ground mobile robots”. In: **2025 IEEE International Conference on Autonomous Robot Systems and Competitions (ICARSC)**. Funchal, Portugal, 2025, pp. 104–111. doi: 10.1109/ICARSC65809.2025.10970176.
  - R.B. Sousa, G. Grisetti, H.M. Sobreira, C.A. Silva, and A.P. Moreira. “Unsigned Distance Maps on 2D Point Cloud Registration”. In: **9th Iberian Robotics Conference (ROBOT2026)**. Barcelona, Spain, 2026. *Accepted for publication*.
  - R.B. Sousa, G. Grisetti, H.M. Sobreira, C.A. Silva, and A.P. Moreira. “RicoSLAM: Framework for Localization and Mapping in Dynamic Environments”. **Paper in preparation for the IEEE Robotics and Automation Letters (RA-L)**.
- **Complementary contributions**
  - J.D. Ribeiro, R.B. Sousa, J.G. Martins, A.S. Aguiar, F.N. dos Santos, and H.M. Sobreira. “Indoor benchmark of 3D LiDAR SLAM at iilab – industry and innovation laboratory”. In: **IEEE Access** 13 (2025), pp. 212421–212442. doi: 10.1109/ACCESS.2025.3643753.
  - M.S. Lopes, A. Cordeiro, R.B. Sousa, J.A. Beça, P. Costa, J.P.C. Souza, and M.F. Silva. “CARGO: container-unloading autonomous robot with generalized operation”. In: **2026 12th International Conference on Automation, Robotics and Applications (ICARA)**. Istanbul, Turkey, 2026, pp. 274–279. doi: 10.1109/ICARA69401.2026.11480416.
  - M.S. Lopes, A. Cordeiro, R.B. Sousa, J.A. Beça, P. Costa, J.P.C. Souza, and M.F. Silva. “Reconfigurable Pipeline for Autonomous Container Unloading with Supervisory Velocity Filtering and Warehouse Validation”. **Under review at IEEE Transactions on Automation Science and Engineering (T-ASE)**.

Contributions: Datasets & Code





- **Public Dataset**
  - J.D. Ribeiro, R.B. Sousa, J.G. Martins, A.S. Aguiar, F.N. dos Santos, and H.M. Sobreira. **IILABS 3D: iilab indoor LiDAR-based SLAM dataset**. iilab – industry and innovation lab: INESC TEC – Institute for Systems, Computer Engineering, Technology, and Science, 2025. doi: 10.25747/VHNJ-WM80.
- **Code Repositories**
  - https://github.com/INESCTEC/ricoslam



  - https://github.com/INESCTEC/inesctec_mrdt_line_fit_based_corner_detector_2d_laser
  - https://github.com/INESCTEC/inesctec_mrdt_hangfa_discovery_q2
  - https://github.com/sousarbarb/slr-ltlm-mr
  - https://github.com/sousarbarb/hector_slam (offline mode that feeds ROS bag messages into the callbacks)
  - https://github.com/sousarbarb/slam_gmapping (offline mode that feeds ROS bag messages into the callbacks)
  - https://github.com/sousarbarb/stage_ros (compatibility with ROS 1 and ROS 2 versions natively)
  - https://gitlab.com/sousarbarb_srrg-software/srrg_system (CMake compatibility with ROS 1 and ROS 2 versions)

CONTRIBUTIONS: INDUSTRIAL R&D PROJECTS

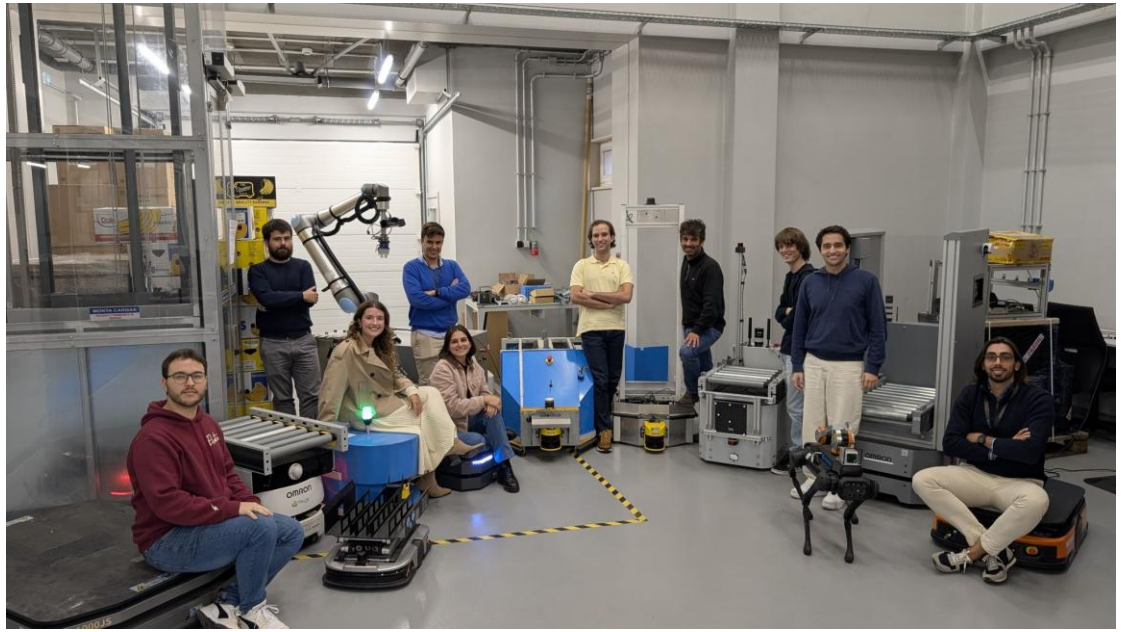
- **GreenAuto: Green Innovation for the Automotive Industry**
  - WP10 – Automated logistics for the automotive industry
    - PPS18 – 3D navigation system for mobile robotic equipment

- **PRODUTECH R3 (Recovery-Resilience-Reindustrialization): Production Technologies Cluster Mobilizing Agenda for Reindustrialization**
  - WP6 – Flexible Robotic Solutions: Intelligent, Compact and Flexible End of Line Robotic Solutions
    - PPS16 – CompactPack: Compact, Flexible and Portable Robotic Cells for End of Line
  - WP9 – SmartIL: Intelligent interoperable robotic intralogistics and quality control platform and solutions for industry
    - PPS29 – Mobile Robotic Solutions for Moving Products on the Shop Floor

Future Work

- **Front-End Perception:**
  - Covariance estimation for the corner-like features
  - Hybrid raw-scan / feature / beacon SLAM on the same SE(2) back-end
  - Improve precomputed point-to-plane formulation
  - Deeper analysis on uncertainty estimation for 2D point cloud registration
  - Sliding window approach to reduce online front-end localization drift
- **SLAM & Dynamics Removal:**
  - Online dynamics removal with multi-return model
  - Semi-static changes & Long-term / Lifelong SLAM
- **Mapping Global Consistency:**
  - GBA integration and real-world validation
  - Simultaneous pose & 2D scene structure optimization for global map refinement

Acknowledgements



Mobile Robotics Development Team (MRDT) @ iilab – industry and innovation laboratory

- Datasets
  - Flowbotic (20251024, 20251027, 20251028)

FLOWBOTIC

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- Special thanks to Giorgio for the Visiting Research opportunity @ Sapienza University of Rome (February – July 2024)



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- This work has also been co-financed by Component 5 – Capitalisation and Business Innovation, integrated in the Resilience Dimension of the Recovery and Resilience Plan within the scope of the Recovery and Resilience Mechanism (MRR) of the European Union (EU), framed in the Next Generation EU, for the period 2021–2026, within project GreenAuto, with reference 54.





PRR
Plano de Recuperação e Resiliência



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RicoSLAM: Localization and Mapping in Dynamic Environments

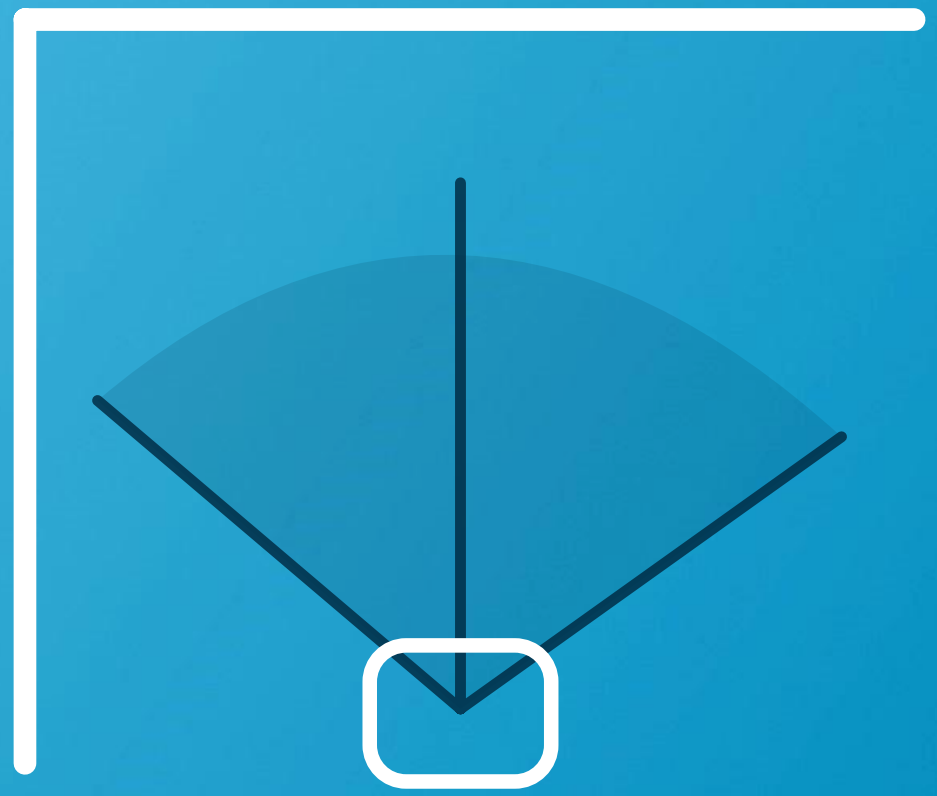


PhD Thesis | Ricardo Barbosa Sousa (https://sousarbarb.github.io/)
DEEC, I025, September 18, 2026

Doctoral Program in Electrical and Computer Engineering (PDEEC)
Faculty of Engineering, University of Porto (FEUP)
INESC TEC – Institute for Systems and Computer Engineering, Technology and Science

Supervisors: Prof. Dr. António Paulo Moreira (FEUP), Héber Miguel Sobreira, PhD (INESC TEC)

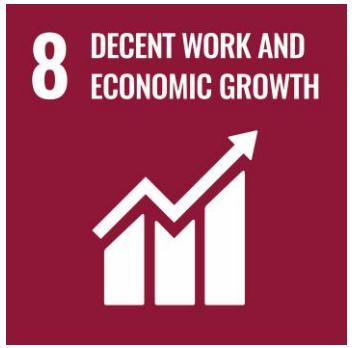
Introduction



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GOALS





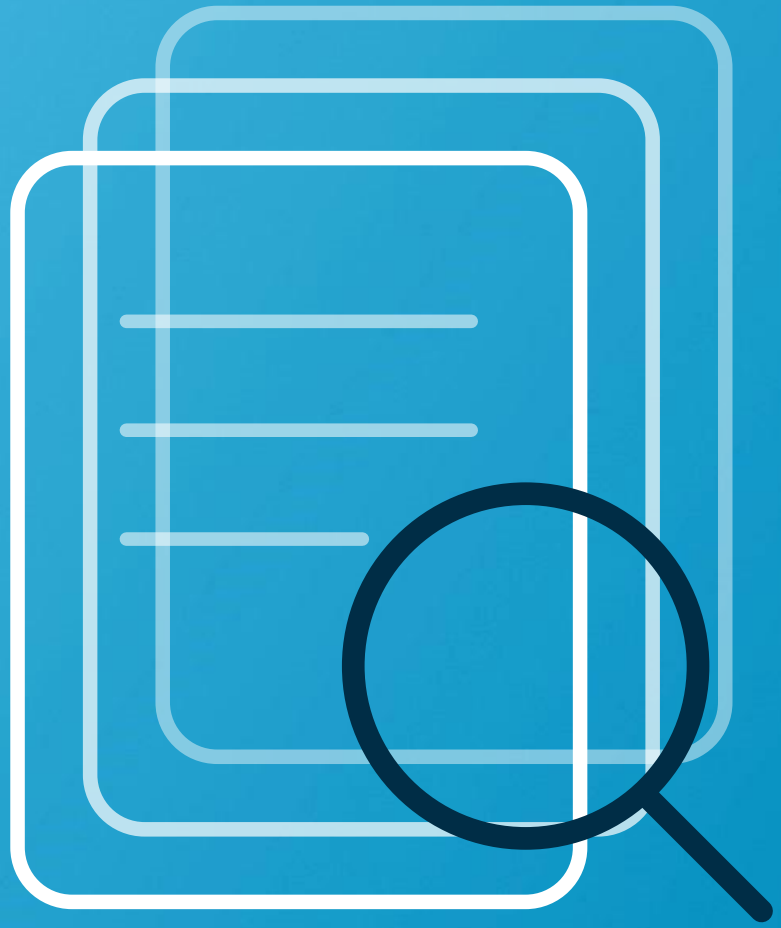




- **Advance SLAM with wheeled odometry and 2D lasers for industrial dynamic environments, under a planar-floor assumption**

- **G1 – Acquire:** modify mobile platforms for multimodal data acquisition & benchmarking
- **G2 – Perceive:** extract geometric information from 2D scans (corner-like detector)
- **G3 – Register:** 2D point cloud registration on Unsigned Distance Fields (UDF)
- **G4 – Integrate:** full distance map-based graph SLAM framework
- **G5 – Update:** ray tracing to handle dynamic observations in the map representation
- **G6 – Refine:** offline distance map-based GBA to remove map distortions

Literature Review



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SLR: Literature Reviews, Surveys, Tutorials

TABLE 1 Existent literature reviews, surveys, and tutorials on Simultaneous Localization and Mapping (SLAM).

Year	Topic	Reference
2006	Probabilistic formulations (EKF, particle filter)	Bailey and Durrant-Whyte (2006), Durrant-Whyte and Bailey (2006)
2008	Probabilistic and pose graph formulations	Thrun (2008)
2010	GraphSLAM	Grisetti et al. (2010)
2011	Observability, convergence, consistency (feature-based SLAM)	Dissanayake et al. (2011)
2012	Visual odometry	Fraundorfer and Scaramuzza (2012), Scaramuzza and Fraundorfer (2011)
2014	Underwater navigation and localization	Paull et al. (2014)
2015	Visual SLAM	Yousif et al. (2015)
2016	Observability, convergence, consistency (feature and graph-based SLAM)	Huang and Dissanayake (2016)
	Multirobot SLAM	Saeedi et al. (2016)
	Visual place recognition	Lowry et al. (2016)
	SLAM literature overview	Cadena et al. (2016)
2017	Autonomous vehicles	Bresson et al. (2017)
2018		Kuutti et al. (2018)
2018	AI for long-term autonomy	Kunze et al. (2018)
	Long-term sensorization	Zaffar et al. (2018)
2020	Deep learning	Fayyad et al. (2020)
	Multirobot search and rescue	Queralta et al. (2020)
2021	Self-driving vehicles	Badue et al. (2021)
2022	Underground navigation	Ebadi et al. (2022)

SLR: Exclusion Criteria

E#	Criteria	Statement
E1	Index	Papers not indexed in a scientific publication venue
E2	Language	Full-text of the papers not published in English
E3	Subject Area	Papers not classified in the databases as Computer Science, Engineering, Mathematics, or Multidisciplinary
E4	Short Papers	Papers classified as short papers accordingly to the publication venue
E5	Gray, Secondary, and Tertiary Literature	Books, preprints, reports, reviews, thesis, ...
E6	Availability	Full-text of the papers not available in digital libraries
E7	Data set	Papers that focus only on data collection
E8	Coverage	Papers using only odometry for localization
E9	Scope	Papers that focus on different and not related subjects

TABLE 2 Exclusion criteria for the selection process.

SLR: Quality Evaluation Criteria

TABLE 3 Quality evaluation criteria and score range.

QE#	Criteria	Score
QE1	Does the paper have an updated state of the art on long-term localization and mapping?	{0.0, 0.5, 1.0}
QE2	Is the methodology appropriate and detailed?	{0.0, 0.5, 1.0}
QE3	Does the methodology consider both localization and mapping problems?	{0.0, 1.0, 2.0}
QE4	Is the hardware and/or software used in the experiments detailed?	{0.0, 0.5, 1.0}
QE5	Does the paper presents any kind of long-term experimental results?	{0.0, 2.0}
QE6	Does the paper presents comparative results with other methods and/or ground-truth data?	{0.0, 1.0}
QE7	Does the work's implementation and/or the data used in the experiments are publicly available?	{0.0, 1.0}
QE8	Is the discussion of the results and conclusions appropriate and detailed?	{0.0, 0.5, 1.0}

SLR: Quality Evaluation Assessment Results

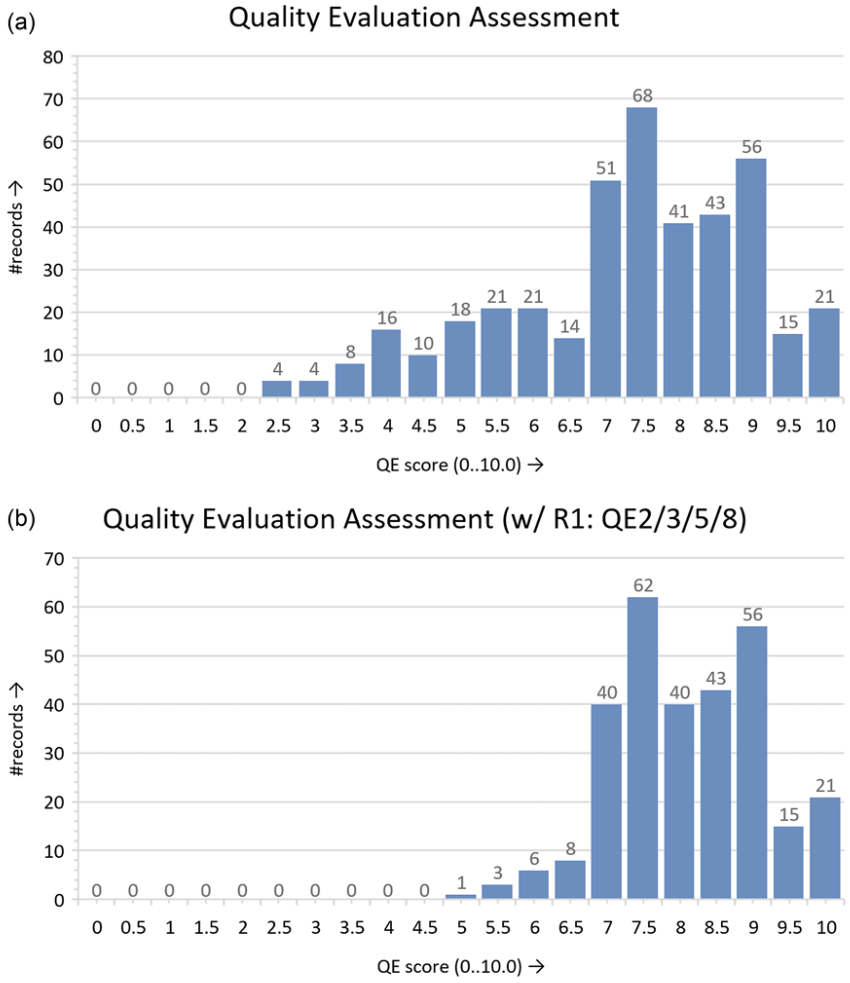


FIGURE 2 Distribution of the quality evaluation scores obtained from assessing the eligible works considered in the review: (a) all eligible works; (b) works that pass the rejection criterion during the QE assessment related to QE2/3/5/8 = 0.0 (no compliance). [Color figure can be viewed at wileyonlinelibrary.com]

SLR: Pairwise Coverage Analysis of Data Sources

TABLE 4 Pairwise coverage analysis of the data sources considered in the review over the 142 included records: (a) identification only on both pairwise sources (#{$A \cap B$}/min{#A, #B}); (b) on either ones (#{$A \cup B$}/#records).

(a) $A \cap B$	acm	dim	ieee	insp	scop	wos
acm	–	96.0%	44.0%	88.0%	96.0%	96.0%
dim	–	–	68.7%	77.4%	97.6%	96.4%
ieee	–	–	–	89.6%	91.0%	74.6%
insp	–	–	–	–	87.3%	69.6%
scp	–	–	–	–	–	89.5%
wos	–	–	–	–	–	–
(b) $A \cup B$	**acm**	**dim**	**ieee**	**insp**	**scop**	**wos**
acm	–	59.9%	57.0%	73.9%	85.2%	74.6%
dim	–	–	73.9%	85.2%	85.9%	76.1%
ieee	–	–	–	76.8%	88.7%	85.9%
insp	–	–	–	–	93.7%	95.8%
scp	–	–	–	–	–	92.3%
wos	–	–	–	–	–	–

Abbreviations: dim, Dimensions (n.d.); ieee, IEEE Xplore (n.d.); insp, INSPEC (n.d.); scop, Scopus (n.d.); wos, Web of Science (n.d.).

SLR: Keywords Co-Occurrence Results

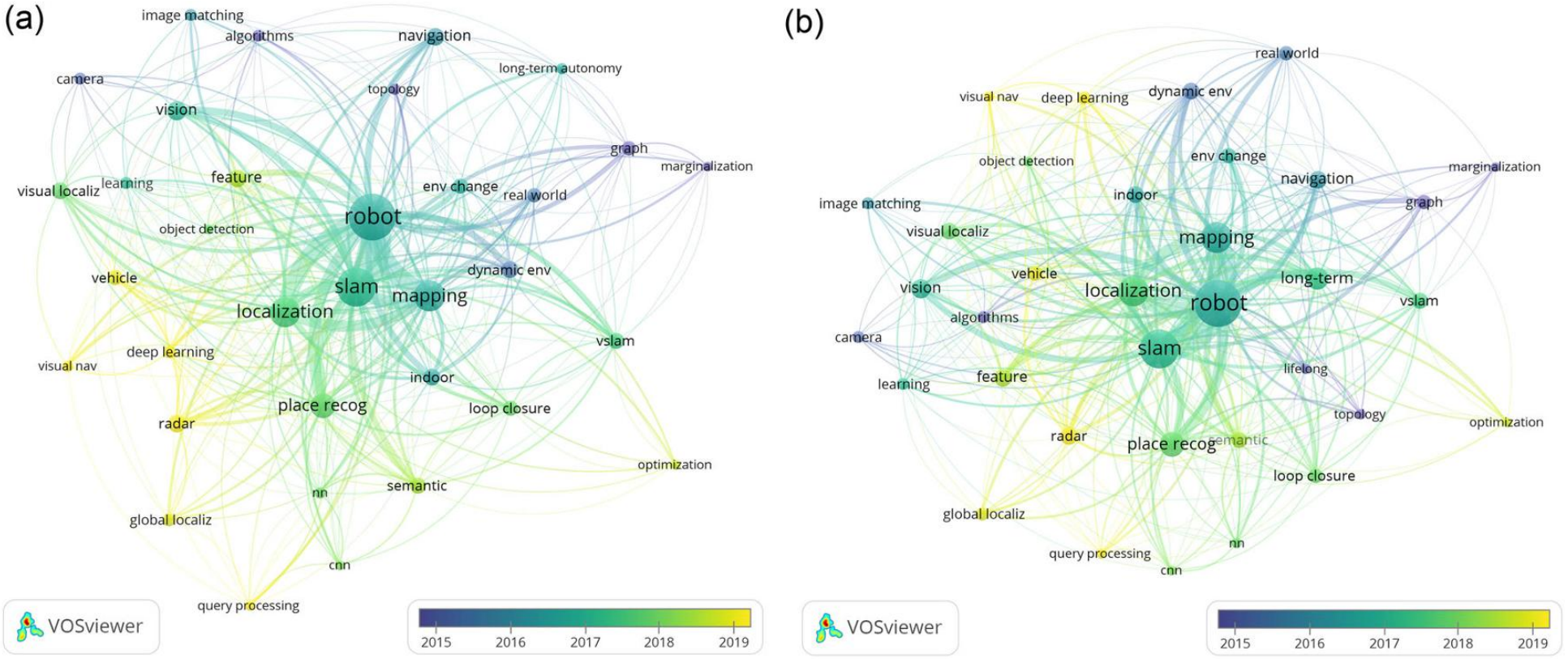


FIGURE 3 Keywords co-occurrence analysis on the 142 included records generated by VOSviewer with overlay visualization by the average publication year: (a) original keywords; (b) all keywords containing long-term and lifelong summarized by the terms themselves. Parameters used for generating the co-occurrence network: minimum number of occurrences = 5, attraction = 2, repulsion = 0, scale = 1.49, circles size variation = 0.5, lines size validation = 1.0. `cnn`, Convolutional Neural Networks; `env`, environment; `localiz`, localization, `nav`, navigation; `nn`, Neural Networks, `recog`, recognition; `vslam`, visual SLAM. [Color figure can be viewed at wileyonlinelibrary.com]

SLR: Co-Authorship Analysis Results

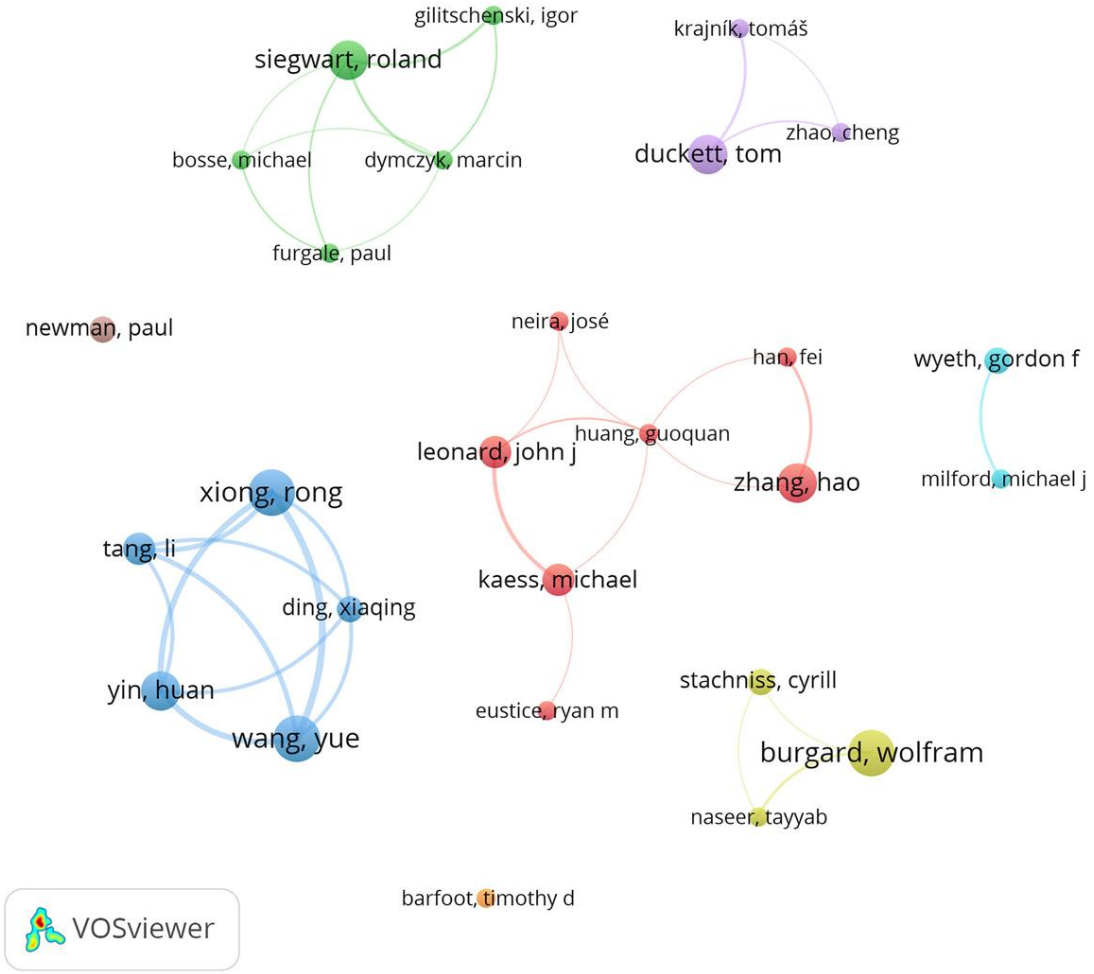


FIGURE 4 Co-authorship analysis on the 142 included records generated by VOSviewer. Parameters used for generating the co-occurrence network: minimum number of occurrences = 3, attraction = 4, repulsion = −2, scale = 1.49, circles size variation = 1.0, lines size validation = 1.0. [Color figure can be viewed at wileyonlinelibrary.com]

SLR: Publication Venues

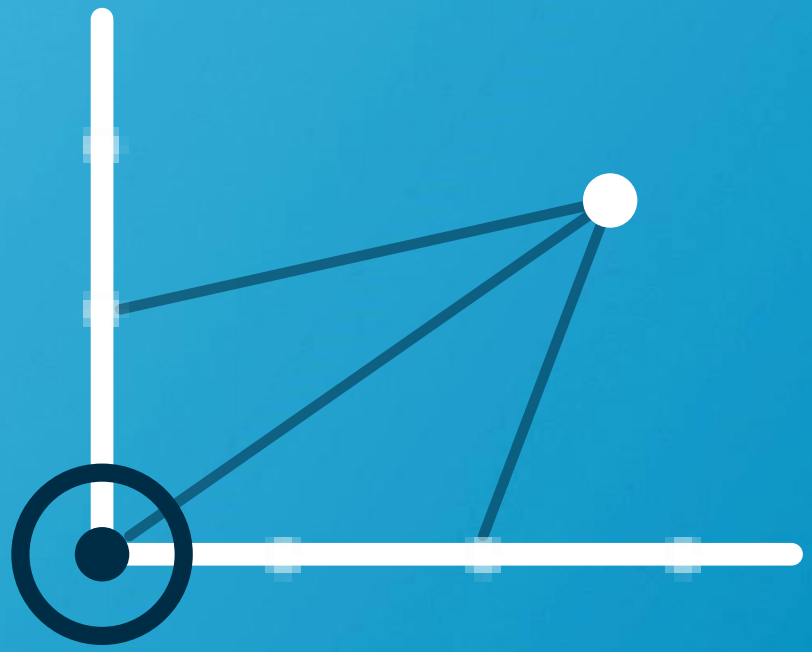
TABLE 5 Publication venues of the included records in this review with more than one record published in the venue: (a) journals; (b) conferences.

(a)				
	Year			
Journal	μ	σ	Max	#
Robotics and Autonomous Systems	2016	3.9	2021	13
IEEE Robotics and Automation Letters	2019	1.7	2022	12
International Journal of Robotics Research	2014	3.2	2022	11
Journal of Field Robotics	2017	3.5	2022	8
Autonomous Robots	2017	2.2	2020	7
IEEE Transactions on Intelligent Transportation Systems	2021	0.8	2022	4
Sensors	2019	0.8	2020	4
IEEE Transactions on Robotics	2017	3.1	2022	4
IEEE Sensors Journal	2020	1.5	2021	2
International Journal of Advanced Robotic Systems	2020	1.5	2021	2

(b)				
	Year			
Conference	μ	σ	Max	#
IEEE International Conference on Robotics and Automation (ICRA)	2016	3.9	2021	22
IEEE/RSJ International Conference on Intelligent Robots and Systems (IROS)	2017	3.6	2021	17
IEEE International Conference on Robotics and Biomimetics (ROBIO)	2019	2.1	2021	3
IEEE International Intelligent Transportation Systems Conference (ITSC)	2018	2.4	2021	3
European Conference on Mobile Robots (ECMR)	2014	0.9	2015	3
IEEE Intelligent Vehicles Symposium (IV)	2019	0.5	2019	2
International Conference on 3D Vision (3DV)	2018	1.5	2019	2
International Conference on Advanced Robotics (ICAR)	2011	2.0	2013	2

Abbreviations: μ, average year of publication; σ, standard deviation of the publication year; max, maximum year of publication; #, number of records published at a certain venue.

Corner-Like Detector for 2D Lasers



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CORNER-LIKE FEATURE DETECTOR FORMULATION

$$\gamma_0 + \gamma_1 x + \gamma_2 y = 0$$

$$y = \beta_0 + \beta_1 x$$

$$\hat{\beta}_1 = \frac{\sum_{i=1}^{n} (x_i - \bar{x})(y_i - \bar{y})}{\sum_{i=1}^{n} (x_i - \bar{x})^2}$$

$$\hat{\beta}_0 = \bar{y} - \hat{\beta}_1 \bar{x}$$

$$x = \alpha_0 + \alpha_1 y$$

$$\hat{\alpha}_1 = \frac{\sum_{i=1}^{n} (x_i - \bar{x})(y_i - \bar{y})}{\sum_{i=1}^{n} (y_i - \bar{y})^2}$$

$$\hat{\alpha}_0 = \bar{x} - \hat{\alpha}_1 \bar{y}$$

$$\varepsilon_{RMSE,\{\hat{\beta}_0,\hat{\beta}_1\}} = \sqrt{\frac{1}{n}\sum_{i=1}^{n}\left(y_i - \hat{\beta}_0 - \hat{\beta}_1 x_i\right)^2}$$

$$\varepsilon_{RMSE,\{\hat{\alpha}_0,\hat{\alpha}_1\}} = \sqrt{\frac{1}{n}\sum_{i=1}^{n}\left(x_i - \hat{\alpha}_0 - \hat{\alpha}_1 y_i\right)^2}$$

- $y = \beta_0^{\text{arr},k} + \beta_1^{\text{arr},k} x \;;\; y = \beta_0^{\text{dep},k} + \beta_1^{\text{dep},k} x$

$$\begin{cases} x_{j,\text{cand},k} = \left(\hat{\beta}_0^{\text{dep},k} - \hat{\beta}_0^{\text{arr},k}\right) / \left(\hat{\beta}_1^{\text{arr},k} - \hat{\beta}_1^{\text{dep},k}\right) \\ y_{j,\text{cand},k} = \hat{\beta}_0^{\text{arr},k} + \hat{\beta}_1^{\text{arr},k} \cdot x_{j,\text{cand},k} \end{cases}$$

- $x = \alpha_0^{\text{arr},k} + \alpha_1^{\text{arr},k} y \;;\; y = \beta_0^{\text{dep},k} + \beta_1^{\text{dep},k} x$

$$\begin{cases} y_{j,\text{cand},k} = \left(\hat{\alpha}_0^{\text{arr},k}\hat{\beta}_1^{\text{dep},k} + \hat{\beta}_0^{\text{dep},k}\right) / \left(1 - \hat{\alpha}_1^{\text{arr},k}\hat{\beta}_1^{\text{dep},k}\right) \\ x_{j,\text{cand},k} = \hat{\alpha}_0^{\text{arr},k} + \hat{\alpha}_1^{\text{arr},k} \cdot y_{j,\text{cand},k} \end{cases}$$

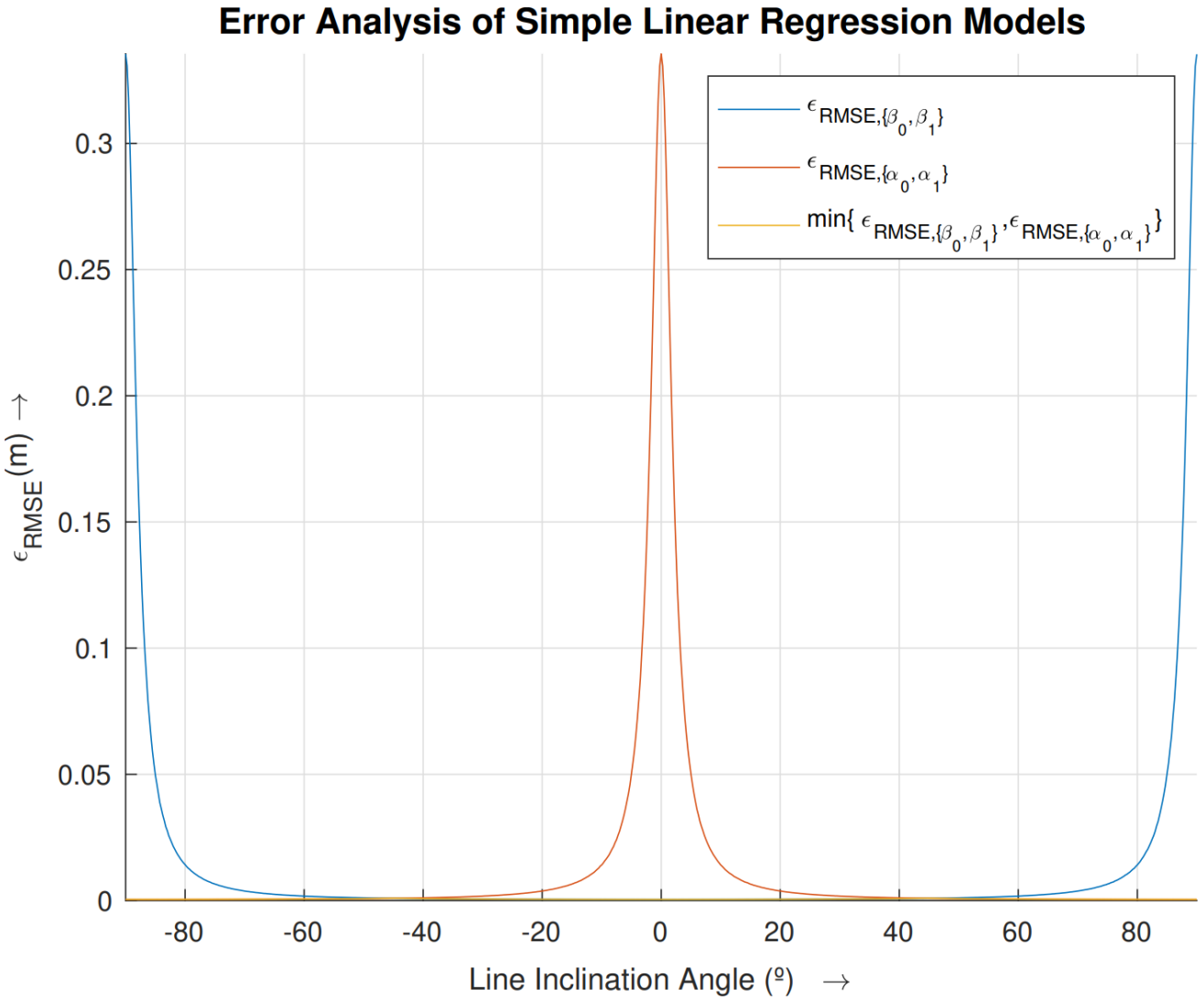
- $y = \beta_0^{\text{arr},k} + \beta_1^{\text{arr},k} x \;;\; x = \alpha_0^{\text{dep},k} + \alpha_1^{\text{dep},k} y$

$$\begin{cases} y_{j,\text{cand},k} = \left(\hat{\alpha}_0^{\text{dep},k}\hat{\beta}_1^{\text{arr},k} + \hat{\beta}_0^{\text{arr},k}\right) / \left(1 - \hat{\alpha}_1^{\text{dep},k}\hat{\beta}_1^{\text{arr},k}\right) \\ x_{j,\text{cand},k} = \hat{\alpha}_0^{\text{dep},k} + \hat{\alpha}_1^{\text{dep},k} \cdot y_{j,\text{cand},k} \end{cases}$$

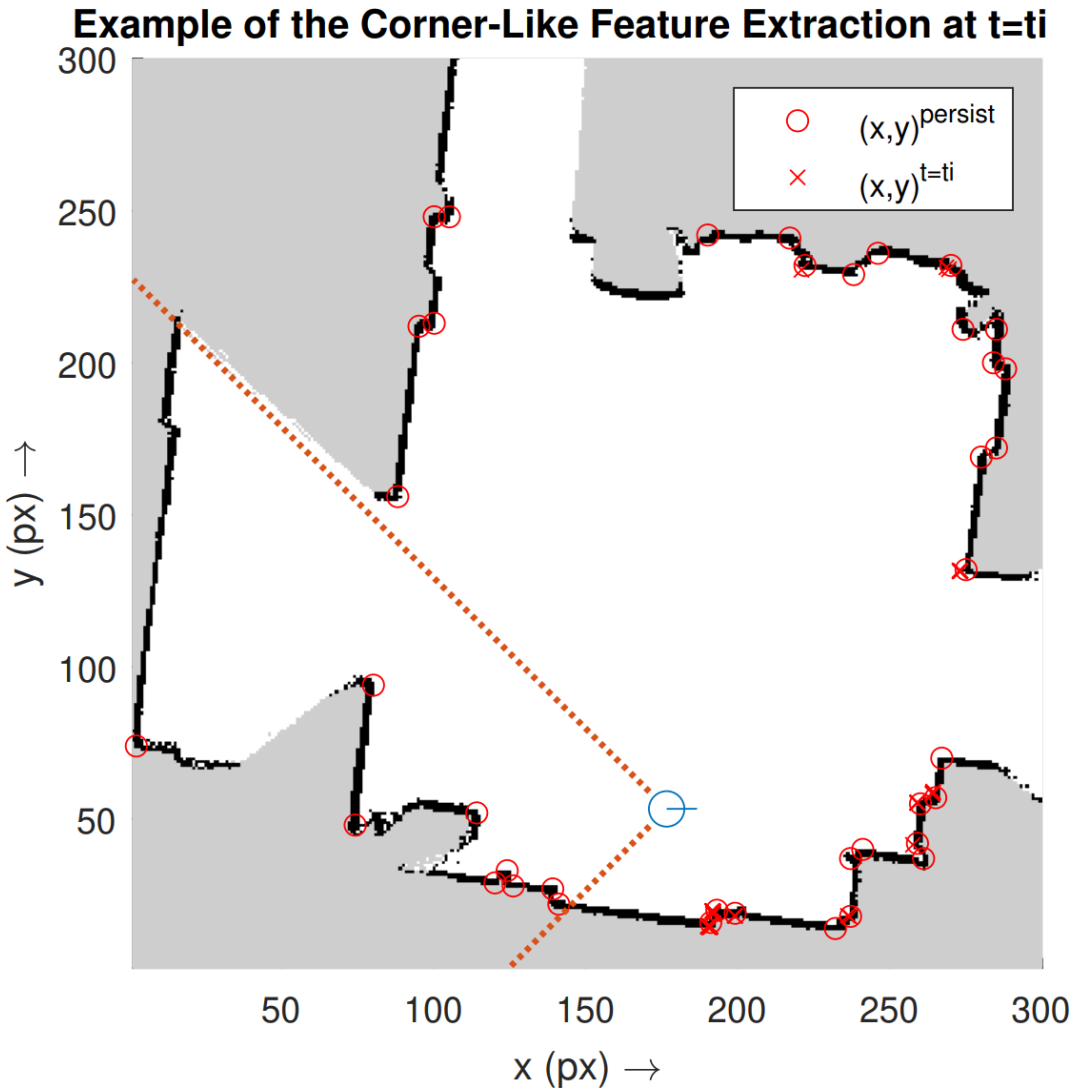
- $x = \alpha_0^{\text{arr},k} + \alpha_1^{\text{arr},k} y \;;\; x = \alpha_0^{\text{dep},k} + \alpha_1^{\text{dep},k} y$

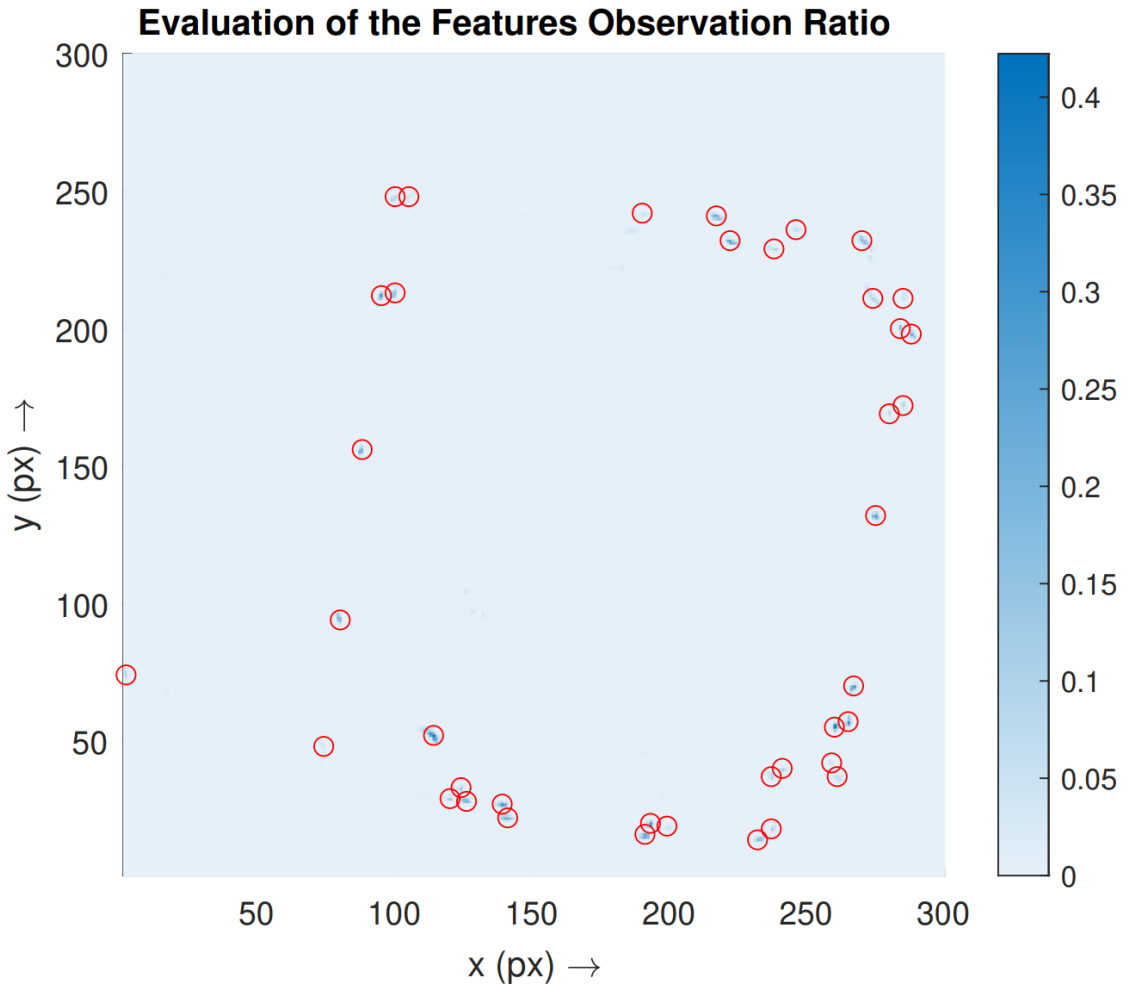
$$\begin{cases} y_{j,\text{cand},k} = \left(\hat{\alpha}_0^{\text{dep},k} - \hat{\alpha}_0^{\text{arr},k}\right) / \left(\hat{\alpha}_1^{\text{arr},k} - \hat{\alpha}_1^{\text{dep},k}\right) \\ x_{j,\text{cand},k} = \hat{\alpha}_0^{\text{arr},k} + \hat{\alpha}_1^{\text{arr},k} \cdot y_{j,\text{cand},k} \end{cases}$$

Error Analysis of Line Regression Models

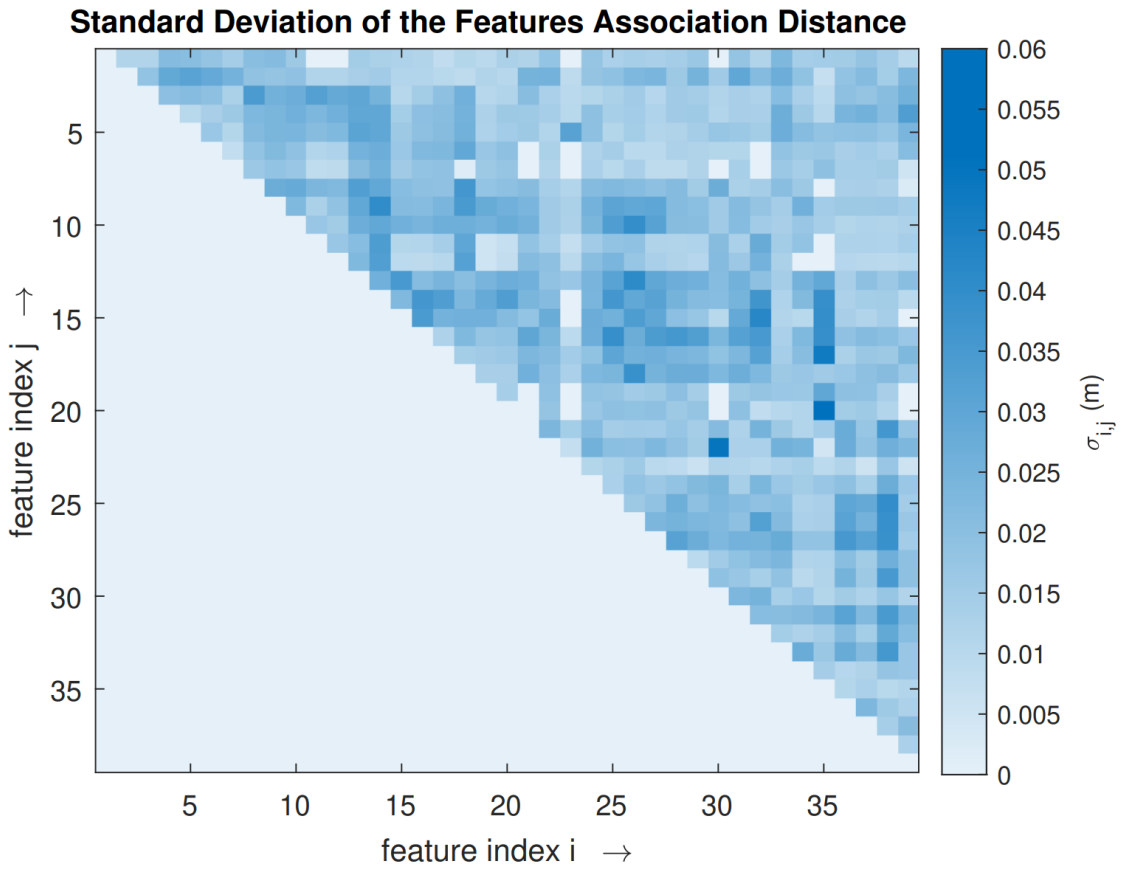


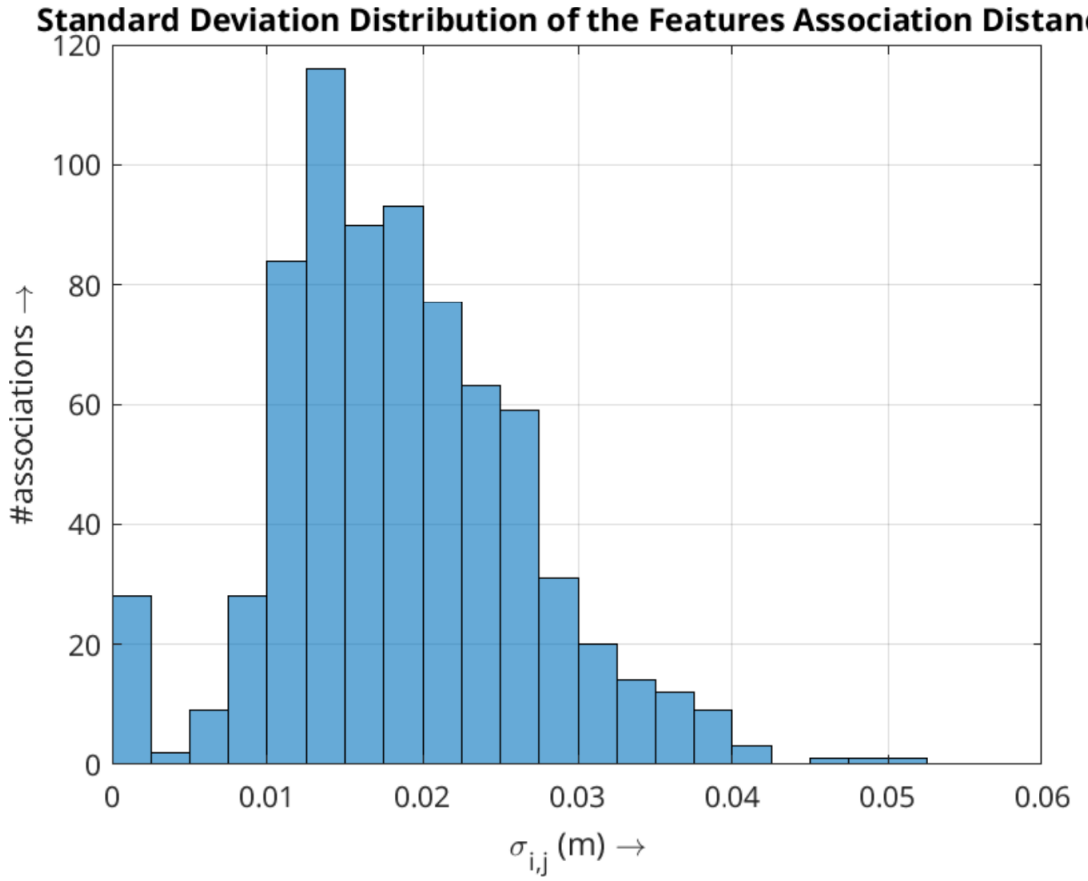
EXPERIMENTAL RESULTS: FEATURE EXTRACTION



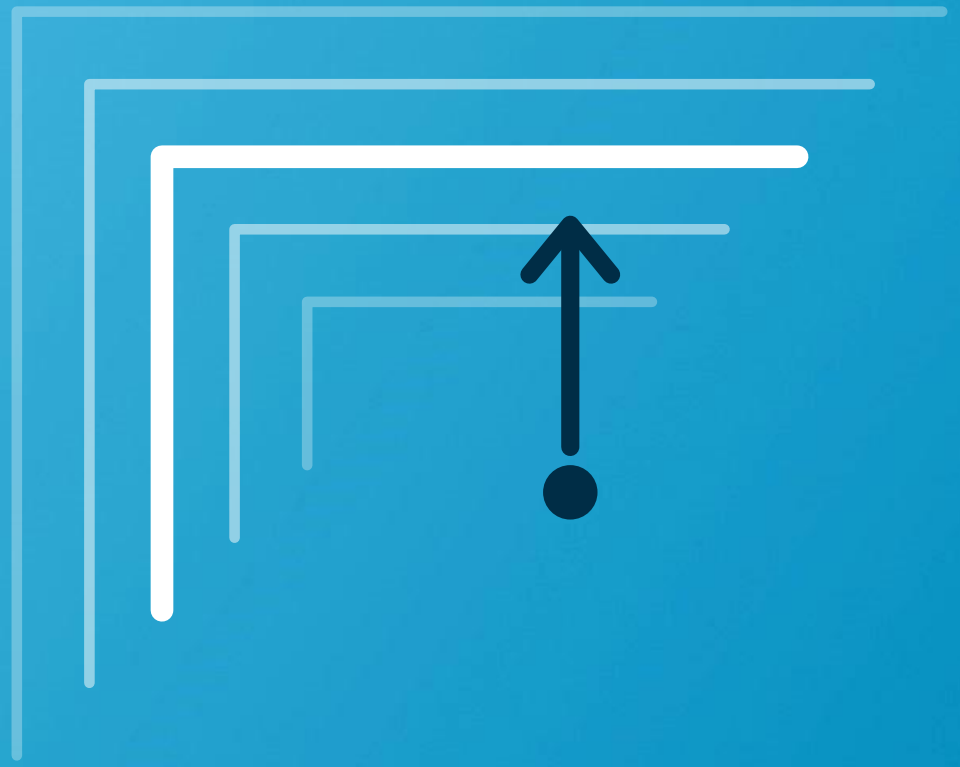


EXPERIMENTAL RESULTS: FEATURE ASSOCIATION DISTANCE





UDF-Based Front-End



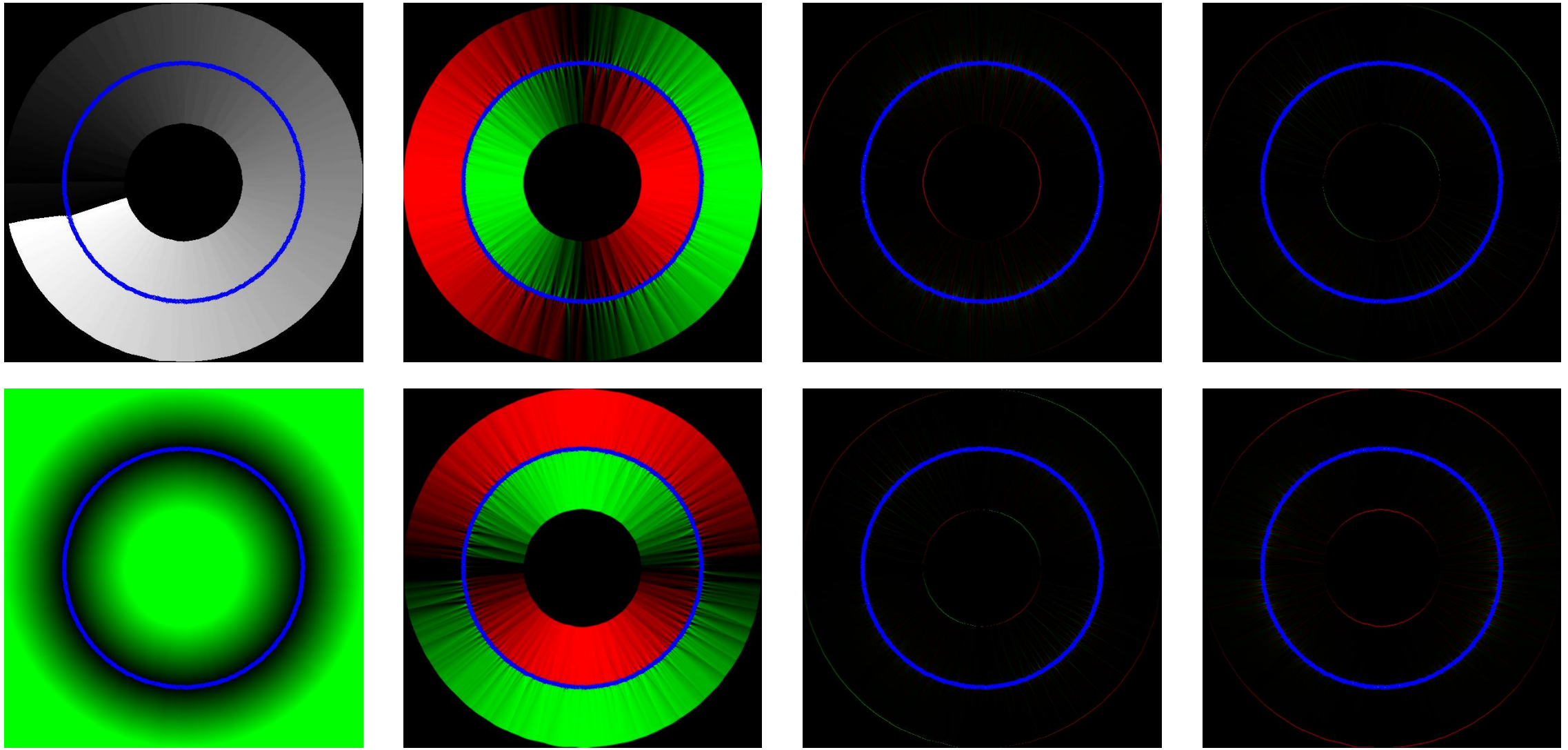
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Common M-Estimator Loss Functions

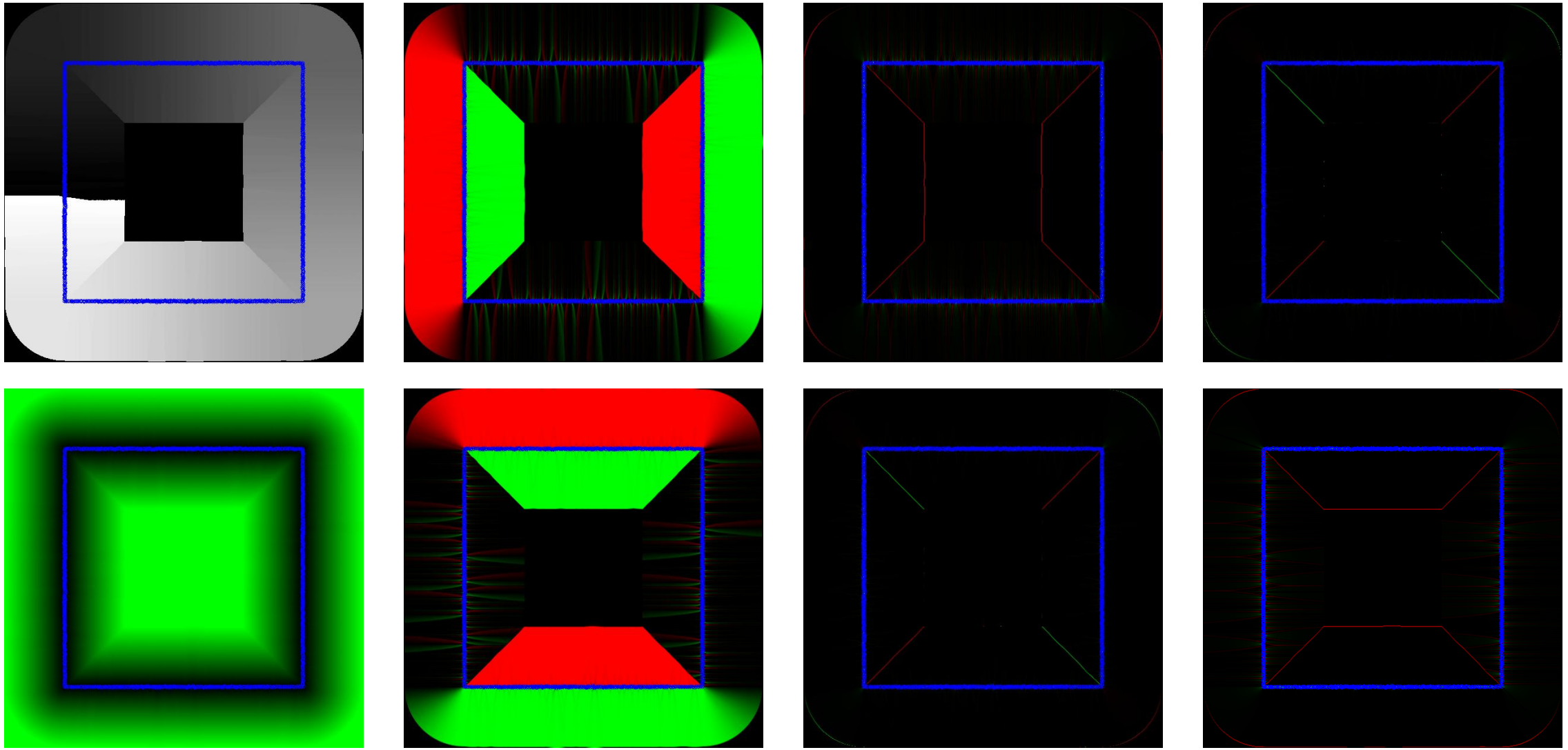
Table 4.1: Common M-estimator loss functions $\rho(s)$, their first-order derivative $\psi(s) = \rho'(s)$, and weight functions $w(s) = \psi(s)/(2s)$. The tuning constant c controls the inlier/outlier threshold [184].

Estimator	$\rho(s)$	$\psi(s)$	$w(s)$
L^2	s^2	$2s$	1
L^1	$\lvert s\rvert$	$s/\lvert s\rvert$	$1/(2\lvert s\rvert)$
L^2_{trunc}	$\begin{cases} s^2 & \text{if } \lvert s\rvert < c \\ c^2 & \text{otherwise} \end{cases}$	$\begin{cases} 2s & \text{if } \lvert s\rvert < c \\ 0 & \text{otherwise} \end{cases}$	$\begin{cases} 1 & \text{if } \lvert s\rvert < c \\ 0 & \text{otherwise} \end{cases}$
L^1_{trunc}	$\begin{cases} \lvert s\rvert & \text{if } \lvert s\rvert < c \\ c & \text{otherwise} \end{cases}$	$\begin{cases} s/\lvert s\rvert & \text{if } \lvert s\rvert < c \\ 0 & \text{otherwise} \end{cases}$	$\begin{cases} 1/(2\lvert s\rvert) & \text{if } \lvert s\rvert < c \\ 0 & \text{otherwise} \end{cases}$
Huber	$\begin{cases} s^2 & \text{if } \lvert s\rvert < c \\ 2c\lvert s\rvert - c^2 & \text{otherwise} \end{cases}$	$\begin{cases} 2s & \text{if } \lvert s\rvert < c \\ 2c \cdot s/\lvert s\rvert & \text{otherwise} \end{cases}$	$\begin{cases} 1 & \text{if } \lvert s\rvert < c \\ c/\lvert s\rvert & \text{otherwise} \end{cases}$
Cauchy	$c^2 \ln\left(1 + \frac{s^2}{c^2}\right)$	$\frac{2s}{1 + (s/c)^2}$	$\frac{1}{1 + (s/c)^2}$
Tukey	$\begin{cases} c^2\left[1 - \left(1 - \frac{s^2}{c^2}\right)^3\right] & \text{if } \lvert s\rvert < c \\ c^2 & \text{otherwise} \end{cases}$	$\begin{cases} 6s\left(1 - \frac{s^2}{c^2}\right)^2 & \text{if } \lvert s\rvert < c \\ 0 & \text{otherwise} \end{cases}$	$\begin{cases} 3\left(1 - \frac{s^2}{c^2}\right)^2 & \text{if } \lvert s\rvert < c \\ 0 & \text{otherwise} \end{cases}$
Geman-McClure	$\frac{s^2}{c^2 + s^2}$	$2s \cdot \frac{c^2}{(c^2 + s^2)^2}$	$\frac{c^2}{(c^2 + s^2)^2}$

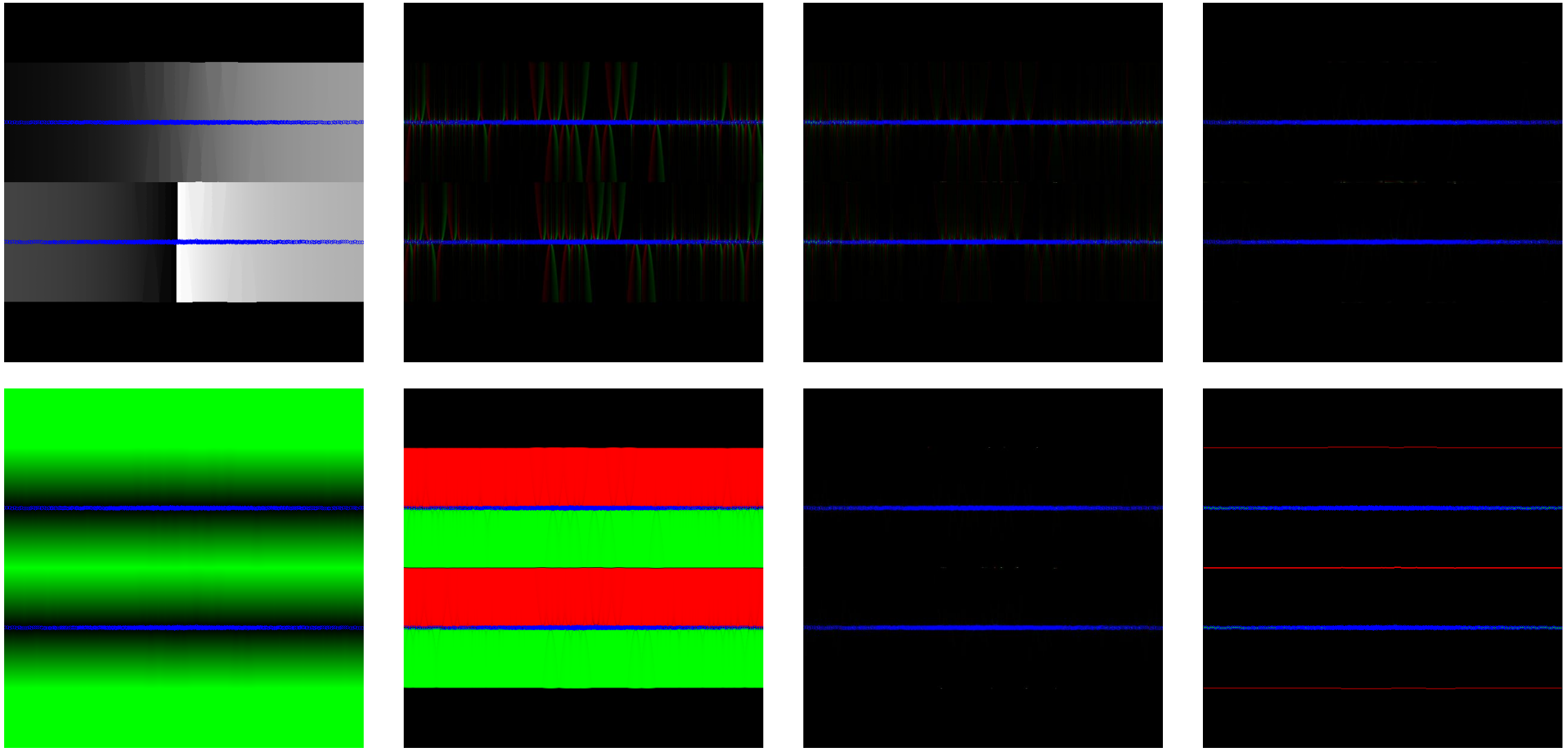
Distance Map Computation – Synthetic Benchmark



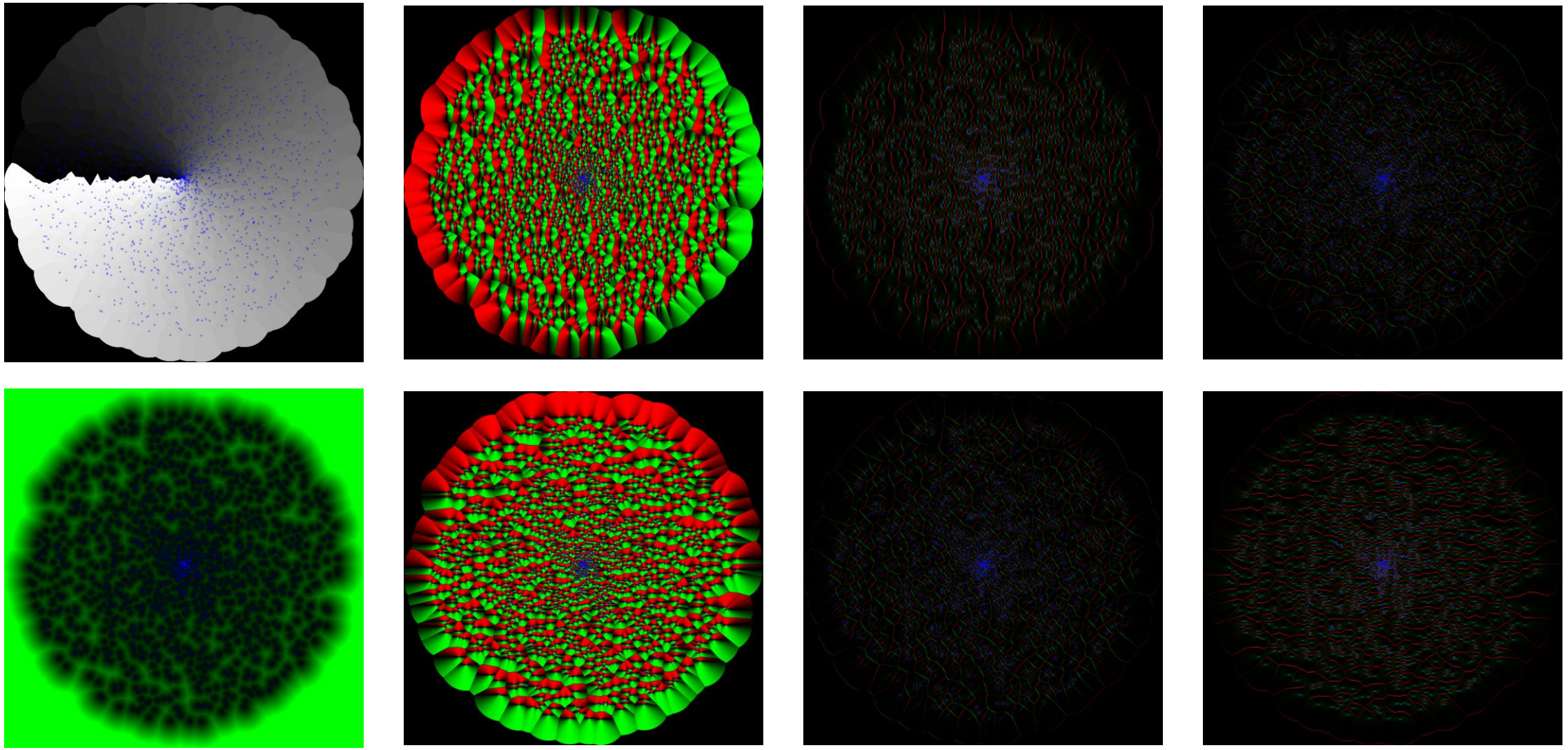
Distance Map Computation – Synthetic Benchmark



Distance Map Computation – Synthetic Benchmark



Distance Map Computation – Synthetic Benchmark



Experimental Results: Synthetic Benchmark

Table 4.3: Synthetic benchmark results for point cloud alignment using distance map-based error formulations and distance map-guided Iterative Closest Point (ICP) baselines [65, 171] (cell resolution $\delta = 0.01$ m; no noise added to the point cloud data). Legend: distance map representation level ($\mathcal{M}_l$); point cloud voxelization (V); sub-pixelization (S) (bi-linear interpolation); enabled (✓), disabled (✗), not applicable (–); translation error ε_t (m); rotation error ε_θ (°); Mahalanobis-squared consistency metric $\chi^2_\varepsilon = \varepsilon^T \mathbf{\Omega}\, \varepsilon$ (dimensionless). For the circle scene, only the translation error is reported, as rotational alignment is degenerate. Similarly, only the observable translation component ε_y and rotational alignment error ε_θ are reported in the corridor scene, as the x axis is degenerate.

				Circle		Square			Corridor			Random		
Meth.	$\mathcal{M}_l$	**V?**	**S?**	ε_t (m)	χ^2_ε	ε_t (m)	ε_θ (°)	χ^2_ε	ε_y (m)	ε_θ (°)	χ^2_ε	ε_t (m)	ε_θ (°)	χ^2_ε
$\mathcal{D}_{\text{p2p}}$	$\mathcal{M}_0$	✗	–	0.001	6404.92	0.001	0.02	0.01	0.004	0.11	543.09	0.001	**0.00**	**0.00**
		✓	–	0.003	5727.27	0.001	0.02	0.01	0.004	0.10	531.00	0.002	**0.00**	**0.00**
	$\mathcal{M}_1$	✗	✗	0.003	2741.96	0.008	0.02	0.04	0.006	0.16	190.41	0.009	**0.00**	0.02
		✗	✓	0.010	1684.46	0.002	0.02	0.01	0.004	0.17	154.93	0.010	**0.00**	0.05
		✓	✗	0.003	2734.18	0.008	0.02	0.04	0.006	0.16	190.54	0.009	**0.00**	0.02
		✓	✓	0.010	1684.46	0.002	0.02	0.01	0.004	0.17	154.93	0.010	**0.00**	0.05
$\mathcal{D}_{\text{p2pl}}$	$\mathcal{M}_0$	✗	–	**0.000**	54.04	**0.000**	**0.00**	**0.00**	**0.000**	**0.00**	**5.72**	–	–	–
		✓	–	**0.000**	**53.98**	**0.000**	**0.00**	**0.00**	**0.000**	**0.00**	**5.72**	–	–	–
	$\mathcal{M}_2$	✗	✗	**0.000**	545.13	0.007	0.02	0.06	0.004	0.01	37.56	–	–	–
		✗	✓	0.007	322.07	0.002	**0.00**	0.01	0.003	**0.00**	31.27	–	–	–
		✓	✗	**0.000**	545.38	0.007	0.02	0.06	0.003	0.01	32.89	–	–	–
		✓	✓	0.007	322.22	0.002	**0.00**	0.01	0.003	**0.00**	31.26	–	–	–
$\text{ICP}_{\mathcal{D},\,\text{p2p}}$ ([65])	$\mathcal{M}_0$	✗	–	0.001	6665.84	0.001	0.02	0.02	**0.000**	0.01	833.41	**0.000**	**0.00**	**0.00**
		✓	–	0.003	6568.31	0.001	0.02	0.02	**0.000**	0.01	824.97	**0.000**	**0.00**	**0.00**
$\text{ICP}_{\mathcal{D},\,\text{p2pl}}$ ([171])	$\mathcal{M}_0$	✗	–	**0.000**	54.04	**0.000**	**0.00**	**0.00**	**0.000**	**0.00**	**5.72**	–	–	–
		✓	–	**0.000**	**53.98**	**0.000**	**0.00**	**0.00**	**0.000**	**0.00**	**5.72**	–	–	–

Experimental Results: Synthetic Benchmark

Table 4.4: Synthetic benchmark results for point cloud alignment using distance map-based error formulations and distance map-guided Iterative Closest Point (ICP) baselines [65, 171] (cell resolution $\delta = 0.01\,\text{m}$; $\sigma = 0.03\,\text{m}$ Gaussian noise added to the point cloud data). Legend: distance map representation level ($\mathcal{M}_l$); point cloud voxelization (V); sub-pixelization (S) (bi-linear interpolation); enabled (✓), disabled (✗), not applicable (–); translation error ε_t (m); rotation error ε_θ (°); Mahalanobis-squared consistency metric $\chi^2_\varepsilon = \varepsilon^T \boldsymbol{\Omega}\, \varepsilon$ (dimensionless). For the circle scene, only the translation error is reported, as rotational alignment is degenerate. Similarly, only the observable translation component ε_y and rotational alignment error ε_θ are reported in the corridor scene, as the x axis is degenerate.

				Circle		**Square**			**Corridor**			**Random**		
Meth.	$\mathcal{M}_l$	**V?**	**S?**	ε_t (m)	χ^2_ε	ε_t (m)	ε_θ (°)	χ^2_ε	ε_y (m)	ε_θ (°)	χ^2_ε	ε_t (m)	ε_θ (°)	χ^2_ε
$\mathcal{D}_{\text{p2p}}$	$\mathcal{M}_0$	✗	–	**0.003**	2980.76	0.003	0.02	0.02	0.003	0.13	435.73	0.005	0.02	0.04
		✓	–	**0.003**	2990.83	0.003	0.02	0.02	0.003	0.13	435.21	0.005	0.02	0.04
	$\mathcal{M}_1$	✗	✗	**0.003**	2118.12	0.003	0.02	**0.01**	0.004	0.16	302.40	0.009	0.03	0.10
		✗	✓	0.008	1810.30	0.007	0.02	0.03	0.007	0.17	263.15	0.015	0.03	0.21
		✓	✗	**0.003**	2118.27	0.003	0.02	**0.01**	0.004	0.16	302.57	0.009	0.03	0.10
		✓	✓	0.008	1810.30	0.007	0.02	0.03	0.007	0.17	263.15	0.015	0.03	0.21
$\mathcal{D}_{\text{p2pl}}$	$\mathcal{M}_0$	✗	–	**0.003**	191.62	**0.002**	**0.01**	**0.01**	**0.001**	**0.01**	**14.04**	–	–	–
		✓	–	**0.003**	**190.76**	**0.002**	**0.01**	**0.01**	**0.001**	**0.01**	14.07	–	–	–
	$\mathcal{M}_2$	✗	✗	0.004	5832.76	0.004	0.15	1.96	**0.001**	**0.01**	445.03	–	–	–
		✗	✓	0.008	4527.51	0.008	0.04	0.14	0.005	**0.01**	337.25	–	–	–
		✓	✗	0.004	5828.14	0.004	0.05	0.16	**0.001**	**0.01**	444.22	–	–	–
		✓	✓	0.008	4527.48	0.008	0.03	0.12	0.005	**0.01**	337.27	–	–	–
$\text{ICP}_{\mathcal{D},\,\text{p2p}}$ ([65])	$\mathcal{M}_0$	✗	–	**0.003**	6584.09	0.003	0.02	0.04	**0.001**	**0.01**	856.26	**0.001**	**0.00**	**0.00**
		✓	–	**0.003**	6586.87	0.003	0.02	0.04	**0.001**	**0.01**	857.53	**0.001**	**0.00**	**0.00**
$\text{ICP}_{\mathcal{D},\,\text{p2pl}}$ ([171])	$\mathcal{M}_0$	✗	–	**0.003**	191.62	**0.002**	**0.01**	**0.01**	**0.001**	**0.01**	**14.04**	–	–	–
		✓	–	**0.003**	**190.76**	**0.002**	**0.01**	**0.01**	**0.001**	**0.01**	14.07	–	–	–

Experimental Results: Synthetic Benchmark

Table 4.5: Synthetic benchmark results for point cloud alignment using distance map-based error formulations and distance map-guided Iterative Closest Point (ICP) baselines [65, 171] (cell resolution $\delta = 0.03$ m; no noise added to the point cloud data). Legend: distance map representation level ($\mathcal{M}_l$); point cloud voxelization (V); sub-pixelization (S) (bi-linear interpolation); enabled (✓), disabled (✗), not applicable (–); translation error ε_t (m); rotation error ε_θ (°); Mahalanobis-squared consistency metric $\chi^2_\varepsilon = \varepsilon^T \mathbf{\Omega}\, \varepsilon$ (dimensionless). For the circle scene, only the translation error is reported, as rotational alignment is degenerate. Similarly, only the observable translation component ε_y and rotational alignment error ε_θ are reported in the corridor scene, as the x axis is degenerate.

				Circle		Square			Corridor			Random		
Meth.	$\mathcal{M}_l$	**V?**	**S?**	ε_t (m)	χ^2_ε	ε_t (m)	ε_θ (°)	χ^2_ε	ε_y (m)	ε_θ (°)	χ^2_ε	ε_t (m)	ε_θ (°)	χ^2_ε
$\mathcal{D}_{\text{p2p}}$	$\mathcal{M}_0$	✗	–	0.001	6234.68	0.001	0.19	0.94	0.007	0.11	540.06	0.009	**0.00**	**0.00**
		✓	–	0.009	4827.02	0.012	0.01	0.10	0.009	0.10	462.70	0.009	**0.00**	**0.00**
	$\mathcal{M}_1$	✗	✗	0.018	586.23	0.015	0.02	0.07	0.008	0.21	88.20	0.025	0.01	0.15
		✗	✓	0.029	257.09	0.051	0.02	0.32	0.011	0.22	70.90	0.030	0.01	0.40
		✓	✗	0.018	584.83	0.015	0.02	0.07	0.008	0.20	98.33	0.027	0.01	0.17
		✓	✓	0.029	257.08	0.051	0.02	0.32	0.011	0.22	75.83	0.030	0.01	0.40
$\mathcal{D}_{\text{p2pl}}$	$\mathcal{M}_0$	✗	–	**0.000**	**52.86**	**0.000**	**0.00**	**0.00**	**0.000**	**0.00**	5.72	–	–	–
		✓	–	**0.000**	54.10	**0.000**	**0.00**	**0.00**	**0.000**	**0.00**	5.23	–	–	–
	$\mathcal{M}_2$	✗	✗	0.009	463.03	0.004	0.03	0.01	0.009	0.01	**4.93**	–	–	–
		✗	✓	0.023	264.87	0.023	0.08	0.33	0.012	0.01	8.66	–	–	–
		✓	✗	0.008	462.02	0.004	0.03	0.01	0.009	0.02	5.09	–	–	–
		✓	✓	0.023	265.03	0.024	0.07	0.36	0.012	0.01	11.40	–	–	–
$\text{ICP}_{\mathcal{D},\text{p2p}}$ ([65])	$\mathcal{M}_0$	✗	–	**0.000**	6432.74	0.001	0.19	2.07	**0.000**	0.01	834.29	**0.000**	**0.00**	**0.00**
		✓	–	0.009	6578.50	0.013	0.01	0.25	**0.000**	0.01	775.70	**0.000**	**0.00**	**0.00**
$\text{ICP}_{\mathcal{D},\text{p2pl}}$ ([171])	$\mathcal{M}_0$	✗	–	**0.000**	**52.86**	**0.000**	**0.00**	**0.00**	**0.000**	**0.00**	5.72	–	–	–
		✓	–	**0.000**	54.10	**0.000**	**0.00**	**0.00**	**0.000**	**0.00**	5.23	–	–	–

Experimental Results: Synthetic Benchmark

Table 4.6: Synthetic benchmark results for point cloud alignment using distance map-based error formulations and distance map-guided Iterative Closest Point (ICP) baselines [65, 171] (cell resolution $\delta = 0.03\,\text{m}$; $\sigma = 0.03\,\text{m}$ Gaussian noise added to the point cloud data). Legend: distance map representation level ($\mathcal{M}_l$); point cloud voxelization (V); sub-pixelization (S) (bi-linear interpolation); enabled (✓), disabled (✗), not applicable (–); translation error ε_t (m); rotation error ε_θ (°); Mahalanobis-squared consistency metric $\chi^2_\varepsilon = \varepsilon^T \boldsymbol{\Omega}\, \varepsilon$ (dimensionless). For the circle scene, only the translation error is reported, as rotational alignment is degenerate. Similarly, only the observable translation component ε_y and rotational alignment error ε_θ are reported in the corridor scene, as the x axis is degenerate.

				Circle		Square			Corridor			Random		
Meth.	$\mathcal{M}_l$	**V?**	**S?**	ε_t (m)	χ^2_ε	ε_t (m)	ε_θ (°)	χ^2_ε	ε_y (m)	ε_θ (°)	χ^2_ε	ε_t (m)	ε_θ (°)	χ^2_ε
$\mathcal{D}_{\text{p2p}}$	$\mathcal{M}_0$	✗	–	**0.003**	3100.86	0.003	0.08	0.19	0.003	0.12	453.71	0.005	0.02	0.03
		✓	–	0.010	3123.95	0.006	0.02	0.04	0.003	0.11	461.08	0.005	0.02	0.03
	$\mathcal{M}_1$	✗	✗	**0.003**	934.92	0.003	**0.01**	**0.00**	0.005	0.20	158.98	0.022	0.03	0.40
		✗	✓	0.021	536.29	0.021	**0.01**	0.12	0.023	0.21	113.04	0.039	0.03	1.02
		✓	✗	**0.003**	937.83	0.003	**0.01**	**0.00**	0.005	0.20	158.77	0.022	0.03	0.40
		✓	✓	0.021	536.49	0.021	**0.01**	0.12	0.024	0.20	112.46	0.039	0.03	1.03
$\mathcal{D}_{\text{p2pl}}$	$\mathcal{M}_0$	✗	–	**0.003**	**189.38**	**0.002**	0.02	0.01	**0.001**	0.02	13.83	–	–	–
		✓	–	0.004	190.31	**0.002**	**0.01**	0.01	0.002	**0.01**	**12.94**	–	–	–
	$\mathcal{M}_2$	✗	✗	**0.003**	1716.85	0.003	0.02	0.01	**0.001**	**0.01**	129.01	–	–	–
		✗	✓	0.021	1120.07	0.021	**0.01**	0.30	0.015	**0.01**	83.51	–	–	–
		✓	✗	**0.003**	1716.63	0.003	0.02	0.01	**0.001**	**0.01**	126.92	–	–	–
		✓	✓	0.021	1119.63	0.021	**0.01**	0.30	0.015	**0.01**	82.42	–	–	–
$\text{ICP}_{\mathcal{D},\,\text{p2p}}$ ([65])	$\mathcal{M}_0$	✗	–	**0.003**	6479.07	0.003	0.08	0.39	**0.001**	0.02	854.64	**0.001**	**0.00**	**0.00**
		✓	–	0.010	6579.53	0.006	0.02	0.08	**0.001**	**0.01**	817.46	**0.001**	**0.00**	**0.00**
$\text{ICP}_{\mathcal{D},\,\text{p2pl}}$ ([171])	$\mathcal{M}_0$	✗	–	**0.003**	**189.38**	**0.002**	0.02	0.01	**0.001**	0.02	13.83	–	–	–
		✓	–	0.004	190.31	**0.002**	**0.01**	0.01	0.002	**0.01**	**12.94**	–	–	–

Experimental Results: Synthetic Benchmark

Table 4.7: Synthetic benchmark results for point cloud alignment using distance map-based error formulations and distance map-guided Iterative Closest Point (ICP) baselines [65, 171] (cell resolution $\delta = 0.05$ m; no noise added to the point cloud data). Legend: distance map representation level ($\mathcal{M}_l$); point cloud voxelization (V); sub-pixelization (S) (bi-linear interpolation); enabled (✓), disabled (✗), not applicable (–); translation error ε_t (m); rotation error ε_θ (°); Mahalanobis-squared consistency metric $\chi^2_\varepsilon = \varepsilon^T \mathbf{\Omega}\, \varepsilon$ (dimensionless). For the circle scene, only the translation error is reported, as rotational alignment is degenerate. Similarly, only the observable translation component ε_y and rotational alignment error ε_θ are reported in the corridor scene, as the x axis is degenerate.

				Circle		Square			Corridor			Random		
Meth.	$\mathcal{M}_l$	**V?**	**S?**	ε_t (m)	χ^2_ε	ε_t (m)	ε_θ (°)	χ^2_ε	ε_y (m)	ε_θ (°)	χ^2_ε	ε_t (m)	ε_θ (°)	χ^2_ε
$\mathcal{D}_{\text{p2p}}$	$\mathcal{M}_0$	✗	–	0.001	6198.44	0.001	0.10	0.35	0.018	0.07	454.93	0.019	**0.00**	**0.00**
		✓	–	0.004	4953.51	0.005	0.01	0.02	0.013	0.11	379.95	0.023	**0.00**	0.01
	$\mathcal{M}_1$	✗	✗	0.022	480.88	0.027	0.05	0.03	0.019	0.22	69.63	0.044	0.01	0.40
		✗	✓	0.044	186.61	0.028	0.17	0.16	0.009	0.24	44.35	0.054	0.03	1.30
		✓	✗	0.022	422.14	0.027	0.05	0.03	0.026	0.21	76.78	0.044	0.01	0.42
		✓	✓	0.044	162.18	0.027	0.16	0.16	0.021	0.24	45.47	0.054	0.03	1.30
$\mathcal{D}_{\text{p2pl}}$	$\mathcal{M}_0$	✗	–	**0.000**	51.30	**0.000**	**0.00**	**0.00**	**0.000**	**0.00**	5.72	–	–	–
		✓	–	**0.000**	**46.18**	**0.000**	**0.00**	**0.00**	**0.000**	**0.00**	**4.08**	–	–	–
	$\mathcal{M}_2$	✗	✗	0.017	261.79	0.030	0.06	0.08	0.016	0.05	20.86	–	–	–
		✗	✓	0.038	120.63	0.006	0.12	0.10	0.006	0.09	15.29	–	–	–
		✓	✗	0.017	235.66	0.030	0.06	0.09	0.018	0.06	17.17	–	–	–
		✓	✓	0.037	106.47	0.007	0.12	0.10	0.007	0.10	12.93	–	–	–
$\text{ICP}_{\mathcal{D},\text{p2p}}$ ([65])	$\mathcal{M}_0$	✗	–	**0.000**	6431.42	0.001	0.10	0.61	**0.000**	0.03	828.40	**0.000**	**0.00**	**0.00**
		✓	–	0.005	5754.85	0.005	0.01	0.04	**0.000**	0.01	635.54	**0.000**	**0.00**	**0.00**
$\text{ICP}_{\mathcal{D},\text{p2pl}}$ ([171])	$\mathcal{M}_0$	✗	–	**0.000**	51.30	**0.000**	**0.00**	**0.00**	**0.000**	**0.00**	5.72	–	–	–
		✓	–	**0.000**	**46.18**	**0.000**	**0.00**	**0.00**	**0.000**	**0.00**	**4.08**	–	–	–

Experimental Results: Synthetic Benchmark

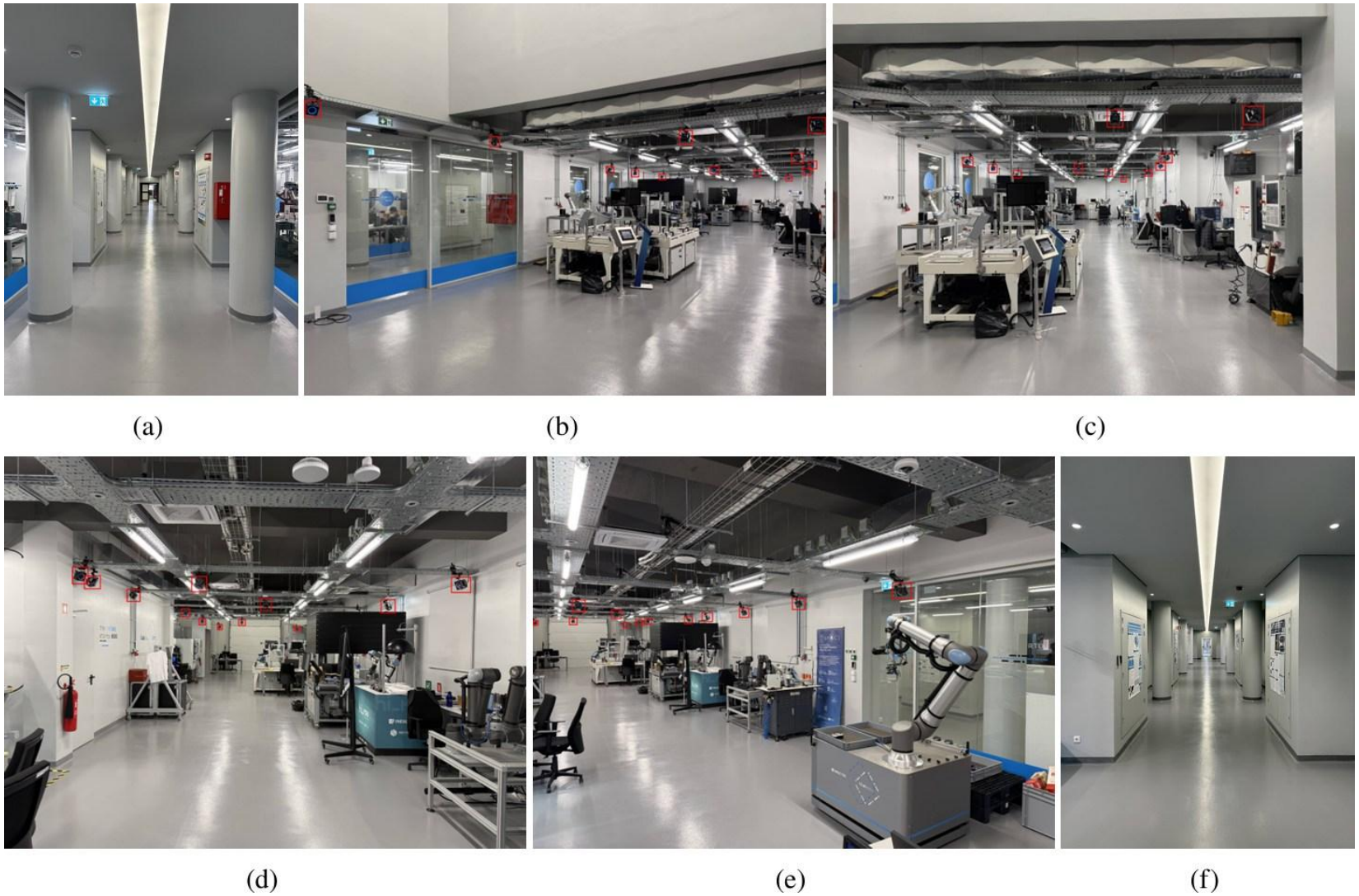
Table 4.8: Synthetic benchmark results for point cloud alignment using distance map-based error formulations and distance map-guided Iterative Closest Point (ICP) baselines [65, 171] (cell resolution $\delta = 0.05\,\text{m}$; $\sigma = 0.03\,\text{m}$ Gaussian noise added to the point cloud data). Legend: distance map representation level ($\mathcal{M}_l$); point cloud voxelization (V); sub-pixelization (S) (bi-linear interpolation); enabled (✓), disabled (✗), not applicable (–); translation error ε_t (m); rotation error ε_θ (°); Mahalanobis-squared consistency metric $\chi^2_\varepsilon = \varepsilon^T \boldsymbol{\Omega}\, \varepsilon$ (dimensionless). For the circle scene, only the translation error is reported, as rotational alignment is degenerate. Similarly, only the observable translation component ε_y and rotational alignment error ε_θ are reported in the corridor scene, as the x axis is degenerate.

				Circle		Square			Corridor			Random		
Meth.	$\mathcal{M}_l$	**V?**	**S?**	ε_t (m)	χ^2_ε	ε_t (m)	ε_θ (°)	χ^2_ε	ε_y (m)	ε_θ (°)	χ^2_ε	ε_t (m)	ε_θ (°)	χ^2_ε
$\mathcal{D}_{\text{p2p}}$	$\mathcal{M}_0$	✗	–	**0.003**	3913.39	0.003	0.29	2.15	0.003	0.14	492.12	0.005	0.02	0.03
		✓	–	0.026	3200.64	0.019	0.03	0.29	0.004	0.13	522.25	0.004	0.02	0.03
	$\mathcal{M}_1$	✗	✗	**0.003**	703.70	0.004	0.02	**0.01**	0.008	0.22	110.43	0.034	0.05	0.89
		✗	✓	0.039	306.68	0.047	**0.01**	0.48	0.036	0.23	71.13	0.063	0.05	2.35
		✓	✗	**0.003**	688.86	0.004	**0.01**	**0.01**	0.011	0.22	106.45	0.034	0.05	0.90
		✓	✓	0.040	304.36	0.048	0.02	0.50	0.039	0.21	69.34	0.063	0.05	2.35
$\mathcal{D}_{\text{p2pl}}$	$\mathcal{M}_0$	✗	–	**0.003**	183.62	**0.002**	0.02	**0.01**	**0.001**	0.02	13.80	–	–	–
		✓	–	0.007	**179.93**	0.004	**0.01**	**0.01**	0.003	**0.01**	**12.01**	–	–	–
	$\mathcal{M}_2$	✗	✗	**0.003**	846.01	**0.002**	**0.01**	**0.01**	**0.001**	0.02	56.29	–	–	–
		✗	✓	0.036	445.88	0.036	**0.01**	0.58	0.026	0.04	33.44	–	–	–
		✓	✗	**0.003**	824.85	**0.002**	**0.01**	**0.01**	**0.001**	0.02	53.47	–	–	–
		✓	✓	0.036	441.65	0.035	**0.01**	0.57	0.025	0.05	31.53	–	–	–
$\text{ICP}_{\mathcal{D},\,\text{p2p}}$ ([65])	$\mathcal{M}_0$	✗	–	**0.003**	6541.08	0.003	0.29	4.60	**0.001**	0.04	844.22	**0.001**	**0.00**	**0.00**
		✓	–	0.026	6248.71	0.020	0.03	0.58	0.002	**0.01**	766.20	**0.001**	**0.00**	**0.00**
$\text{ICP}_{\mathcal{D},\,\text{p2pl}}$ ([171])	$\mathcal{M}_0$	✗	–	**0.003**	183.62	**0.002**	0.02	**0.01**	**0.001**	0.02	13.80	–	–	–
		✓	–	0.007	**179.93**	0.004	**0.01**	**0.01**	0.003	**0.01**	**12.01**	–	–	–

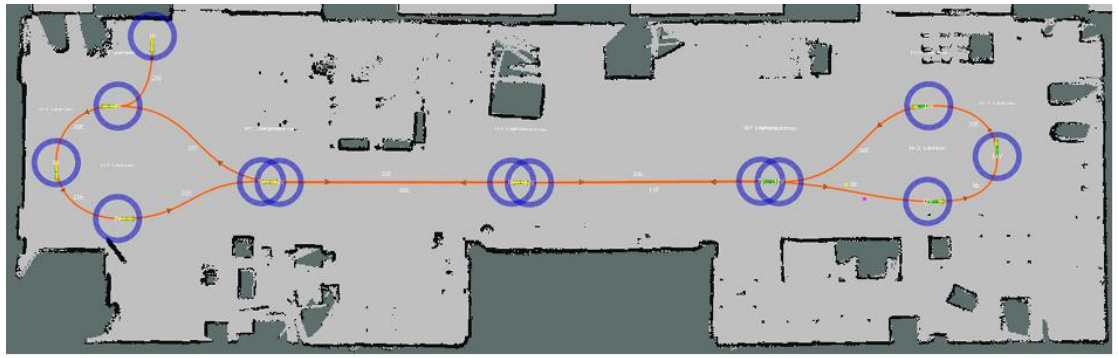
IILABS 3D DATASET

Indoor Benchmark of 3D LiDAR SLAM at iilab - Industry and Innovation Laboratory (https://youtu.be/OwyDBEl3O3M)

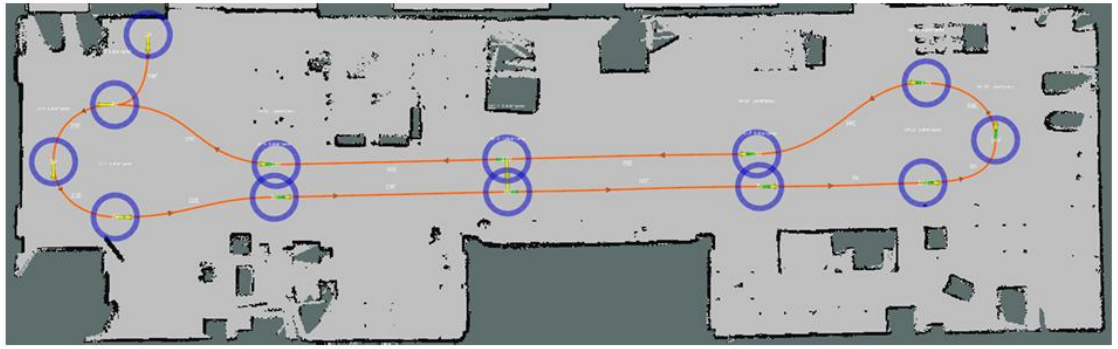
Experimental Results: IILABS 3D – Environment



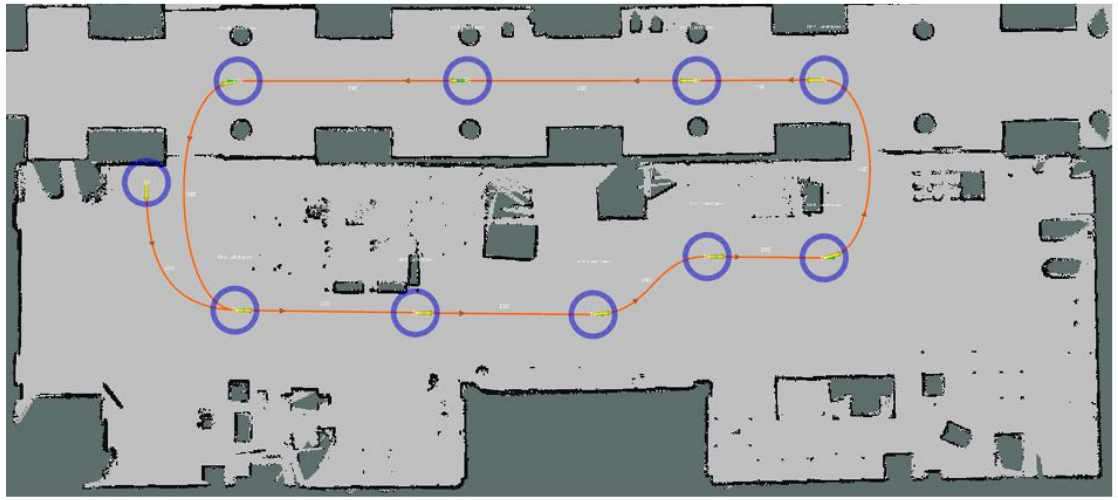
Experimental Results: IILABS 3D – Trajectories



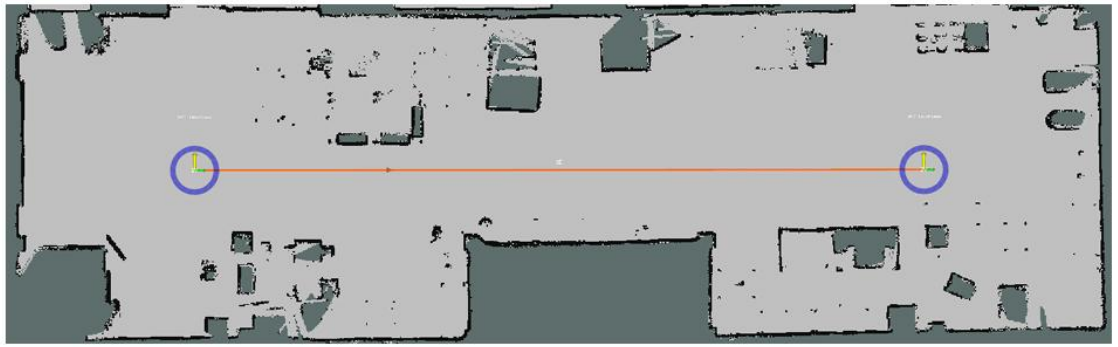
(a) Nav A Diff



(b) Nav A Omni



(c) Loop



(d) Slippage

Experimental Results: IILABS 3D – Laser Odometry

Table 4.9: Real-world laser odometry drift on the IILABS 3D [2] dataset using distance map-based formulations and distance map-guided ICP baselines for the 2D point cloud registration problem (r_{max} = 10 m, $\mathcal{M}_0$ representation level). RTE (%) and RRE (° m^{-1}) are reported as the mean over 10 m segments across four sequences (Nav A Diff, Nav A Omni, Loop, Slippage). Bold entries indicate the lowest drift overall within a tolerance of 0.05 % for RTE and 0.005 ° m^{-1} for RRE. Underlined entries indicate the lowest drift within the pairwise method–splitting criterion for RTE and RRE.

Method	Splitting Criteria	Thresh.	Nav A Diff RTE (%)	Nav A Diff RRE (° m^{-1})	Nav A Omni RTE (%)	Nav A Omni RRE (° m^{-1})	Loop RTE (%)	Loop RRE (° m^{-1})	Slippage RTE (%)	Slippage RRE (° m^{-1})
$\mathcal{D}_{p2p}(\mathcal{M}_0)$	*Translation* (m)	0.50 m	3.13	0.155	2.58	0.134	7.28	0.353	1.27	0.105
		1.00 m	2.20	0.146	2.00	0.101	5.23	0.281	1.01	0.080
		1.50 m	1.79	0.147	1.55	0.096	4.23	0.244	0.90	0.075
		2.00 m	1.56	0.139	1.36	0.091	2.95	0.179	0.70	0.062
		2.50 m	1.41	0.130	1.25	0.100	4.01	0.226	0.93	0.072
		3.00 m	1.66	0.143	1.23	0.085	4.39	0.254	0.82	**0.033**
	Inliers Ratio (%)	40%	1.36	0.115	1.36	0.106	2.56	0.167	0.88	0.058
		50%	1.41	0.130	1.42	0.123	3.68	0.199	0.92	**0.038**
		60%	1.61	0.141	1.47	0.098	3.66	0.216	1.04	0.077
		70%	1.95	0.137	1.90	0.112	4.92	0.248	1.05	0.086
		80%	2.50	0.145	2.35	0.100	6.15	0.299	1.29	0.104
		90%	5.25	0.221	4.12	0.159	10.62	0.390	2.61	0.075
$\mathcal{D}_{p2pl}(\mathcal{M}_0)$	*Translation* (m)	0.50 m	1.55	0.143	1.37	0.105	4.79	0.310	1.45	0.132
		1.00 m	1.45	0.119	1.27	0.097	3.06	0.236	1.09	0.100
		1.50 m	1.37	0.144	1.21	0.103	3.48	0.244	0.83	0.115
		2.00 m	1.20	0.129	1.13	0.097	2.79	0.192	0.81	0.095
		2.50 m	1.16	0.113	0.97	0.117	3.27	0.206	0.68	0.066
		3.00 m	1.72	0.126	1.25	0.107	2.95	0.157	0.73	0.086
	Inliers Ratio (%)	40%	1.13	**0.104**	1.16	0.125	**1.81**	0.155	1.01	0.119
		50%	1.09	**0.104**	0.97	0.098	2.43	**0.143**	1.01	0.107
		60%	**1.03**	0.114	1.03	0.089	3.01	0.190	0.91	0.118
		70%	1.18	0.113	1.00	0.093	3.32	0.208	0.71	0.078
		80%	1.87	0.151	2.30	0.119	5.04	0.231	4.02	0.078
		90%	8.45	0.233	9.23	0.230	8.32	**0.138**	7.12	0.167
$\text{ICP}_{\mathcal{D},\,p2p}$ ([65])	*Translation* (m)	0.50 m	3.14	0.147	2.60	0.108	6.91	0.338	1.38	0.104
		1.00 m	2.20	0.138	1.77	0.109	5.15	0.276	0.60	0.088
		1.50 m	1.69	0.134	1.46	0.112	3.93	0.235	**0.48**	0.059
		2.00 m	1.62	0.125	1.31	0.092	2.75	0.180	**0.45**	0.054
		2.50 m	1.34	0.109	1.16	0.096	4.19	0.231	0.77	0.045
		3.00 m	1.57	0.139	1.37	0.102	3.62	0.222	0.68	0.039
	Inliers Ratio (%)	40%	1.34	0.114	1.19	0.106	2.89	0.176	0.87	0.061
		50%	1.45	0.132	1.21	0.106	3.33	0.197	0.64	0.051
		60%	1.56	0.136	1.41	0.112	3.56	0.213	0.56	0.053
		70%	1.85	0.129	1.66	0.108	4.80	0.239	0.75	0.079
		80%	2.45	0.130	2.13	**0.079**	6.02	0.303	1.06	0.097
		90%	5.00	0.216	4.07	0.163	10.93	0.407	2.61	0.093
$\text{ICP}_{\mathcal{D},\,p2pl}$ ([171])	*Translation* (m)	0.50 m	1.55	0.142	1.35	0.104	4.80	0.309	1.47	0.134
		1.00 m	1.45	0.119	1.29	0.096	3.06	0.237	1.09	0.100
		1.50 m	1.38	0.141	1.20	0.097	3.44	0.246	0.83	0.115
		2.00 m	1.23	0.133	1.07	0.094	2.79	0.190	0.69	0.084
		2.50 m	1.15	0.111	1.01	0.108	3.41	0.208	0.68	0.066
		3.00 m	1.78	0.133	1.34	0.115	2.83	0.151	0.64	0.063
	Inliers Ratio (%)	40%	1.17	**0.102**	1.18	0.118	**1.79**	0.153	1.01	0.119
		50%	1.13	0.113	**0.91**	0.089	2.41	0.144	1.01	0.107
		60%	**1.02**	**0.105**	**0.92**	0.090	3.12	0.190	0.88	0.108
		70%	1.19	0.115	**0.95**	0.087	3.30	0.209	0.73	0.080
		80%	1.88	0.151	2.33	0.126	5.04	0.230	4.08	0.074
		90%	8.44	0.231	9.18	0.237	8.28	**0.139**	7.10	0.168

Table 4.10: Real-world laser odometry drift on the IILABS 3D [2] dataset using distance map-based formulations ($\mathcal{M}_1$ and $\mathcal{M}_2$ for point-to-point and point-to-plane, respectively) and distance map-guided ICP baselines ($\mathcal{M}_0$ representation level) for the 2D point cloud registration problem (r_{max} = 10 m). RTE (%) and RRE (° m^{-1}) are reported as the mean over 10 m segments across four sequences (Nav A Diff, Nav A Omni, Loop, Slippage). Bold entries indicate the lowest drift overall within a tolerance of 0.05 % for RTE and 0.005 ° m^{-1} for RRE. Underlined entries indicate the lowest drift within the pairwise method–splitting criterion for RTE and RRE.

Method	Splitting Criteria	Thresh.	Nav A Diff RTE (%)	Nav A Diff RRE (° m^{-1})	Nav A Omni RTE (%)	Nav A Omni RRE (° m^{-1})	Loop RTE (%)	Loop RRE (° m^{-1})	Slippage RTE (%)	Slippage RRE (° m^{-1})
$\mathcal{D}_{p2p}(\mathcal{M}_1)$	*Translation* (m)	0.50 m	1.68	0.129	1.39	**0.076**	3.34	0.222	1.30	0.155
		1.00 m	1.35	0.113	1.06	0.088	3.51	0.225	0.93	0.089
		1.50 m	1.16	0.116	1.05	**0.078**	3.44	0.221	1.10	0.112
		2.00 m	1.20	**0.104**	0.97	0.093	3.55	0.219	0.58	0.071
		2.50 m	1.25	**0.105**	0.99	**0.079**	3.58	0.220	0.83	0.088
		3.00 m	1.37	0.126	1.33	0.124	4.04	0.220	0.86	0.085
	Inliers Ratio (%)	40%	1.09	**0.101**	1.08	0.108	2.81	0.177	0.66	0.065
		50%	1.11	0.108	**0.88**	0.086	3.19	0.183	0.74	0.086
		60%	1.12	0.112	**0.92**	0.098	3.25	0.211	0.63	0.084
		70%	1.30	0.117	1.01	0.084	3.35	0.207	0.98	0.110
		80%	1.43	0.117	1.13	**0.077**	3.29	0.212	1.10	0.137
		90%	2.09	0.146	1.77	0.093	4.42	0.187	2.40	0.182
$\mathcal{D}_{p2pl}(\mathcal{M}_2)$	*Translation* (m)	0.50 m	1.37	0.139	1.18	0.099	4.50	0.286	0.89	0.078
		1.00 m	1.35	0.136	1.17	0.117	4.19	0.265	1.22	0.111
		1.50 m	1.36	0.136	1.10	0.099	4.03	0.251	0.80	0.050
		2.00 m	1.38	0.133	1.16	0.108	3.23	0.202	0.96	0.077
		2.50 m	1.33	0.122	1.20	0.106	3.79	0.213	0.81	0.065
		3.00 m	1.23	0.117	1.46	0.118	2.71	0.187	0.89	0.081
	Inliers Ratio (%)	40%	1.21	0.121	1.33	0.132	2.13	0.169	1.34	0.131
		50%	1.28	0.119	1.33	0.122	2.60	**0.143**	1.03	0.116
		60%	1.31	0.121	1.11	0.089	3.35	0.204	1.27	0.132
		70%	1.36	0.136	1.15	0.105	4.19	0.234	0.75	0.070
		80%	1.36	0.153	1.42	0.096	4.50	0.246	5.52	0.133
		90%	13.15	0.659	12.81	0.445	9.10	0.292	16.29	0.504
$\text{ICP}_{\mathcal{D},\,p2p}$ ([65])	*Translation* (m)	0.50 m	3.14	0.147	2.60	0.108	6.91	0.338	1.38	0.104
		1.00 m	2.20	0.138	1.77	0.109	5.15	0.276	0.60	0.088
		1.50 m	1.69	0.134	1.46	0.112	3.93	0.235	**0.48**	0.059
		2.00 m	1.62	0.125	1.31	0.092	2.75	0.180	**0.45**	0.054
		2.50 m	1.34	0.109	1.16	0.096	4.19	0.231	0.77	0.045
		3.00 m	1.57	0.139	1.37	0.102	3.62	0.222	0.68	**0.039**
	Inliers Ratio (%)	40%	1.34	0.114	1.19	0.106	2.89	0.176	0.87	0.061
		50%	1.45	0.132	1.21	0.106	3.33	0.197	0.64	0.051
		60%	1.56	0.136	1.41	0.112	3.56	0.213	0.56	0.053
		70%	1.85	0.129	1.66	0.108	4.80	0.239	0.75	0.079
		80%	2.45	0.130	2.13	**0.079**	6.02	0.303	1.06	0.097
		90%	5.00	0.216	4.07	0.163	10.93	0.407	2.61	0.093
$\text{ICP}_{\mathcal{D},\,p2pl}$ ([171])	*Translation* (m)	0.50 m	1.55	0.142	1.35	0.104	4.80	0.309	1.47	0.134
		1.00 m	1.45	0.119	1.29	0.096	3.06	0.237	1.09	0.100
		1.50 m	1.38	0.141	1.20	0.097	3.44	0.246	0.83	0.115
		2.00 m	1.23	0.133	1.07	0.094	2.79	0.190	0.69	0.084
		2.50 m	1.15	0.111	1.01	0.108	3.41	0.208	0.68	0.066
		3.00 m	1.78	0.133	1.34	0.115	2.83	0.151	0.64	0.063
	Inliers Ratio (%)	40%	1.17	**0.102**	1.18	0.118	**1.79**	0.153	1.01	0.119
		50%	1.13	0.113	**0.91**	0.089	2.41	**0.144**	1.01	0.107
		60%	**1.02**	**0.105**	**0.92**	0.090	3.12	0.190	0.88	0.108
		70%	1.19	0.115	0.95	0.087	3.30	0.209	0.73	0.080
		80%	1.88	0.151	2.33	0.126	5.04	0.230	4.08	0.074
		90%	8.44	0.231	9.18	0.237	8.28	**0.139**	7.10	0.168

Experimental Results: IILABS 3D – Laser Odometry

Table 4.11: Real-world laser odometry drift on the IILABS 3D [2] dataset using distance map-based formulations and distance map-guided ICP baselines for the 2D point cloud registration problem (r_{max} = 20 m, $\mathcal{M}_0$ representation level). RTE (%) and RRE (° m^{-1}) are reported as the mean over 10 m segments across four sequences (Nav A Diff, Nav A Omni, Loop, Slippage). Bold entries indicate the lowest drift overall within a tolerance of 0.05 % for RTE and 0.005 ° m^{-1} for RRE. Underlined entries indicate the lowest drift within the pairwise method–splitting criterion for RTE and RRE.

Method	Splitting Criteria	Thresh.	Nav A Diff RTE (%)	Nav A Diff RRE (° m^{-1})	Nav A Omni RTE (%)	Nav A Omni RRE (° m^{-1})	Loop RTE (%)	Loop RRE (° m^{-1})	Slippage RTE (%)	Slippage RRE (° m^{-1})
$\mathcal{D}_{p2p}(\mathcal{M}_0)$	*Translation* (m)	0.50 m	3.01	0.188	2.62	0.148	6.03	0.325	1.05	0.070
		1.00 m	2.04	0.161	1.94	0.102	4.48	0.235	0.58	0.065
		1.50 m	1.78	0.144	1.59	0.090	2.92	0.179	0.54	0.050
		2.00 m	1.50	0.130	1.42	0.090	3.20	0.199	0.52	**0.038**
		2.50 m	1.30	0.111	1.28	0.093	3.60	0.197	0.53	**0.036**
		3.00 m	1.55	0.126	1.41	0.119	4.60	0.226	0.54	0.057
	Inliers Ratio (%)	40%	1.17	0.114	1.16	0.088	2.58	0.154	0.54	0.053
		50%	1.38	0.110	1.34	0.092	2.99	0.141	0.57	0.051
		60%	1.60	0.127	1.51	0.093	3.19	0.170	0.58	**0.039**
		70%	1.86	0.141	1.65	0.091	4.43	0.217	0.60	0.052
		80%	2.45	0.163	2.19	0.117	5.46	0.266	0.93	0.076
		90%	5.38	0.311	4.02	0.186	9.76	0.410	2.47	0.070
$\mathcal{D}_{p2pl}(\mathcal{M}_0)$	*Translation* (m)	0.50 m	1.65	0.158	1.35	0.103	4.19	0.285	1.13	0.120
		1.00 m	1.33	0.114	1.18	0.105	3.80	0.236	0.74	0.093
		1.50 m	1.28	0.122	1.05	0.083	2.94	0.210	0.79	0.115
		2.00 m	1.11	0.119	**0.87**	**0.070**	2.75	0.165	0.69	0.095
		2.50 m	**1.05**	**0.091**	1.06	0.089	2.92	0.184	0.76	0.099
		3.00 m	1.53	0.109	1.30	0.113	3.29	0.149	0.71	0.075
	Inliers Ratio (%)	40%	**1.06**	**0.090**	0.97	0.087	**1.75**	**0.121**	0.79	0.078
		50%	1.15	0.100	0.94	0.089	2.27	0.146	0.65	0.083
		60%	1.10	0.104	**0.84**	0.079	2.68	0.150	0.76	0.118
		70%	1.23	0.117	**0.84**	0.079	3.66	0.216	0.67	0.076
		80%	3.04	0.158	5.02	0.214	6.84	0.255	5.08	0.079
		90%	7.97	0.215	9.50	0.211	8.29	0.137	6.99	0.242
$\text{ICP}_{\mathcal{D},\,p2p}$ ([65])	*Translation* (m)	0.50 m	3.01	0.175	2.54	0.124	5.95	0.308	1.19	0.083
		1.00 m	2.05	0.154	1.80	0.125	4.33	0.234	0.60	0.092
		1.50 m	1.69	0.147	1.47	0.104	3.13	0.188	**0.50**	0.053
		2.00 m	1.42	0.124	1.32	0.115	2.84	0.194	**0.49**	0.057
		2.50 m	1.31	0.114	1.09	0.087	3.83	0.200	**0.47**	0.057
		3.00 m	1.55	0.117	1.54	0.103	3.66	0.190	0.78	0.062
	Inliers Ratio (%)	40%	1.10	0.100	0.97	0.080	3.16	0.179	0.62	0.067
		50%	1.32	0.119	1.02	0.087	2.84	0.142	0.86	0.087
		60%	1.49	0.135	1.15	0.101	3.11	0.168	**0.47**	0.045
		70%	1.87	0.152	1.52	0.098	4.39	0.210	0.55	0.052
		80%	2.46	0.162	2.03	0.111	5.41	0.253	0.91	0.061
		90%	5.13	0.303	4.06	0.178	10.32	0.420	2.40	0.075
$\text{ICP}_{\mathcal{D},\,p2pl}$ ([171])	*Translation* (m)	0.50 m	1.63	0.153	1.34	0.101	4.20	0.285	1.13	0.120
		1.00 m	1.38	0.115	1.18	0.106	3.81	0.236	0.74	0.093
		1.50 m	1.32	0.127	1.04	0.081	2.88	0.209	0.79	0.115
		2.00 m	1.10	0.114	0.88	**0.071**	2.74	0.161	0.69	0.095
		2.50 m	**1.04**	**0.090**	1.02	0.083	2.91	0.184	0.76	0.099
		3.00 m	1.51	0.115	1.24	0.108	3.55	0.154	0.71	0.075
	Inliers Ratio (%)	40%	1.14	**0.091**	0.93	0.097	**1.74**	**0.123**	0.79	0.078
		50%	**1.06**	0.102	**0.84**	0.091	2.13	0.137	0.65	0.083
		60%	**1.05**	0.111	**0.84**	0.081	2.72	0.151	0.76	0.115
		70%	1.19	0.121	**0.83**	0.080	3.82	0.213	0.67	0.076
		80%	3.03	0.158	4.98	0.230	7.04	0.259	5.04	0.079
		90%	7.96	0.219	9.48	0.211	8.28	0.135	6.94	0.241

Table 4.12: Real-world laser odometry drift on the IILABS 3D [2] dataset using distance map-based formulations ($\mathcal{M}_1$ and $\mathcal{M}_2$ for point-to-point and point-to-plane, respectively) and distance map-guided ICP baselines ($\mathcal{M}_0$ representation level) for the 2D point cloud registration problem (r_{max} = 20 m). RTE (%) and RRE (° m^{-1}) are reported as the mean over 10 m segments across four sequences (Nav A Diff, Nav A Omni, Loop, Slippage). Bold entries indicate the lowest drift overall within a tolerance of 0.05 % for RTE and 0.005 ° m^{-1} for RRE. Underlined entries indicate the lowest drift within the pairwise method–splitting criterion for RTE and RRE.

Method	Splitting Criteria	Thresh.	Nav A Diff RTE (%)	Nav A Diff RRE (° m^{-1})	Nav A Omni RTE (%)	Nav A Omni RRE (° m^{-1})	Loop RTE (%)	Loop RRE (° m^{-1})	Slippage RTE (%)	Slippage RRE (° m^{-1})
$\mathcal{D}_{p2p}(\mathcal{M}_1)$	*Translation* (m)	0.50 m	1.53	0.107	1.29	**0.066**	2.62	0.150	1.58	0.154
		1.00 m	1.24	0.109	0.98	**0.063**	3.14	0.160	1.07	0.129
		1.50 m	1.10	0.101	0.92	0.076	3.03	0.161	1.07	0.102
		2.00 m	1.07	**0.087**	0.88	0.085	2.58	0.150	0.87	0.114
		2.50 m	1.10	0.091	0.91	0.080	3.19	0.165	0.80	0.081
		3.00 m	1.23	0.108	1.27	0.113	2.72	0.145	0.79	0.079
	Inliers Ratio (%)	40%	**1.01**	**0.085**	1.04	0.088	2.29	0.135	0.76	0.073
		50%	**0.98**	**0.085**	**0.84**	**0.067**	2.57	0.136	0.71	0.091
		60%	1.16	0.103	**0.87**	0.086	2.51	**0.123**	0.72	0.070
		70%	1.23	0.106	1.04	0.077	2.73	0.157	0.89	0.115
		80%	1.34	0.115	1.24	0.073	3.09	0.168	1.04	0.107
		90%	1.96	0.132	1.56	0.075	4.12	0.140	2.17	0.160
$\mathcal{D}_{p2pl}(\mathcal{M}_2)$	*Translation* (m)	0.50 m	1.33	0.129	1.09	0.102	4.48	0.281	0.90	0.081
		1.00 m	1.21	0.120	1.09	0.108	4.18	0.254	1.19	0.136
		1.50 m	1.23	0.114	1.18	0.110	3.80	0.227	0.77	0.051
		2.00 m	1.15	0.105	1.07	0.105	3.20	0.188	0.76	**0.042**
		2.50 m	1.18	0.103	1.05	0.088	3.48	0.188	0.70	**0.045**
		3.00 m	1.26	0.121	0.90	0.084	2.45	0.161	1.02	0.116
	Inliers Ratio (%)	40%	**0.98**	**0.086**	**0.87**	0.089	1.91	**0.121**	0.67	0.065
		50%	1.17	0.111	0.94	0.085	2.90	0.140	0.83	0.059
		60%	1.26	0.118	1.10	0.096	3.10	0.152	0.89	0.068
		70%	1.36	0.125	1.07	0.095	3.86	0.223	1.19	0.072
		80%	2.09	0.155	2.54	0.209	7.03	0.237	7.73	0.252
		90%	14.08	0.628	12.10	0.578	11.22	0.622	16.12	0.461
$\text{ICP}_{\mathcal{D},\,p2p}$ ([65])	*Translation* (m)	0.50 m	3.01	0.175	2.54	0.124	5.95	0.308	1.19	0.083
		1.00 m	2.05	0.154	1.80	0.125	4.33	0.234	0.60	0.092
		1.50 m	1.69	0.147	1.47	0.104	3.13	0.188	**0.50**	0.053
		2.00 m	1.42	0.124	1.32	0.115	2.84	0.194	**0.49**	0.057
		2.50 m	1.31	0.114	1.09	0.087	3.83	0.200	**0.47**	0.057
		3.00 m	1.55	0.117	1.54	0.103	3.66	0.190	0.78	0.062
	Inliers Ratio (%)	40%	1.10	0.100	0.97	0.080	3.16	0.179	0.62	0.067
		50%	1.32	0.119	1.02	0.087	2.84	0.142	0.86	0.087
		60%	1.49	0.135	1.15	0.101	3.11	0.168	**0.47**	**0.045**
		70%	1.87	0.152	1.52	0.098	4.39	0.210	0.55	0.052
		80%	2.46	0.162	2.03	0.111	5.41	0.253	0.91	0.061
		90%	5.13	0.303	4.06	0.178	10.32	0.420	2.40	0.075
$\text{ICP}_{\mathcal{D},\,p2pl}$ ([171])	*Translation* (m)	0.50 m	1.63	0.153	1.34	0.101	4.20	0.285	1.13	0.120
		1.00 m	1.38	0.115	1.18	0.106	3.81	0.236	0.74	0.093
		1.50 m	1.32	0.127	1.04	0.081	2.88	0.209	0.79	0.115
		2.00 m	1.10	0.114	0.88	0.071	2.74	0.161	0.69	0.095
		2.50 m	1.04	**0.090**	1.02	0.083	2.91	0.184	0.76	0.099
		3.00 m	1.51	0.115	1.24	0.108	3.55	0.154	0.71	0.075
	Inliers Ratio (%)	40%	1.14	0.091	0.93	0.097	**1.74**	**0.123**	0.79	0.078
		50%	1.06	0.102	**0.84**	0.091	2.13	0.137	0.65	0.083
		60%	1.05	0.111	**0.84**	0.081	2.72	0.151	0.76	0.115
		70%	1.19	0.121	**0.83**	0.080	3.82	0.213	0.67	0.076
		80%	3.03	0.158	4.98	0.230	7.04	0.259	5.04	0.079
		90%	7.96	0.219	9.48	0.211	8.28	0.135	6.94	0.241

Experimental Results: IILABS 3D – Laser Odometry

Table 4.13: Real-world laser odometry drift on the IILABS 3D [2] dataset using distance map-based formulations and distance map-guided ICP baselines for the 2D point cloud registration problem (r_{max} = 30 m, $\mathcal{M}_0$ representation level). RTE (%) and RRE (° m^{-1}) are reported as the mean over 10 m segments across four sequences (Nav A Diff, Nav A Omni, Loop, Slippage). Bold entries indicate the lowest drift overall within a tolerance of 0.05 % for RTE and 0.005 ° m^{-1} for RRE. Underlined entries indicate the lowest drift within the pairwise method–splitting criterion for RTE and RRE.

Method	Splitting Criteria	Thresh.	Nav A Diff RTE (%)	Nav A Diff RRE (° m^{-1})	Nav A Omni RTE (%)	Nav A Omni RRE (° m^{-1})	Loop RTE (%)	Loop RRE (° m^{-1})	Slippage RTE (%)	Slippage RRE (° m^{-1})
$\mathcal{D}_{p2p}(\mathcal{M}_0)$	*Translation* (m)	0.50 m	2.91	0.182	2.64	0.134	5.80	0.310	1.28	0.059
		1.00 m	2.05	0.161	1.94	0.108	4.35	0.236	0.96	0.074
		1.50 m	1.76	0.146	1.63	0.098	3.03	0.183	0.78	0.056
		2.00 m	1.51	0.127	1.50	0.102	3.13	0.202	0.97	0.049
		2.50 m	1.35	0.126	1.23	0.088	3.86	0.199	0.68	**0.033**
		3.00 m	1.47	0.120	1.45	0.121	3.91	0.198	0.92	0.060
	Inliers Ratio (%)	40%	1.12	0.098	1.09	0.090	2.54	0.158	0.71	**0.034**
		50%	1.35	0.122	1.31	0.089	3.00	0.144	0.64	0.052
		60%	1.58	0.129	1.39	0.090	3.12	0.175	0.63	0.052
		70%	1.83	0.141	1.66	0.098	4.37	0.217	0.95	0.082
		80%	2.53	0.174	2.17	0.127	5.34	0.256	1.24	0.064
		90%	5.34	0.292	3.98	0.187	9.65	0.399	2.80	0.093
$\mathcal{D}_{p2pl}(\mathcal{M}_0)$	*Translation* (m)	0.50 m	1.60	0.145	1.35	0.103	4.13	0.275	1.48	0.149
		1.00 m	1.36	0.115	1.30	0.106	3.75	0.239	1.01	0.091
		1.50 m	1.25	0.126	1.19	0.096	2.88	0.210	0.77	0.077
		2.00 m	1.16	0.115	0.88	**0.064**	2.39	0.163	0.71	0.097
		2.50 m	1.08	0.103	1.05	0.105	2.91	0.189	0.66	0.081
		3.00 m	1.67	0.114	1.25	0.108	3.59	0.140	0.75	0.090
	Inliers Ratio (%)	40%	**0.92**	0.094	1.09	0.104	**1.41**	**0.107**	0.84	0.073
		50%	1.06	0.104	**0.84**	0.086	2.05	0.139	0.72	0.084
		60%	1.10	0.111	**0.85**	0.084	2.77	0.156	0.68	0.056
		70%	1.24	0.124	1.06	0.090	3.69	0.212	0.69	0.056
		80%	3.38	0.154	5.40	0.233	7.10	0.253	5.01	0.084
		90%	8.02	0.216	9.48	0.214	8.24	0.139	7.06	0.251
ICP$_{\mathcal{D},\,p2p}$ ([65])	*Translation* (m)	0.50 m	3.03	0.180	2.56	0.133	5.63	0.297	1.20	0.070
		1.00 m	2.01	0.154	1.76	0.117	4.46	0.235	**0.57**	0.067
		1.50 m	1.56	0.133	1.50	0.107	2.97	0.180	0.70	0.052
		2.00 m	1.40	0.123	1.19	0.100	2.56	0.180	0.67	0.056
		2.50 m	1.24	0.110	1.13	0.082	3.55	0.181	0.67	0.045
		3.00 m	1.33	0.113	1.59	0.095	3.60	0.186	0.67	**0.038**
	Inliers Ratio (%)	40%	1.10	0.099	1.07	0.092	3.36	0.192	**0.56**	0.055
		50%	1.32	0.116	1.16	0.082	2.95	0.158	0.72	0.065
		60%	1.48	0.133	1.21	0.100	3.15	0.169	0.64	0.048
		70%	1.80	0.147	1.60	0.110	4.32	0.209	0.88	0.070
		80%	2.54	0.169	2.14	0.111	5.24	0.247	0.83	0.057
		90%	5.14	0.298	4.01	0.177	10.01	0.403	2.63	0.099
ICP$_{\mathcal{D},\,p2pl}$ ([171])	*Translation* (m)	0.50 m	1.61	0.146	1.42	0.111	4.14	0.275	1.48	0.149
		1.00 m	1.37	0.114	1.28	0.103	3.71	0.238	1.06	0.095
		1.50 m	1.25	0.122	1.19	0.097	2.82	0.207	0.77	0.077
		2.00 m	1.13	0.115	0.91	0.070	2.45	0.164	0.71	0.097
		2.50 m	1.09	0.095	1.02	0.105	3.02	0.187	0.66	0.081
		3.00 m	1.68	0.112	1.22	0.103	3.55	0.140	0.75	0.090
	Inliers Ratio (%)	40%	**0.88**	**0.082**	1.06	0.106	**1.37**	**0.106**	0.78	0.073
		50%	1.01	0.097	**0.83**	0.093	2.07	0.140	0.72	0.084
		60%	1.08	0.109	**0.82**	0.075	2.71	0.154	0.82	0.070
		70%	1.24	0.123	0.93	0.090	3.91	0.207	0.69	0.056
		80%	3.34	0.155	5.39	0.231	7.32	0.257	4.92	0.081
		90%	7.98	0.214	9.45	0.217	8.26	0.136	6.97	0.251

Table 4.14: Real-world laser odometry drift on the IILABS 3D [2] dataset using distance map-based formulations ($\mathcal{M}_1$ and $\mathcal{M}_2$ for point-to-point and point-to-plane, respectively) and distance map-guided ICP baselines ($\mathcal{M}_0$ representation level) for the 2D point cloud registration problem (r_{max} = 30 m). RTE (%) and RRE (° m^{-1}) are reported as the mean over 10 m segments across four sequences (Nav A Diff, Nav A Omni, Loop, Slippage). Bold entries indicate the lowest drift overall within a tolerance of 0.05 % for RTE and 0.005 ° m^{-1} for RRE. Underlined entries indicate the lowest drift within the pairwise method–splitting criterion for RTE and RRE.

Method	Splitting Criteria	Thresh.	Nav A Diff RTE (%)	Nav A Diff RRE (° m^{-1})	Nav A Omni RTE (%)	Nav A Omni RRE (° m^{-1})	Loop RTE (%)	Loop RRE (° m^{-1})	Slippage RTE (%)	Slippage RRE (° m^{-1})
$\mathcal{D}_{p2p}(\mathcal{M}_1)$	*Translation* (m)	0.50 m	1.62	0.109	1.38	0.080	2.87	0.140	1.47	0.088
		1.00 m	1.12	0.096	1.01	0.077	3.09	0.160	0.89	0.088
		1.50 m	1.11	0.098	0.98	0.080	2.81	0.154	1.04	0.142
		2.00 m	1.06	0.091	0.92	0.079	2.82	0.155	**0.55**	0.056
		2.50 m	1.12	0.091	0.91	0.080	3.07	0.166	0.85	0.086
		3.00 m	1.09	0.091	1.36	0.123	3.03	0.156	0.88	0.069
	Inliers Ratio (%)	40%	1.01	0.090	0.89	0.087	2.14	0.134	0.66	0.076
		50%	0.95	**0.078**	**0.81**	**0.065**	2.33	0.123	0.64	0.070
		60%	1.10	0.089	0.89	0.080	2.80	0.143	0.85	0.087
		70%	1.15	0.094	0.97	**0.063**	3.01	0.155	0.81	0.080
		80%	1.37	0.112	1.15	0.075	3.18	0.160	0.89	0.083
		90%	1.92	0.131	1.61	**0.067**	3.99	0.133	2.05	0.144
$\mathcal{D}_{p2pl}(\mathcal{M}_2)$	*Translation* (m)	0.50 m	1.34	0.128	1.07	0.097	4.43	0.279	0.90	0.084
		1.00 m	1.30	0.120	1.11	0.109	3.93	0.246	0.92	0.097
		1.50 m	1.24	0.121	1.12	0.102	3.83	0.227	0.78	0.051
		2.00 m	1.13	0.103	1.05	0.103	3.21	0.188	1.00	0.053
		2.50 m	1.13	0.106	0.98	0.093	3.50	0.189	0.81	0.060
		3.00 m	1.22	0.123	1.24	0.118	2.26	0.156	0.97	0.112
	Inliers Ratio (%)	40%	1.03	0.087	1.00	0.085	2.09	0.124	0.92	0.098
		50%	1.02	0.090	0.92	0.087	3.18	0.148	1.06	0.133
		60%	1.24	0.108	1.11	0.104	3.24	0.168	0.98	0.050
		70%	1.37	0.129	1.11	0.106	3.87	0.225	1.16	0.062
		80%	2.60	0.174	2.62	0.216	7.61	0.250	7.99	0.240
		90%	14.02	0.630	12.10	0.577	11.33	0.611	16.03	0.472
ICP$_{\mathcal{D},\,p2p}$ ([65])	*Translation* (m)	0.50 m	3.03	0.180	2.56	0.133	5.63	0.297	1.20	0.070
		1.00 m	2.01	0.154	1.76	0.117	4.46	0.235	**0.57**	0.067
		1.50 m	1.56	0.133	1.50	0.107	2.97	0.180	0.70	0.052
		2.00 m	1.40	0.123	1.19	0.100	2.56	0.180	0.67	0.056
		2.50 m	1.24	0.110	1.13	0.082	3.55	0.181	0.67	0.045
		3.00 m	1.33	0.113	1.59	0.095	3.60	0.186	0.67	**0.038**
	Inliers Ratio (%)	40%	1.10	0.099	1.07	0.092	3.36	0.192	**0.56**	0.055
		50%	1.32	0.116	1.16	0.082	2.95	0.158	0.72	0.065
		60%	1.48	0.133	1.21	0.100	3.15	0.169	0.64	0.048
		70%	1.80	0.147	1.60	0.110	4.32	0.209	0.88	0.070
		80%	2.54	0.169	2.14	0.111	5.24	0.247	0.83	0.057
		90%	5.14	0.298	4.01	0.177	10.01	0.403	2.63	0.099
ICP$_{\mathcal{D},\,p2pl}$ ([171])	*Translation* (m)	0.50 m	1.61	0.146	1.42	0.111	4.14	0.275	1.48	0.149
		1.00 m	1.37	0.114	1.28	0.103	3.71	0.238	1.06	0.095
		1.50 m	1.25	0.122	1.19	0.097	2.82	0.207	0.77	0.077
		2.00 m	1.13	0.115	0.91	0.070	2.45	0.164	0.71	0.097
		2.50 m	1.09	0.095	1.02	0.105	3.02	0.187	0.66	0.081
		3.00 m	1.68	0.112	1.22	0.103	3.55	0.140	0.75	0.090
	Inliers Ratio (%)	40%	**0.88**	**0.082**	1.06	0.106	**1.37**	**0.106**	0.78	0.073
		50%	1.01	0.097	**0.83**	0.093	2.07	0.140	0.72	0.084
		60%	1.08	0.109	**0.82**	0.075	2.71	0.154	0.82	0.070
		70%	1.24	0.123	0.93	0.090	3.91	0.207	0.69	0.056
		80%	3.34	0.155	5.39	0.231	7.32	0.257	4.92	0.081
		90%	7.98	0.214	9.45	0.217	8.26	0.136	6.97	0.251

ROBOT2026: METHODOLOGY

$$\mathcal{P} = \{\mathbf{p}_1, \ldots, \mathbf{p}_n\} \subset \mathbb{R}^2$$

$$\mathbf{p}^*(\mathbf{y}) = \operatorname{argmin}_{\mathbf{p}_i \in \mathcal{P}} \|\mathbf{y} - \mathbf{p}_i\|_2$$

$$\mathcal{D}(\mathbf{y}) = \{\mathbb{R}^2 \to \mathbb{R}_0^+ : \|\mathbf{y} - \mathbf{p}^*(\mathbf{y})\|_2\}$$

$$\boldsymbol{\nabla}\mathcal{D}(\mathbf{y}) = \frac{\mathbf{y} - \mathbf{p}^*(\mathbf{y})}{\|\mathbf{y} - \mathbf{p}^*(\mathbf{y})\|_2}, \quad \|\boldsymbol{\nabla}\mathcal{D}(\mathbf{y})\|_2 = 1$$

$$\mathbf{H}_{\mathcal{D}}(\mathbf{y}) = \begin{bmatrix} \frac{(y_2 - p_y^*(\mathbf{y}))^2}{\|\mathbf{y} - \mathbf{p}^*(\mathbf{y})\|_2^3} & -\frac{(y_1 - p_x^*(\mathbf{y}))(y_2 - p_y^*(\mathbf{y}))}{\|\mathbf{y} - \mathbf{p}^*(\mathbf{y})\|_2^3} \\ -\frac{(y_1 - p_x^*(\mathbf{y}))(y_2 - p_y^*(\mathbf{y}))}{\|\mathbf{y} - \mathbf{p}^*(\mathbf{y})\|_2^3} & \frac{(y_1 - p_x^*(\mathbf{y}))^2}{\|\mathbf{y} - \mathbf{p}^*(\mathbf{y})\|_2^3} \end{bmatrix}$$

Variant	Storage	Values/cell	Query	$\mathbf{d}_{\max}$?
Nano ($\mathcal{M}_{0,\text{dense}}$)	Dense array	1 (int)	$O(1)$	optional
Standard ($\mathcal{M}_{1\|2,\text{dense}}$)	Dense array	3–8*	$O(1)$	optional
Sparse ($\mathcal{M}_{1\|2,\text{sparse}}$)	Hash map	3–8*	$O(1)^\dagger$	advisable

* Depends on the representation level: 3 values for $\mathcal{M}_1$, 7 for $\mathcal{M}_2$, +1 if the optional nearest-neighbor index $k^*(\mathbf{y})$ is stored.

† average $O(1)$; worst-case $O(N_d)$ under hash collision, mitigated by the post-propagation rehash, where $N_d \leq N_G$ represent the number of populated cells within $d_{\max}$ and N_G the total number of cells.

$$\mathbf{x} = [\mathbf{R} \mid \mathbf{t}]$$

$$\mathbf{x}_{k+1} = \mathbf{x}_k \oplus \boldsymbol{\Delta}\mathbf{x} = \mathbf{x}_k \cdot \exp(\boldsymbol{\Delta}\mathbf{x})$$

$$\mathbf{z}_i \in \mathbb{R}^2, \, i = 1, \ldots, M$$

$$\tilde{\mathbf{z}}_i = \mathbf{T}_R^S \, \mathbf{x} \, \mathbf{T}_S^R \, \mathbf{z}_i, \qquad \mathbf{z}_{R,i} = \mathbf{T}_S^R \, \mathbf{z}_i$$

$$e_i^{\text{p2p}}(\mathbf{x}) = \mathcal{D}(\tilde{\mathbf{z}}_i) = \mathcal{D}\big(\mathbf{T}_R^S \cdot \mathbf{x} \cdot \mathbf{T}_S^R \mathbf{z}_i\big) \in \mathbb{R}$$

$$\mathbf{J}_i^{\text{p2p}} = \boldsymbol{\nabla}\mathcal{D}(\tilde{\mathbf{z}}_i)^T \cdot \underbrace{\mathbf{R}_R^S \mathbf{R}(\theta) \cdot (\mathbf{I}_{2\times 2} \mid \lfloor \mathbf{z}_{R,i} \rfloor)}_{\mathbf{J}_i^{\text{p2p,ICP}}} \in \mathbb{R}^{1\times 3}$$

$$e_i^{\text{p2pl}}(\mathbf{x}) = (\mathbf{R}(\theta)\, \mathbf{n}_{z_i})^T \, \hat{\boldsymbol{\nabla}}\mathcal{D}(\tilde{\mathbf{z}}_i) \, \mathcal{D}(\tilde{\mathbf{z}}_i)$$

$$\begin{aligned} \mathbf{J}_i^{\text{p2pl}} = \mathbf{n}_{z_i}^T \Big(& \mathbf{0}_{2\times 2} \mid \lfloor -\mathbf{R}(\theta)^T \hat{\boldsymbol{\nabla}}\mathcal{D}(\tilde{\mathbf{z}}_i) \, \mathcal{D}(\tilde{\mathbf{z}}_i) \rfloor \Big) \\ & + (\mathbf{R}(\theta)\mathbf{n}_{z_i})^T \, \mathbf{H}_{\mathcal{D}}(\tilde{\mathbf{z}}_i) \, \mathcal{D}(\tilde{\mathbf{z}}_i) \, \mathbf{J}_i^{\text{p2p,ICP}} \\ & + (\mathbf{R}(\theta)\mathbf{n}_{z_i})^T \, \hat{\boldsymbol{\nabla}}\mathcal{D}(\tilde{\mathbf{z}}_i) \, \boldsymbol{\nabla}\mathcal{D}(\tilde{\mathbf{z}}_i)^T \mathbf{J}_i^{\text{p2p,ICP}} \end{aligned}$$

ROBOT2026: Results – Synthetic Benchmark

Synthetic Benchmark Results for Point Cloud Alignment using Proposed Distance Map-based Error Formulations and Distance Map-guided ICP Baselines [1], [2] (Cell Resolution $\delta = 0.03\,\mathrm{m}$)

Meth.	$\mathcal{M}_l$	V?	S?	Circle $\sigma = 0.00\,\mathrm{m}$ ε_t (m)	Circle $\sigma = 0.00\,\mathrm{m}$ $\chi^2_{\boldsymbol{\varepsilon}}$	Circle $\sigma = 0.03\,\mathrm{m}$ ε_t (m)	Circle $\sigma = 0.03\,\mathrm{m}$ $\chi^2_{\boldsymbol{\varepsilon}}$	Square $\sigma = 0.00\,\mathrm{m}$ ε_t (m)	Square $\sigma = 0.00\,\mathrm{m}$ ε_θ (°)	Square $\sigma = 0.00\,\mathrm{m}$ $\chi^2_{\boldsymbol{\varepsilon}}$	Square $\sigma = 0.03\,\mathrm{m}$ ε_t (m)	Square $\sigma = 0.03\,\mathrm{m}$ ε_θ (°)	Square $\sigma = 0.03\,\mathrm{m}$ $\chi^2_{\boldsymbol{\varepsilon}}$	Corridor $\sigma = 0.00\,\mathrm{m}$ ε_y (m)	Corridor $\sigma = 0.00\,\mathrm{m}$ ε_θ (°)	Corridor $\sigma = 0.00\,\mathrm{m}$ $\chi^2_{\boldsymbol{\varepsilon}}$	Corridor $\sigma = 0.03\,\mathrm{m}$ ε_y (m)	Corridor $\sigma = 0.03\,\mathrm{m}$ ε_θ (°)	Corridor $\sigma = 0.03\,\mathrm{m}$ $\chi^2_{\boldsymbol{\varepsilon}}$
$\mathcal{D}_{\mathrm{p2p}}$	$\mathcal{M}_0$	✗	–	0.001	6234.68	**0.003**	3100.86	0.001	0.19	0.94	0.003	0.08	0.19	0.007	0.11	540.06	0.003	0.12	453.71
		✓	–	0.009	4827.02	0.010	3123.95	0.012	0.01	0.10	0.006	0.02	0.04	0.009	0.10	462.70	0.003	0.11	461.08
	$\mathcal{M}_1$	✗	✗	0.018	586.23	**0.003**	934.92	0.015	0.02	0.07	0.003	**0.01**	**0.00**	0.008	0.21	88.20	0.005	0.20	158.98
		✗	✓	0.029	257.09	0.021	536.29	0.051	0.02	0.32	0.021	**0.01**	0.12	0.011	0.22	70.90	0.023	0.21	113.04
		✓	✗	0.018	584.83	**0.003**	937.83	0.015	0.02	0.07	0.003	**0.01**	**0.00**	0.008	0.20	98.33	0.005	0.20	158.77
		✓	✓	0.029	257.08	0.021	536.49	0.051	0.02	0.32	0.021	**0.01**	0.12	0.011	0.22	75.83	0.024	0.20	112.46
$\mathcal{D}_{\mathrm{p2pl}}$	$\mathcal{M}_0$	✗	–	**0.000**	**52.86**	**0.003**	**189.38**	**0.000**	**0.00**	**0.00**	**0.002**	0.02	0.01	**0.000**	**0.00**	5.72	**0.001**	0.02	13.83
		✓	–	**0.000**	54.10	0.004	190.31	**0.000**	**0.00**	**0.00**	**0.002**	**0.01**	0.01	**0.000**	**0.00**	5.23	0.002	**0.01**	**12.94**
	$\mathcal{M}_2$	✗	✗	0.009	463.03	**0.003**	1716.85	0.004	0.03	0.01	0.003	0.02	0.01	0.009	0.01	**4.93**	**0.001**	**0.01**	129.01
		✗	✓	0.023	264.87	0.021	1120.07	0.023	0.08	0.33	0.021	**0.01**	0.30	0.012	0.01	8.66	0.015	**0.01**	83.51
		✓	✗	0.008	462.02	**0.003**	1716.63	0.004	0.03	0.01	0.003	0.02	0.01	0.009	0.02	5.09	**0.001**	**0.01**	126.92
		✓	✓	0.023	265.03	0.021	1119.63	0.024	0.07	0.36	0.021	**0.01**	0.30	0.012	0.01	11.40	0.015	**0.01**	82.42
$\mathrm{ICP}_{\mathcal{D},\,\mathrm{p2p}}$ [1]	$\mathcal{M}_0$	✗	–	**0.000**	6432.74	**0.003**	6479.07	0.001	0.19	2.07	0.003	0.08	0.39	**0.000**	0.01	834.29	**0.001**	0.02	854.64
		✓	–	0.009	6578.50	0.010	6579.53	0.013	0.01	0.25	0.006	0.02	0.08	**0.000**	0.01	775.70	**0.001**	**0.01**	817.46
$\mathrm{ICP}_{\mathcal{D},\,\mathrm{p2pl}}$ [2]	$\mathcal{M}_0$	✗	–	**0.000**	**52.86**	**0.003**	**189.38**	**0.000**	**0.00**	**0.00**	**0.002**	0.02	0.01	**0.000**	**0.00**	5.72	**0.001**	0.02	13.83
		✓	–	**0.000**	54.10	0.004	190.31	**0.000**	**0.00**	**0.00**	**0.002**	**0.01**	0.01	**0.000**	**0.00**	5.23	0.002	**0.01**	**12.94**

Legend: distance map representation level ($\mathcal{M}_l$); point cloud voxelization (V); sub-pixelization (S) (bi-linear interpolation); enabled (✓), disabled (✗), not applicable (–); translation error ε_t (m); rotation error ε_θ (°); pose-error consistency metric $\chi^2_{\boldsymbol{\varepsilon}} = \boldsymbol{\varepsilon}^T \boldsymbol{\Omega}\, \boldsymbol{\varepsilon}$ (dimensionless).

ROBOT2026: Results – Synthetic Benchmark

Synthetic Benchmark Results for Point Cloud Alignment using Proposed Distance Map-based Error Formulations and Distance Map-guided ICP Baselines [1], [2] (Cell Resolution $\delta = 0.05\,\mathrm{m}$)

Meth.	$\mathcal{M}_l$	V?	S?	Circle $\sigma = 0.00\,\mathrm{m}$ ε_t (m)	Circle $\sigma = 0.00\,\mathrm{m}$ χ^2_ε	Circle $\sigma = 0.03\,\mathrm{m}$ ε_t (m)	Circle $\sigma = 0.03\,\mathrm{m}$ χ^2_ε	Square $\sigma = 0.00\,\mathrm{m}$ ε_t (m)	Square $\sigma = 0.00\,\mathrm{m}$ ε_θ (°)	Square $\sigma = 0.00\,\mathrm{m}$ χ^2_ε	Square $\sigma = 0.03\,\mathrm{m}$ ε_t (m)	Square $\sigma = 0.03\,\mathrm{m}$ ε_θ (°)	Square $\sigma = 0.03\,\mathrm{m}$ χ^2_ε	Corridor $\sigma = 0.00\,\mathrm{m}$ ε_y (m)	Corridor $\sigma = 0.00\,\mathrm{m}$ ε_θ (°)	Corridor $\sigma = 0.00\,\mathrm{m}$ χ^2_ε	Corridor $\sigma = 0.03\,\mathrm{m}$ ε_y (m)	Corridor $\sigma = 0.03\,\mathrm{m}$ ε_θ (°)	Corridor $\sigma = 0.03\,\mathrm{m}$ χ^2_ε
$\mathcal{D}_{\mathrm{p2p}}$	$\mathcal{M}_0$	✗	–	0.001	6198.44	**0.003**	3913.39	0.001	0.10	0.35	0.003	0.29	2.15	0.018	0.07	454.93	0.003	0.14	492.12
		✓	–	0.004	4953.51	0.026	3200.64	0.005	0.01	0.02	0.019	0.03	0.29	0.013	0.11	379.95	0.004	0.13	522.25
	$\mathcal{M}_1$	✗	✗	0.022	480.88	**0.003**	703.70	0.027	0.05	0.03	0.004	0.02	**0.01**	0.019	0.22	69.63	0.008	0.22	110.43
		✗	✓	0.044	186.61	0.039	306.68	0.028	0.17	0.16	0.047	**0.01**	0.48	0.009	0.24	44.35	0.036	0.23	71.13
		✓	✗	0.022	422.14	**0.003**	688.86	0.027	0.05	0.03	0.004	**0.01**	**0.01**	0.026	0.21	76.78	0.011	0.22	106.45
		✓	✓	0.044	162.18	0.040	304.36	0.027	0.16	0.16	0.048	0.02	0.50	0.021	0.24	45.47	0.039	0.21	69.34
$\mathcal{D}_{\mathrm{p2pl}}$	$\mathcal{M}_0$	✗	–	**0.000**	51.30	**0.003**	183.62	**0.000**	**0.00**	**0.00**	**0.002**	0.02	**0.01**	**0.000**	**0.00**	5.72	**0.001**	0.02	13.80
		✓	–	**0.000**	**46.18**	0.007	**179.93**	**0.000**	**0.00**	**0.00**	0.004	**0.01**	**0.01**	**0.000**	**0.00**	**4.08**	0.003	**0.01**	**12.01**
	$\mathcal{M}_2$	✗	✗	0.017	261.79	**0.003**	846.01	0.030	0.06	0.08	**0.002**	**0.01**	**0.01**	0.016	0.05	20.86	**0.001**	0.02	56.29
		✗	✓	0.038	120.63	0.036	445.88	0.006	0.12	0.10	0.036	**0.01**	0.58	0.006	0.09	15.29	0.026	0.04	33.44
		✓	✗	0.017	235.66	**0.003**	824.85	0.030	0.06	0.09	**0.002**	**0.01**	**0.01**	0.018	0.06	17.17	**0.001**	0.02	53.47
		✓	✓	0.037	106.47	0.036	441.65	0.007	0.12	0.10	0.035	**0.01**	0.57	0.007	0.10	12.93	0.025	0.05	31.53
$\mathrm{ICP}_{\mathcal{D},\,\mathrm{p2p}}$ [1]	$\mathcal{M}_0$	✗	–	**0.000**	6431.42	**0.003**	6541.08	0.001	0.10	0.61	0.003	0.29	4.60	**0.000**	0.03	828.40	**0.001**	0.04	844.22
		✓	–	0.005	5754.85	0.026	6248.71	0.005	0.01	0.04	0.020	0.03	0.58	**0.000**	0.01	635.54	0.002	**0.01**	766.20
$\mathrm{ICP}_{\mathcal{D},\,\mathrm{p2pl}}$ [2]	$\mathcal{M}_0$	✗	–	**0.000**	51.30	**0.003**	183.62	**0.000**	**0.00**	**0.00**	**0.002**	0.02	**0.01**	**0.000**	**0.00**	5.72	**0.001**	0.02	13.80
		✓	–	**0.000**	**46.18**	0.007	**179.93**	**0.000**	**0.00**	**0.00**	0.004	**0.01**	**0.01**	**0.000**	**0.00**	**4.08**	0.003	**0.01**	**12.01**

Legend: distance map representation level ($\mathcal{M}_l$); point cloud voxelization (V); sub-pixelization (S) (bi-linear interpolation); enabled (✓), disabled (✗), not applicable (–); translation error ε_t (m); rotation error ε_θ (°); pose-error consistency metric $\chi^2_\varepsilon = \boldsymbol{\varepsilon}^T \boldsymbol{\Omega}\, \boldsymbol{\varepsilon}$ (dimensionless).

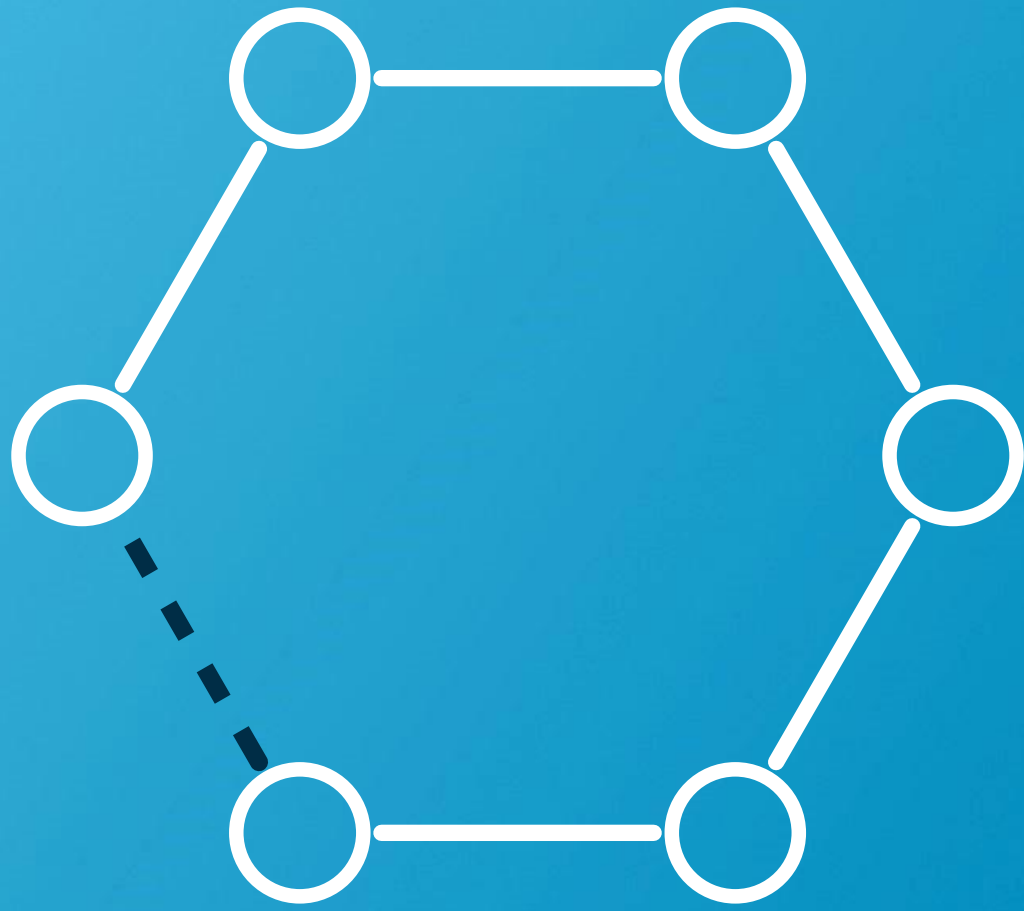
ROBOT2026: RESULTS – IILABS 3D (LASER ODOMETRY)

REAL-WORLD LASER ODOMETRY DRIFT ON THE IILABS 3D [11] DATASET USING PROPOSED DISTANCE MAP-BASED FORMULATIONS AND DISTANCE MAP-GUIDED ICP BASELINES [1], [2] FOR THE 2D POINT CLOUD REGISTRATION PROBLEM

Method	Splitting Criteria	Threshold	Nav A Diff RTE (%)	Nav A Diff RRE (°m⁻¹)	Nav A Omni RTE (%)	Nav A Omni RRE (°m⁻¹)	Loop RTE (%)	Loop RRE (°m⁻¹)	Slippage RTE (%)	Slippage RRE (°m⁻¹)
$\mathcal{D}_{\text{p2p}}(\mathcal{M}_0)$	*Translation* (m)	0.50 m	2.91	0.182	2.64	0.134	5.80	0.310	1.28	0.059
		1.00 m	2.05	0.161	1.94	0.108	4.35	0.236	0.96	0.074
		1.50 m	1.76	0.146	1.63	0.098	3.03	0.183	0.78	0.056
		2.00 m	1.51	0.127	1.50	0.102	3.13	0.202	0.97	0.049
		2.50 m	1.35	0.126	1.23	0.088	3.86	0.199	0.68	**0.033**
		3.00 m	1.47	0.120	1.45	0.121	3.91	0.198	0.92	0.060
	Inliers Ratio (%)	40%	1.12	0.098	1.09	0.090	2.54	0.158	0.71	**0.034**
		50%	1.35	0.122	1.31	0.089	3.00	0.144	0.64	0.052
		60%	1.58	0.129	1.39	0.090	3.12	0.175	0.63	0.052
		70%	1.83	0.141	1.66	0.098	4.37	0.217	0.95	0.082
		80%	2.53	0.174	2.17	0.127	5.34	0.256	1.24	0.064
		90%	5.34	0.292	3.98	0.187	9.65	0.399	2.80	0.093
$\mathcal{D}_{\text{p2p}}(\mathcal{M}_1)$	*Translation* (m)	0.50 m	1.62	0.109	1.38	0.080	2.87	0.140	1.47	0.088
		1.00 m	1.12	0.096	1.01	0.077	3.09	0.160	0.89	0.088
		1.50 m	1.11	0.098	0.98	0.080	2.81	0.154	1.04	0.142
		2.00 m	1.06	0.091	0.92	0.079	2.82	0.155	**0.55**	0.056
		2.50 m	1.12	0.091	0.91	0.080	3.07	0.166	0.85	0.086
		3.00 m	1.09	0.091	1.36	0.123	3.03	0.156	0.88	0.069
	Inliers Ratio (%)	40%	1.01	0.090	0.89	0.087	2.14	0.134	0.66	0.076
		50%	0.95	**0.078**	**0.81**	**0.065**	2.33	0.123	0.64	0.070
		60%	1.10	0.089	0.89	0.080	2.80	0.143	0.85	0.087
		70%	1.15	0.094	0.97	**0.063**	3.01	0.155	0.81	0.080
		80%	1.37	0.112	1.15	0.075	3.18	0.160	0.89	0.083
		90%	1.92	0.131	1.61	**0.067**	3.99	0.133	2.05	0.144
$\mathcal{D}_{\text{p2pl}}(\mathcal{M}_0)$	*Translation* (m)	0.50 m	1.60	0.145	1.35	0.103	4.13	0.275	1.48	0.149
		1.00 m	1.36	0.115	1.30	0.106	3.75	0.239	1.01	0.091
		1.50 m	1.25	0.126	1.19	0.096	2.88	0.210	0.77	0.077
		2.00 m	1.16	0.115	0.88	**0.064**	2.39	0.163	0.71	0.097
		2.50 m	1.08	0.103	1.05	0.105	2.91	0.189	0.66	0.081
		3.00 m	1.67	0.114	1.25	0.108	3.59	0.140	0.75	0.090
	Inliers Ratio (%)	40%	**0.92**	0.094	1.09	0.104	**1.41**	**0.107**	0.84	0.073
		50%	1.06	0.104	**0.84**	0.086	2.05	0.139	0.72	0.084
		60%	1.10	0.111	**0.85**	0.084	2.77	0.156	0.68	0.056
		70%	1.24	0.124	1.06	0.090	3.69	0.212	0.69	0.056
		80%	3.38	0.154	5.40	0.233	7.10	0.253	5.01	0.084
		90%	8.02	0.216	9.48	0.214	8.24	0.139	7.06	0.251
$\mathcal{D}_{\text{p2pl}}(\mathcal{M}_2)$	*Translation* (m)	0.50 m	1.34	0.128	1.07	0.097	4.43	0.279	0.90	0.084
		1.00 m	1.30	0.120	1.11	0.109	3.93	0.246	0.92	0.097
		1.50 m	1.24	0.121	1.12	0.102	3.83	0.227	0.78	0.051
		2.00 m	1.13	0.103	1.05	0.103	3.21	0.188	1.00	0.053
		2.50 m	1.13	0.106	0.98	0.093	3.50	0.189	0.81	0.060
		3.00 m	1.22	0.123	1.24	0.118	2.26	0.156	0.97	0.112
	Inliers Ratio (%)	40%	1.03	0.087	1.00	0.085	2.09	0.124	0.92	0.098
		50%	1.02	0.090	0.92	0.087	3.18	0.148	1.06	0.133
		60%	1.24	0.108	1.11	0.104	3.24	0.168	0.98	0.050
		70%	1.37	0.129	1.11	0.106	3.87	0.225	1.16	0.062
		80%	2.60	0.174	2.62	0.216	7.61	0.250	7.99	0.240
		90%	14.02	0.630	12.10	0.577	11.33	0.611	16.03	0.472
$\text{ICP}_{\mathcal{D},\text{p2p}}$ [1]	*Translation* (m)	0.50 m	3.03	0.180	2.56	0.133	5.63	0.297	1.20	0.070
		1.00 m	2.01	0.154	1.76	0.117	4.46	0.235	**0.57**	0.067
		1.50 m	1.56	0.133	1.50	0.107	2.97	0.180	0.70	0.052
		2.00 m	1.40	0.123	1.19	0.100	2.56	0.180	0.67	0.056
		2.50 m	1.24	0.110	1.13	0.082	3.55	0.181	0.67	0.045
		3.00 m	1.33	0.113	1.59	0.095	3.60	0.186	0.67	**0.038**
	Inliers Ratio (%)	40%	1.10	0.099	1.07	0.092	3.36	0.192	**0.56**	0.055
		50%	1.32	0.116	1.16	0.082	2.95	0.158	0.72	0.065
		60%	1.48	0.133	1.21	0.100	3.15	0.169	0.64	0.048
		70%	1.80	0.147	1.60	0.110	4.32	0.209	0.88	0.070
		80%	2.54	0.169	2.14	0.111	5.24	0.247	0.83	0.057
		90%	5.14	0.298	4.01	0.177	10.01	0.403	2.63	0.099
$\text{ICP}_{\mathcal{D},\text{p2pl}}$ [2]	*Translation* (m)	0.50 m	1.61	0.146	1.42	0.111	4.14	0.275	1.48	0.149
		1.00 m	1.37	0.114	1.28	0.103	3.71	0.238	1.06	0.095
		1.50 m	1.25	0.122	1.19	0.097	2.82	0.207	0.77	0.077
		2.00 m	1.13	0.115	0.91	0.070	2.45	0.164	0.71	0.097
		2.50 m	1.09	0.095	1.02	0.105	3.02	0.187	0.66	0.081
		3.00 m	1.68	0.112	1.22	0.103	3.55	0.140	0.75	0.090
	Inliers Ratio (%)	40%	**0.88**	**0.082**	1.06	0.106	**1.37**	**0.106**	0.78	0.073
		50%	1.01	0.097	**0.83**	0.093	2.07	0.140	0.72	0.084
		60%	1.08	0.109	**0.82**	0.075	2.71	0.154	0.82	0.070
		70%	1.24	0.123	0.93	0.090	3.91	0.207	0.69	0.056
		80%	3.34	0.155	5.39	0.231	7.32	0.257	4.92	0.081
		90%	7.98	0.214	9.45	0.217	8.26	0.136	6.97	0.251

Legend: distance map representation level ($\mathcal{M}_l$); Relative Translational Error (RTE); Relative Rotational Error (RRE); RTE (%) and RRE (°m⁻¹) reported as mean over 10 m segments. Bold entries indicate lowest drift within 0.05 % RTE and 0.005 °m⁻¹ RRE. Underlined entries indicate lowest drift within method–splitting setups.

RicoSLAM Framework



INESC TEC

U.PORTO
FEUP FACULDADE DE ENGENHARIA
UNIVERSIDADE DO PORTO

Why RicoSLAM (naming)?

MRDT Naming Tradition

- Adam & Eve (twin four-wheeled omnidirectional platform)
- Friday (four-wheeled omnidirectional platform)
- Jarvis (original testbed for the navigation stack)
- Moses (Flowbotic platform, Portuguese maker)
- Jack, Rosie, Scuffy, …
- …

Rico (Madagascar)

- Named by my co-supervisor Héber Miguel Sobreira

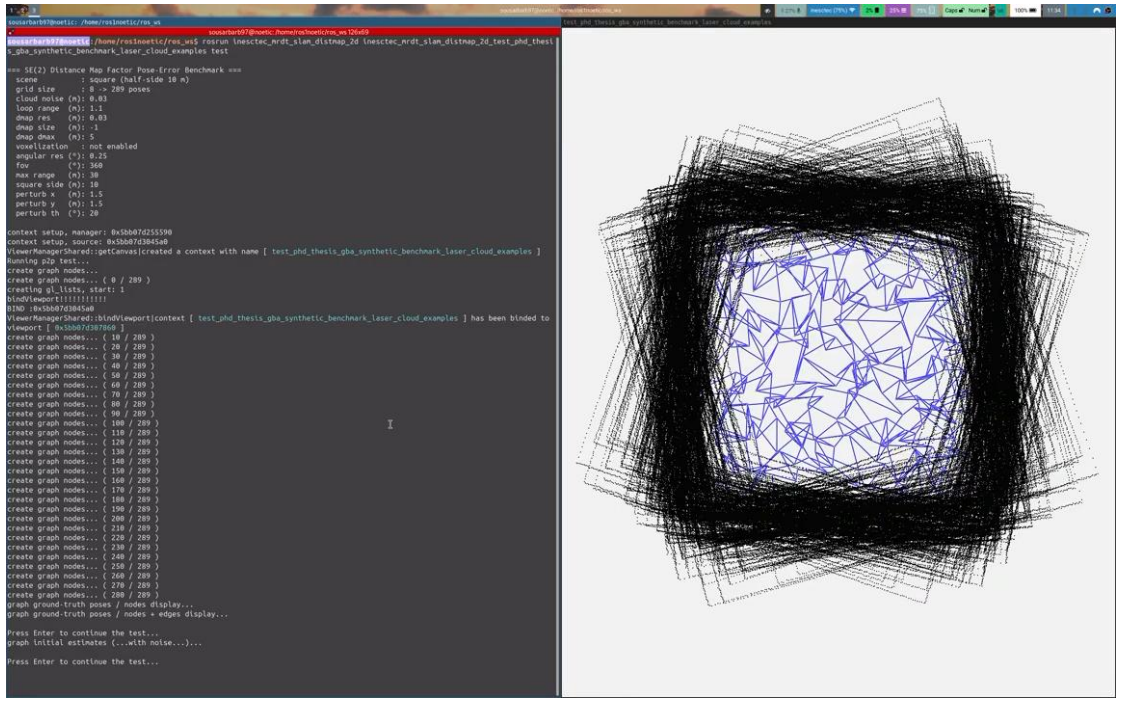


Ricardo B. Sousa

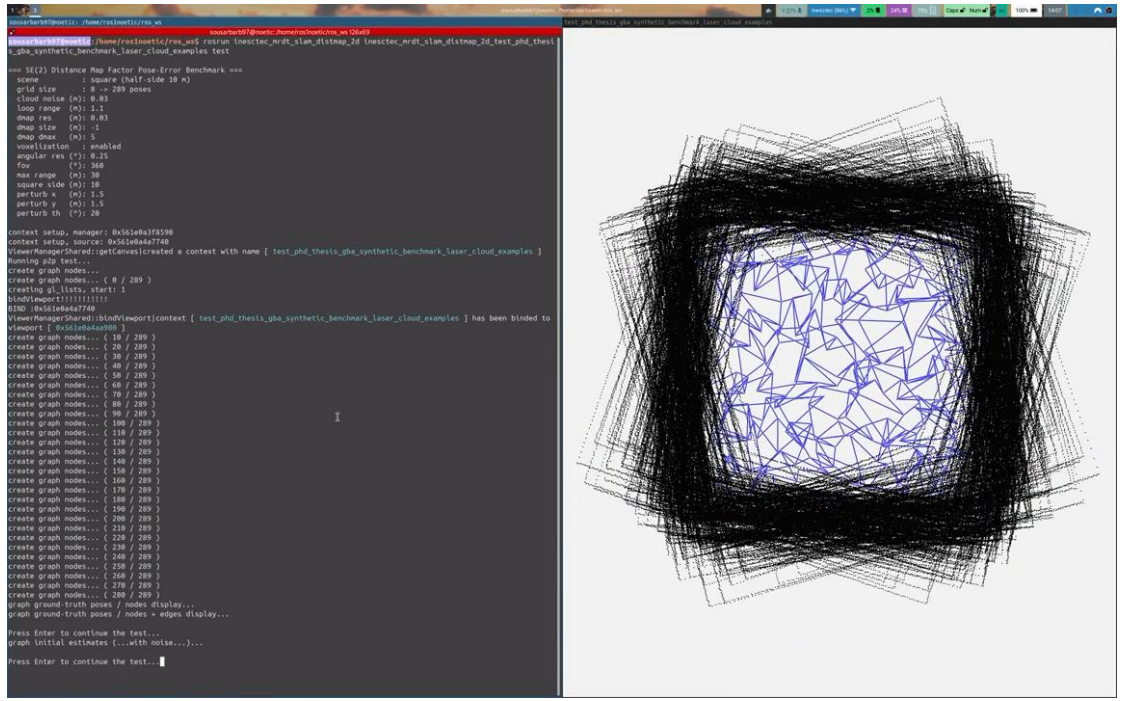
- The thesis is about estimating rotations
- Turns out I've carried a 5º yaw bias since birth on my pinguin-like feet

Rico SLAM

RESULTS: GBA ON SYNTHETIC DATA



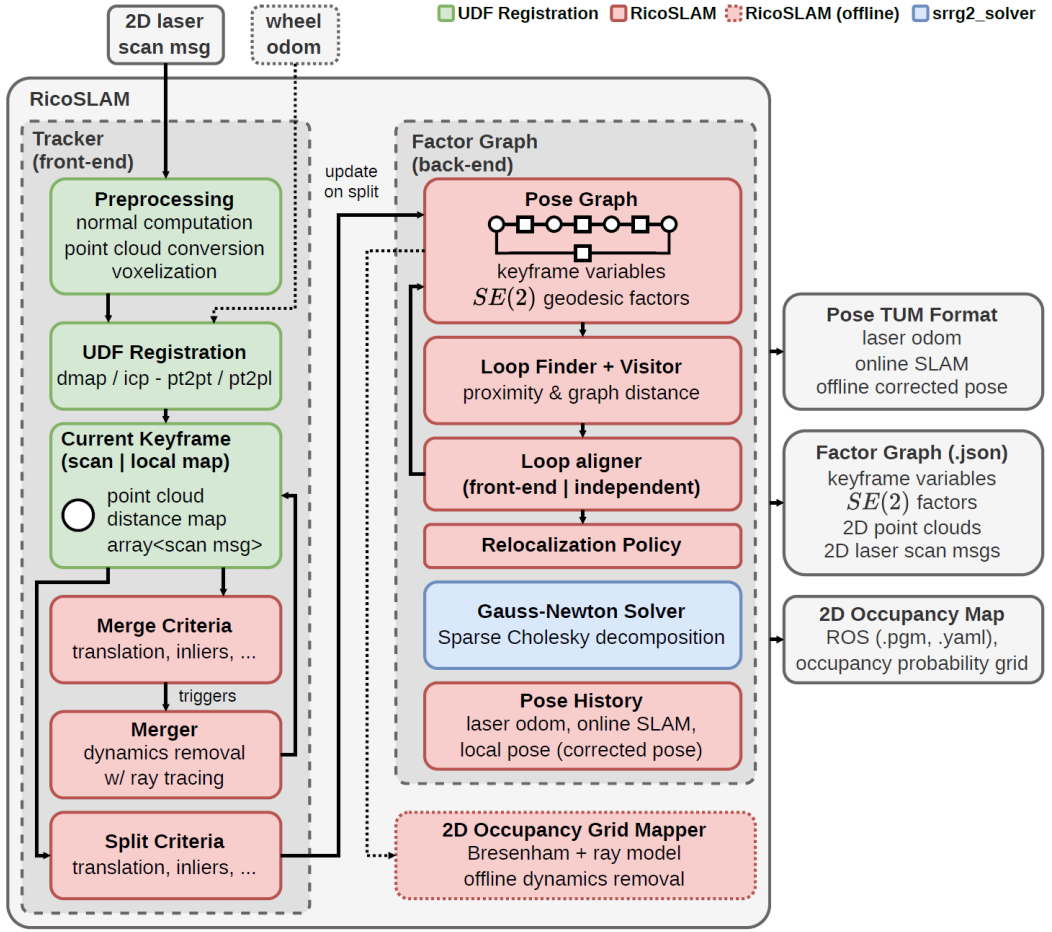
*GBA $\mathcal{D}_{p2p}(\mathcal{M}_0)$ on Square Scene w/ 0.03 m Gaussian noise added to range data **w/o voxelization** (https://youtu.be/gO-Se5RBQEE)*



*GBA $\mathcal{D}_{p2p}(\mathcal{M}_1)$ on Square Scene w/ 0.03 m Gaussian noise added to range data **w/ voxelization** (https://youtu.be/AOuAD12Ga6g)*

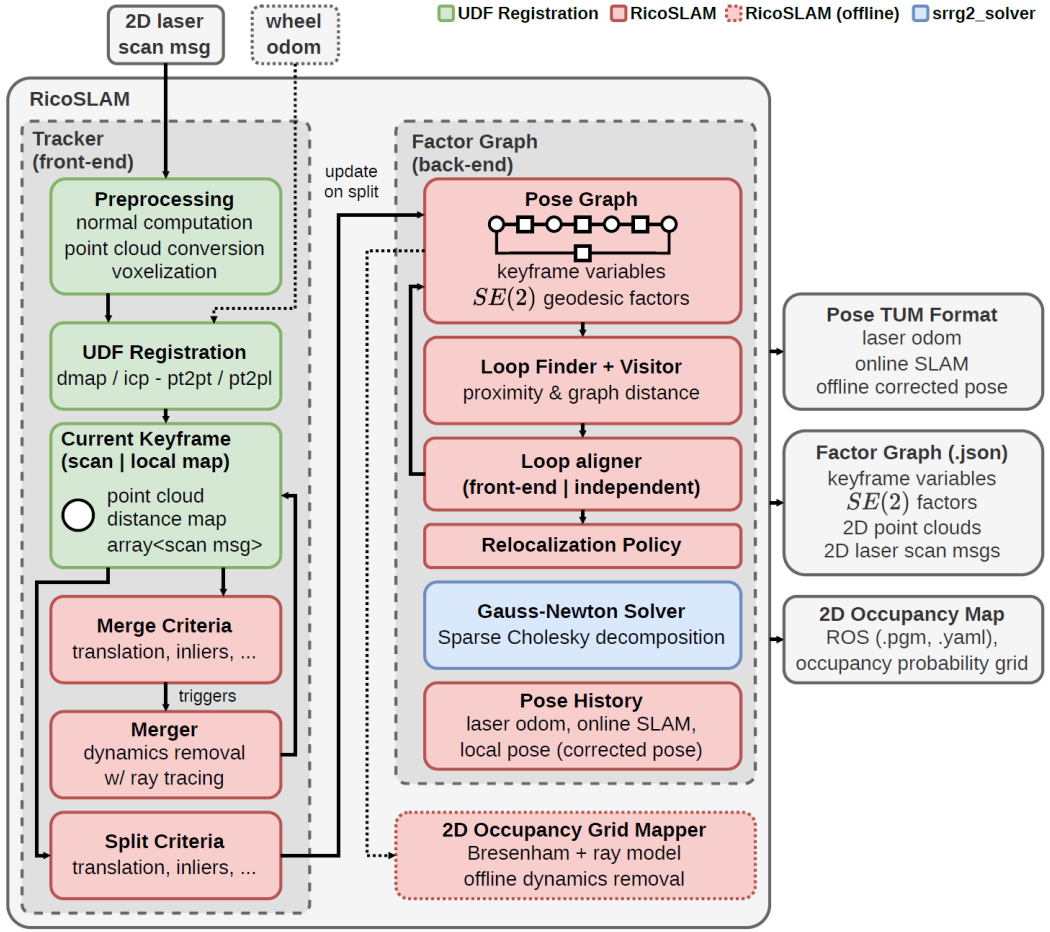
IEEE RA-L Paper

- *"RicoSLAM: Framework for Localization and Mapping in Dynamic Environments"*
- **Main Contributions:**
  - Inlier ratio-based splitting criterion
  - Relocalization policy
  - Ray tracing-based dynamics removal
    - Online on local map
    - Offline before rasterization
- **Key Claims:**
  - < 0.05m ATE on IILABS 3D w/o wheel odometry
  - 1–2 orders of magnitude fewer nodes than baselines
  - Back-end bounded by explored area instead of time
  - Cleaner / more stable occupancy grid representation



IEEE RA-L Paper

- *"RicoSLAM: Framework for Localization and Mapping in Dynamic Environments"*
- **Differences to PhD Thesis:**
  - UDF-based front-end from ROBOT2026 paper
  - Include Cartographer in experimental evaluation
  - Focus on SLAM and dynamics removal
  - Remove global bundle adjustment topic to limit the scope of the paper (also, not yet stable on real-world data)



IEEE RA-L Paper: Trajectory Accuracy Evaluation

Method	Variant	Nav A Diff ATE (m)	Nav A Diff σ_{ATE} (m)	Nav A Diff RTE (%)	Nav A Diff RRE (°/m)	Nav A Omni ATE (m)	Nav A Omni σ_{ATE} (m)	Nav A Omni RTE (%)	Nav A Omni RRE (°/m)	Loop ATE (m)	Loop σ_{ATE} (m)	Loop RTE (%)	Loop RRE (°/m)	Slippage ATE (m)	Slippage σ_{ATE} (m)	Slippage RTE (%)	Slippage RRE (°/m)
GMapping [16]	0.3m 0.3rad	0.148	0.069	1.77	0.111	0.126	0.046	2.16	0.093	0.122	0.075	3.04	0.117	0.055	0.025	1.85	0.310
	0.5m 0.5rad	0.127	0.047	2.01	0.120	0.122	0.038	1.72	0.116	0.114	0.068	3.67	0.129	0.048	0.021	3.49	0.272
	1.0m 1.0rad	0.117	0.030	2.04	0.167	0.118	0.029	2.08	0.170	0.113	0.069	3.25	0.148	0.061	0.031	4.19	0.320
SLAM Toolbox [21]	0.3m	0.075	0.046	**0.86**	0.084	0.081	0.044	1.40	0.105	0.068	0.037	1.69	0.061	0.042	0.021	5.13	0.326
	0.5m	0.053	0.027	0.94	0.121	0.061	0.025	1.89	0.175	0.040	0.021	1.68	0.055	0.064	0.029	6.00	0.487
Cartographer [20] (default)	scan-only	0.113	0.071	1.01	0.078	0.033	0.017	**0.72**	0.071	0.119	0.083	1.42	0.068	**0.015**	**0.005**	0.80	0.111
	scan-only*	0.072	0.050	1.09	0.087	0.029	0.015	0.88	0.073	**0.029**	**0.015**	0.99	0.076	**0.013**	**0.005**	**0.34**	**0.042**
	scan+wheel	0.098	0.060	1.04	0.091	0.030	**0.014**	0.98	0.090	0.114	0.085	1.23	0.059	–	–	–	–
	scan+wheel*	0.069	0.046	1.17	0.104	0.033	0.015	**0.77**	0.070	**0.025**	**0.013**	1.07	0.083	0.055	0.031	0.43	0.097
Cartographer [20] (0.5m 0.5rad)	scan-only	**0.047**	0.025	0.99	0.079	0.034	0.020	0.79	0.076	–	–	–	–	**0.017**	**0.007**	0.70	0.069
	scan-only*	0.060	0.035	0.99	0.081	0.034	0.018	0.86	0.075	0.131	0.106	1.56	0.083	**0.015**	**0.007**	0.48	0.058
	scan+wheel	**0.046**	0.028	1.01	0.098	**0.023**	**0.009**	**0.77**	**0.065**	0.101	0.083	1.32	0.071	–	–	–	–
	scan+wheel*	**0.046**	0.026	**0.84**	**0.074**	0.031	0.017	0.82	0.068	0.111	0.086	1.49	0.074	0.044	0.026	0.49	0.074
Cartographer [20] (1.0m 1.0rad)	scan-only	0.092	0.052	1.10	0.083	0.038	0.019	0.87	0.067	–	–	–	–	0.020	**0.010**	0.79	0.139
	scan-only*	0.091	0.052	1.00	0.083	0.062	0.032	0.84	0.077	0.182	0.106	2.08	0.089	0.020	**0.008**	0.62	0.050
	scan+wheel	0.059	0.029	1.02	0.100	**0.025**	**0.010**	0.78	**0.061**	0.091	0.053	1.14	0.057	–	–	–	–
	scan+wheel*	0.061	0.034	1.03	0.083	0.032	0.015	0.93	0.078	0.154	0.081	1.88	0.078	0.050	0.031	**0.39**	0.061
RicoSLAM	scan	**0.043**	0.026	**0.82**	**0.074**	0.049	0.028	0.92	0.078	**0.029**	**0.017**	0.95	0.056	**0.014**	**0.006**	0.66	0.057
	scan w/reloc.	0.062	0.037	1.02	0.088	0.074	0.049	1.07	0.099	0.079	0.071	1.66	0.081	**0.013**	**0.005**	0.68	0.059
	local	**0.045**	**0.019**	0.91	0.077	0.042	0.018	0.98	0.082	0.039	**0.015**	**0.94**	0.062	**0.015**	**0.006**	0.83	0.083
	local w/upd.	0.056	0.037	0.88	**0.074**	0.030	0.015	0.95	0.076	0.032	**0.014**	**0.89**	0.055	**0.015**	**0.006**	0.63	0.062
	local w/reloc.	0.093	0.046	**0.87**	**0.070**	0.077	0.028	0.92	0.080	0.042	**0.018**	1.24	**0.048**	**0.015**	**0.007**	0.75	0.062
RicoSLAM†	scan	**0.022**	**0.009**	0.85	**0.065**	**0.023**	**0.010**	0.84	0.073	**0.019**	**0.008**	1.13	**0.046**	**0.015**	**0.005**	0.70	0.050
	scan w/reloc.	0.043	0.017	1.08	0.082	0.045	0.020	1.04	0.086	**0.021**	**0.010**	1.09	**0.047**	**0.013**	**0.005**	0.68	0.059
	local	0.034	0.015	**0.84**	0.067	0.032	0.016	0.94	0.071	0.029	**0.010**	1.06	**0.046**	**0.016**	**0.006**	0.69	0.064
	local w/upd.	**0.027**	**0.012**	0.86	**0.063**	**0.026**	**0.012**	0.95	0.068	0.026	**0.009**	1.09	**0.046**	**0.015**	**0.006**	0.70	0.080
	local w/reloc.	0.034	**0.014**	**0.79**	**0.061**	0.031	0.016	0.89	0.077	0.038	0.015	1.28	**0.042**	**0.015**	**0.007**	0.75	0.062

Bold indicates best of online vs offline within 0.005 m (ATE), 0.05 % (RTE), 0.005 ° m^{-1} (RRE). Underlined indicates best within variants.
* enable the real-time correlative scan matcher of Cartographer [20], instead of the default fast version for the matching process.
† offline corrected pose stream of the configurations from the online RicoSLAM results.

IEEE RA-L Paper: Scalability Results

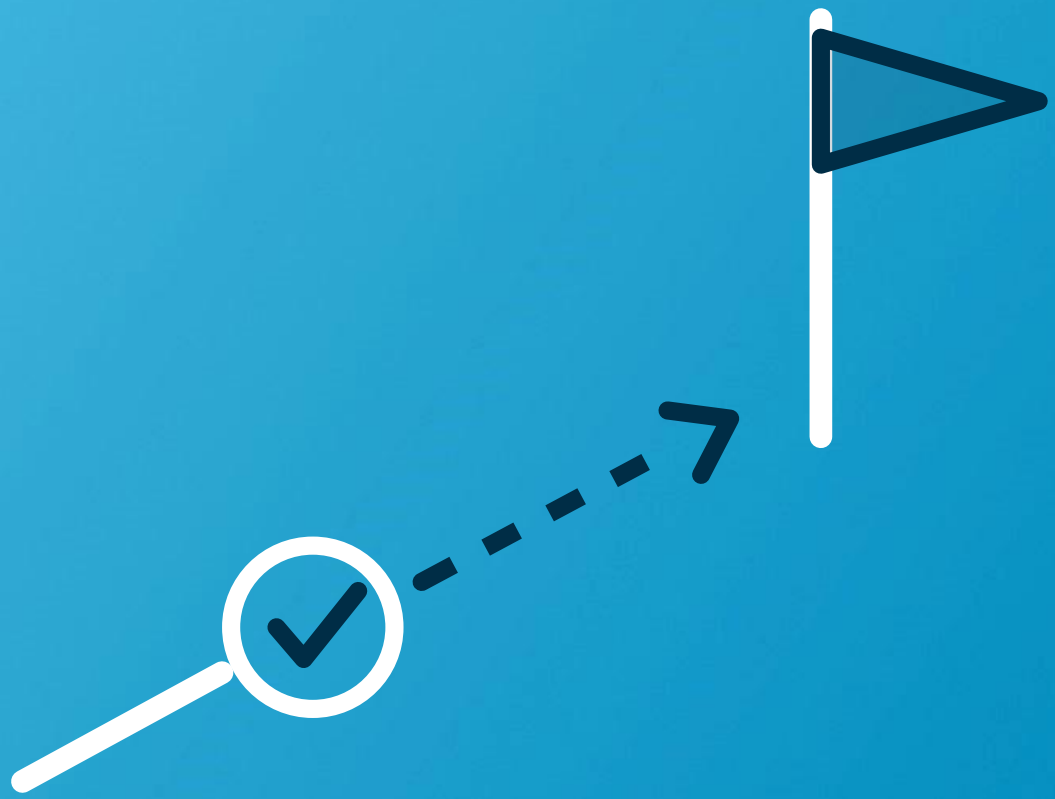
Method	Variant	Nav A Diff #node	Nav A Diff #scan	Nav A Diff #edge	Nav A Omni #node	Nav A Omni #scan	Nav A Omni #edge
GMapping [16]	0.3m 0.3rad	–	897	–	–	373	–
	0.5m 0.5rad	–	544	–	–	226	–
	1.0m 1.0rad	–	273	–	–	113	–
SLAM Toolbox [21]	0.3m	541	541	2488	222	222	409
	0.5m	310	310	470	126	126	163
Cartographer [20] (default)	scan-only	2814	2814	14 812	1629	1629	5836
	scan-only*	2810	2810	14 736	1616	1616	5645
	scan+wheel	2817	2817	14 766	1636	1636	5832
	scan+wheel*	2810	2810	14 669	1625	1625	5844
Cartographer [20] (0.5m 0.5rad)	scan-only	544	544	2622	222	222	664
	scan-only*	544	544	2603	222	222	661
	scan+wheel	543	543	2652	222	222	659
	scan+wheel*	544	544	2627	222	222	656
Cartographer [20] (1.0m 1.0rad)	scan-only	273	273	1128	112	112	290
	scan-only*	273	273	1146	112	112	288
	scan+wheel	273	273	1153	112	112	290
	scan+wheel*	273	273	1152	112	112	286
RicoSLAM	scan	77	77	173	30	30	46
	scan w/reloc.	22	**22**	**22**	18	**18**	**18**
	local	106	376	248	43	158	66
	local w/upd.	106	376	246	43	158	66
	local w/reloc.	**20**	96	29	**16**	79	19

Method	Variant	Loop #node	Loop #scan	Loop #edge	Slippage #node	Slippage #scan	Slippage #edge
GMapping [16]	0.3m 0.3rad	–	757	–	–	141	–
	0.5m 0.5rad	–	459	–	–	84	–
	1.0m 1.0rad	–	231	–	–	42	–
SLAM Toolbox [21]	0.3m	458	458	1291	77	77	111
	0.5m	261	261	393	46	46	59
Cartographer [20] (default)	scan-only	2455	2455	11 213	204	204	326
	scan-only*	2437	2437	10 777	197	197	310
	scan+wheel	2455	2455	11 202	–	–	–
	scan+wheel*	2452	2452	11 111	196	196	307
Cartographer [20] (0.5m 0.5rad)	scan-only	–	–	–	77	77	145
	scan-only*	460	460	2073	77	77	145
	scan+wheel	460	460	2085	–	–	–
	scan+wheel*	460	460	2074	77	77	145
Cartographer [20] (1.0m 1.0rad)	scan-only	–	–	–	39	39	73
	scan-only*	231	231	908	39	39	74
	scan+wheel	231	231	953	–	–	–
	scan+wheel*	231	231	927	39	39	73
RicoSLAM	scan	83	83	155	10	10	11
	scan w/reloc.	19	**19**	19	**8**	**8**	**7**
	local	89	328	168	15	48	17
	local w/upd.	89	328	168	15	48	17
	local w/reloc.	**18**	83	**18**	**8**	34	**7**

IEEE RA-L Paper: Total Computation Time

Method	Variant	Nav A Diff (s)	Nav A Omni (s)	Loop (s)	Slippage (s)
GMapping [16]	0.3m 0.3rad	673.742	278.550	582.317	102.783
	0.5m 0.5rad	407.757	168.363	352.890	62.899
	1.0m 1.0rad	207.233	85.411	177.995	32.533
Cartographer [20] (default)	scan-only	46.666	18.642	43.745	3.792
	scan-only*	139.121	70.536	121.174	19.091
	scan+wheel	49.105	20.261	45.486	–
	scan+wheel*	142.860	74.093	124.660	19.635
Cartographer [20] (0.5m 0.5rad)	scan-only	28.632	13.964	–	3.272
	scan-only*	129.761	67.024	112.056	18.914
	scan+wheel	32.776	16.205	29.196	–
	scan+wheel*	134.169	69.024	116.307	19.597
Cartographer [20] (1.0m 1.0rad)	scan-only	**26.859**	**13.391**	–	**3.257**
	scan-only*	127.548	65.976	109.503	18.612
	scan+wheel	31.217	15.743	**26.563**	–
	scan+wheel*	132.227	68.418	113.216	19.424
RicoSLAM	scan	82.978	41.475	67.406	10.401
	scan w/reloc.	101.336	48.628	75.116	10.099
	local	90.174	41.374	70.339	9.556
	local w/upd.	91.164	41.636	70.915	9.643
	local w/reloc.	113.148	50.272	77.608	10.841
RicoSLAM (no OpenGL)	scan	**20.378**	**11.344**	**16.967**	**2.631**
	scan w/reloc.	48.259	21.872	32.297	3.765
	local	25.320	12.389	20.008	3.084
	local w/upd.	26.110	12.464	20.581	3.125
	local w/reloc.	58.099	22.934	33.771	4.565

Conclusions & Future Work



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Curriculum Vitae

Publications & Data

17 peer-reviewed publications

6 within PhD Thesis scope

5 journal articles

12 conference papers

IILABS 3D public dataset

DOI 10.25747/VHNJ-WM80

Citation Metrics as of September 16, 2026

	#citations	*h-index*
Google Scholar	**273**	**7**
Scopus	**191**	**5**
Web of Science	**155**	**5**

Funding & Projects

FCT PhD Scholarship

DOI 10.54499/2021.04591.BD

4 R&D projects

GreenAuto, PRODUTECH R3, VINCI 7D, 5G & Digital Transformation Test Bed

Academic Experience

Visiting Researcher, Sapienza University of Rome

Prof. Dr. Giorgio Grisetti, Feb – Jul 2024 (6 months)

Invited Assistant, FEUP

2021 – 2023

10 students co-supervised

5 BSc internships, 5 BII fellows

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